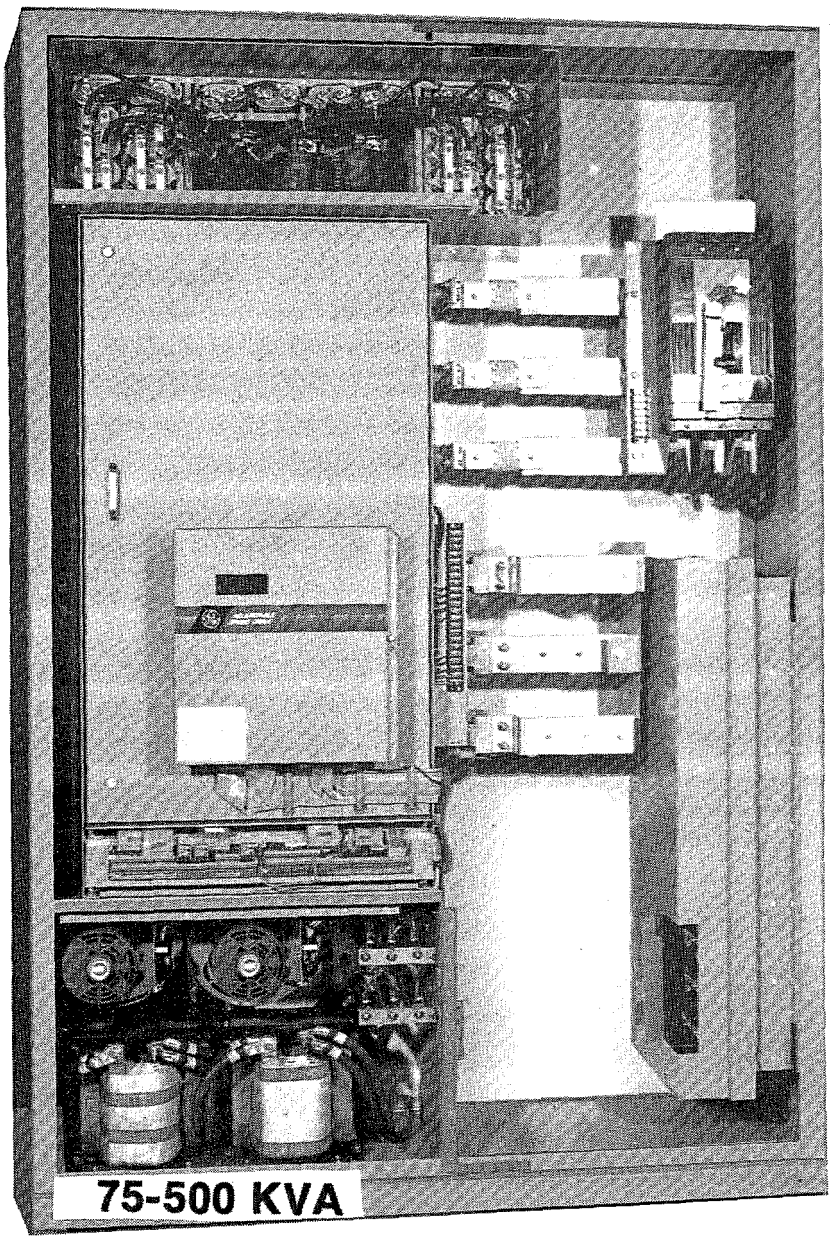
GE Drives

INSTRUCTIONS



AC-500™ Adjustable Speed Drives

TM TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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WARNINGS, CAUTIONS, AND NOTES AS USED ON EQUIPMENT

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

WHEN A WARNING LABEL IS PROVIDED IT WILL BE RED IN COLOR WITH BLACK OR WHITE LETTERING.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF THE EQUIPMENT.

WHEN A CAUTION LABEL IS PROVIDED IT WILL BE AMBER IN COLOR WITH BLACK LETTERING.

NOTE

NOTES MERELY CALL ATTENTION TO INFORMATION THAT IS ESPECIALLY SIGNIFICANT IN UNDERSTANDING AND OPERATING THE EQUIPMENT.

WHEN A NOTE LABEL IS PROVIDED IT WILL BE WHITE IN COLOR WITH BLACK LETTERING.

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS.

WARNINGS

WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT GREAT CAUTION MUST BE USED. THE OSCILLOSCOPES CHASSIS SHOULD BE GROUNDED AND A DIFFERENTIAL AMPLIFIER INPUT SHOULD BE USED. CARE SHOULD BE USED IN THE SELECTION OF PROBES AND LEADS AND IN THE ADJUSTMENT OF THE OSCILLOSCOPE SO THAT ACCURATE READINGS MAY BE MADE. SEE OSCILLOSCOPE MANUFACTURER'S INSTRUCTION BOOK FOR PROPER OPERATION AND ADJUSTMENTS TO THE OSCILLOSCOPE.

SOME CONTROLLERS ARE FURNISHED WITH PARTIAL ENCLOSURES OPEN AT TOP AND BOTTOM. THESE ARE INTENDED ONLY FOR MOUNTING IN ANOTHER ENCLOSURE OR IN A CONTROL ROOM HAVING ACCESS BY QUALIFIED PERSONNEL ONLY.

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE CONTROLLERS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE FLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE CONTROLLERS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH AN AC MOTOR SUITABLE FOR USE IN THESE LOCATIONS.

ALL MOTOR BASES AND EQUIPMENT ENCLOSURE HOUSING SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEM.

CAUTIONS

DO NOT REMOVE CONNECTIONS OR PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

DO NOT REMOVE POWER FROM THE DRIVE UNTIL IT HAS FULLY EXECUTED A STOP SEQUENCE AS THIS CAN DAMAGE THE DRIVE SYSTEM.

INSTALLATION WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND BE CONSISTENT WITH ALL LOCAL CODES. SECONDARIES OF 115 VOLT CONTROL TRANSFORMERS TYPICALLY HAVE ONE SIDE FUSED AND THE OTHER GOUNDED OR AVAILABLE FOR GROUNDING BY THE USER.

CAUTIONS

THE ELEMENTARY DIAGRAMS SUPPLIED WITH THE AC-500 DRIVE CONTROLLER IDENTIFY SOME SIGNAL CONNECTIONS AS "LOW LEVEL ANALOG AND/OR DIGITAL SIGNALS" AND INDICATE SPECIAL WIRING PRACTICES. THESE SIGNALS SHOULD BE ISOLATED FROM ALL OTHER WIRING (PARTICULARLY ANY POWER WIRES) AND RUN IN A SEPARATE CONDUIT OR WIRE RUN TO PREVENT POSSIBLE MISOPERATION OF THE CONTROLLER.

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HI POT WITHOUT CONSULTING THE **GE DRIVES DEPARTMENT.**

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN THOSE SHOWN ON THE ELEMENTARY DIAGRAM. CONNECTION OF DEVICES SUCH AS AMMETERS ON THE CURRENT TRANSFORMERS OR VOLTMETERS ON THE TACHOMETER MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING THE **GE DRIVES DEPARTMENT** DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

DO NOT OPERATE THE DRIVE WITH THE PLASTIC AIR FLOW BAFFLES REMOVED FROM THE POWER SECTION.

DO NOT OPERATE THE DRIVE UNDER LOAD WITH THE POWER SECTION DOOR OPEN.

NOTES

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT. FOLLOW THE STARTUP PROCEDURES STEP BY STEP.

READ AND HEED ALL WARNING, CAUTION AND NOTE LABELS POSTED ON THE EQUIPMENT.

RECEIVING AND STORAGE

RECEIVING

The controller should be placed under adequate cover immediately upon receipt; packaging cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVES DEPARTMENT, ERIE, PA.** When seeking assistance please use serial number, requisition number, model number and drive code found on drive nameplate to identify the equipment.
(Telephone 814-455-3219)

STORAGE

If this controller is not to be installed immediately, it should be stored in **a** clean, dry location at ambient temperature from -20°C (-4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage environment exceeds a 15" C (24°F) drop in temperature at 50% humidity over a 4 hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes can also cause condensation.

INSTALLATION

LOCATION

AC-500 controllers are suitable for most factory areas where industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104") and relative humidities up to 90%. It should be recognized, however, that since life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held down between 20°C (68°F) and 30°C (87°F) as opposed to 40°C (104°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

ENVIRONMENTAL CONSIDERATIONS

Environments which include excessive amounts of one or

ENVIRONMENTAL CONSIDERATIONS

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dust, dirt or foreign matter.
2. Vibration or shock.
3. Moisture or vapors.
4. Rapid temperature excursions or high ambient temperatures.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference or "noise" introduced by:

   a. Radio frequency signals, typically from portable transmitters used in the vicinity of the equipment or its wiring.

   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay/contactor/brake coils as a part of or in the vicinity of drive control circuits.

The controller should be positioned to permit heat radiation from all surfaces.

CAUTION

IF CAUSTIC OR CORROSIVE FUMES ARE PRESENT IN THE AREA OF THE MOTOR, THE CONDUIT BETWEEN THE MOTOR AND CONTROLLER SHOULD BE SEALED.

MOUNTING

Cases may be bolted down using 3/8" diameter mounting bolts or studs. If studs are cast in floor, they should extend 3-1/2" minimum above floor. Conduit entry openings through the base are fitted with removable sheet steel covers. Other conduit entry area is available through the top of the case.

WIRING CONSIDERATIONS

Attention should be given to the National Electrical Code and any applicable local codes when installing any drives. Wire size and insulation type, conduit sizing, enclosures, etc., should be determined per these codes.

Additional consideration should also be given to separation of wiring types to minimize electromagnetic interference problems as follows:

1) Wiring should be run in separate conduits or wireways for each type of wiring: signal, control and power.

   a) Signal --Low level analog and digital signals
   --Speed and/or position signals
   --Power supplies and logic signals
   (See system schematics and notes to identify signal level wires)

   b) Control --AC or DC control circuits

   c) Power --3 phase, input or output leads
   --Brakes, 115V AC circuits

2) Signal wiring and power wiring may cross at right angles with a minimum one inch separation. Avoid parallel runs between signal level wires and power or control wires. If signal wires must be run parallel to control or power wires, a minimum of a four-inch separation must be maintained between the wires.

GROUNDING

1) Drive common (PCOMX) should be grounded at only one point, normally located at 3TB or COMTB, as supplied with the controller enclosure. If not properly grounded, high voltages to ground may be present. If the reference is supplied by a process instrument or other source with a grounded common the drive common should not be grounded separately.

2) If an isolation transformer is used, it is recommended that a high resistance ground on the secondary side of the transformer be used.

3) Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

COMMONS

Connection of common for reference, meter, or any other external device can only be made at 3TB-PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, starters, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volts capacitor can be used.

TIGHTEN ALL CONNECTIONS

1) Incoming AC line connections

2) Power connections to motor

3) 1TB, 3TB, 4TB if furnished

NOTE: THIS PAGE WAS LEFT BLANK INTENTIONALLY.

MAIN CONTROLLER ELEMENTS AND THEIR FUNCTIONS

GENERAL DESCRIPTION

The AC-500 adjustable speed AC drive is a digitally controlled programmable drive utilizing microprocessors to provide digital regulation. Drive parameters such as minimum and maximum speed, acceleration and deceleration rates etc., are stored in electrically erasable read only memory (EEROM), and therefore, are accurate and drift free.

Figures 1.2 and 1.6 show the two package sizes that make up the 75-500KVA range of the AC-500 controller covered in this Instruction Book. The 200KVA package is rated 75 to 200KVA (at 460V AC) ; the 500KVA package is rated from 250 to 500KVA (at 460V AC)

The AC-500 controller provides over 180 diagnostic messages which can be decoded via the LED diagnostic readout or an optional programmer module with alphanumeric readout. Diagnostic messages can also be read via an optional coded 8-wire output or over the optional RS422/232C communication link. Drive operating parameters stored in the EEROM may also be examined and changed via the RS422/232C link. In addition, the AC-500 controller provides diagnostics which can check the SCRs and provide a power-up self check down to the printed circuit board level.

The AC-500 controller is a variable voltage, square-wave inverter utilizing a full-wave converter input that will normally accept a 230, 460, or 575 volt three-phase input without the need for an input or output transformer.

The block or '"flow" diagam in Figure 1.1 shows the basic signals, function and power section of an AC-500 controller. AC input power is fed through an optional circuit breaker and AC fuses into the converter section, where it is converted to regulated DC voltage. This voltage is then filtered by an L-C filter whose output is fed into the inverter section where it is switched into variable frequency three-phase voltage feeding the drive motor.

The AC-500 controller uses four main printed circuit cards independent of KVA (Control card, Power Supply/Interface card, Inverter Firing Generator card and Snubber card). Additionally, each package size utilizes its own Converter and Inverter Pulse Transformer cards. The 500KVA package also uses a Commutation Power Supply card as well.

The Control card provides output voltage/frequency regulating loops, and where higher speed accuracy is a requirement, an outer speed loop can be implemented. Individually timed linear acceleration and deceleration are adjustable with rates up to 999.9 seconds. Optional "S-Curve " reference control is also available.

Output metering signals for motor current, frequency, torque and horsepower are also available from the AC-500 controller. In addition, all signals can be brought out as an analog signal via one of two optional digital-to-analog converters or digitally via the optional RS422/232C serial link. These signals and others can be programmed to activate optional programmable signal level detectors.

Optional cards for the AC-500 include the Process Interface card, Application card and Programmer card. These provide the capability of having additional analog voltage or optional current, frequency or digital inputs as references. Likewise a wide variety of tachometer generators can be interfaced to the drive including both AC or DC analog tachometers.

POWER CIRCUITS (See Figures 1.1 through 1.9)

The various modules and components to be described are physically located in the AC-500 power unit as shown in Figures 1.3, 1.4, 1.7 and 1.8 for the two different KVA ratings. These modules and components are also shown in the controller block diagram of Figure 1.1. Following is a description and operating explanation of each system block, starting with the power blocks and finishing with the control blocks.

CONVERTER SECTION

The converter section is a three phase, full wave controlled rectifier bridge which converts the incoming three phase AC power to variable voltage DC power. The six SCR converter is shown in more detail in the power circuit of Figures 1.5 and 1.9. RC snubber circuts (not shown) and MOVs act to protect the converter SCRs against voltage transients. The converter DC output voltage can be adjusted from zero to maximum output by adjusting rhe firing point of each SCR relative to its AC supply phase voltage. The resultant DC output voltage, therefore, contains a six times AC supply frequency ripple component of voltage. The ripple voltage must be filtered to improve the waveform before being applied to the inverter section.

DC LINK FILTER

Iron core reactors, L1 and L2, and a bank of electrolytic capacitors, Cl, act as an LC filter in the DC link, as shown in Figures 1.5 and 1.9. The L1 and L2 reactor filters the converter current supplied to the Cl capacitor which filters the voltage being applied to the inverter section. The air core reactor, LK, limits di/dt levels in the inverter section and also the Cl discharge current if an inverter fault should occur. The Cl capacitor bank also acts to supply motor reactive power.

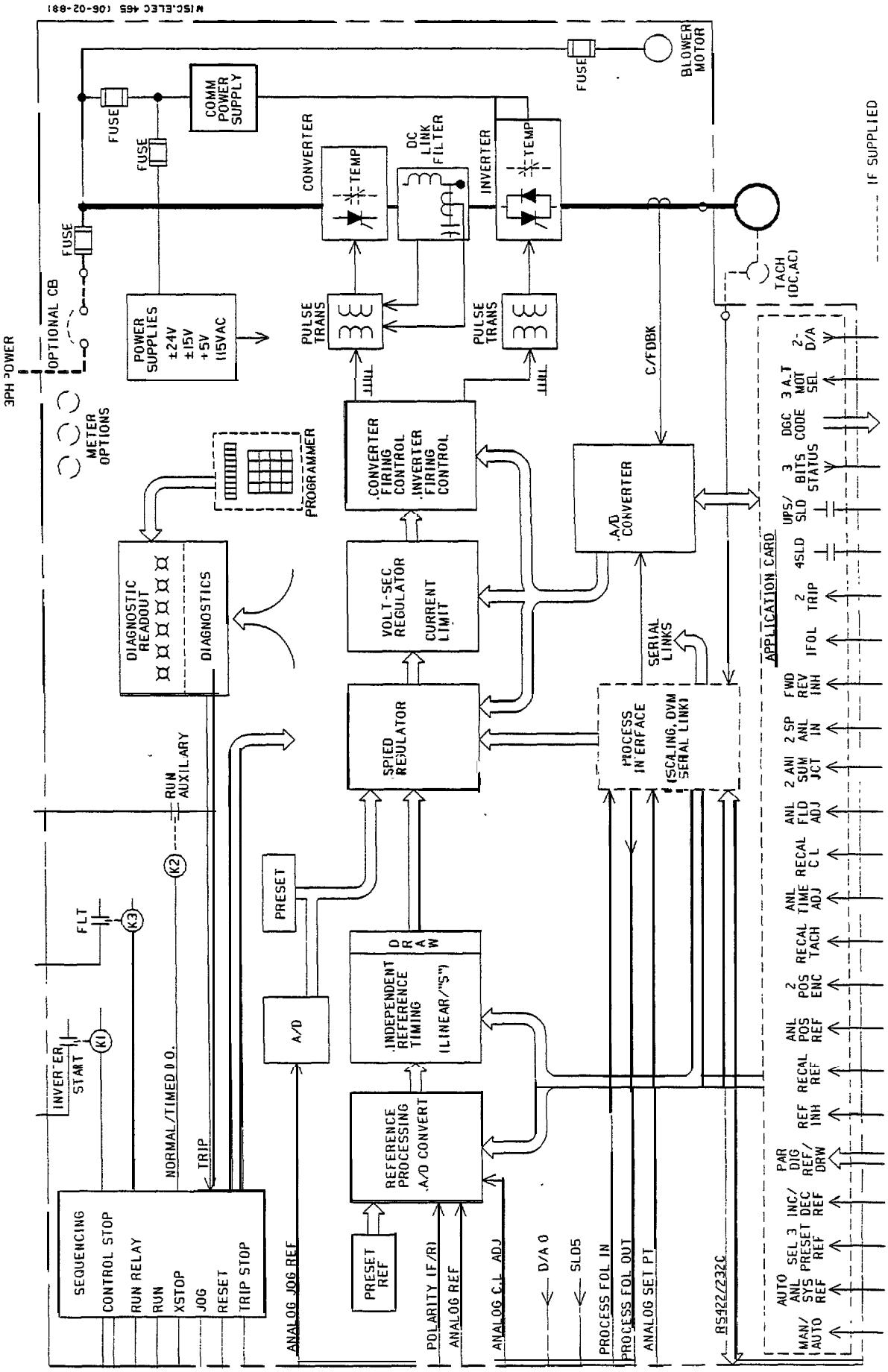


FIGURE 1.1
AC-500 BLOCK DIAGRAM

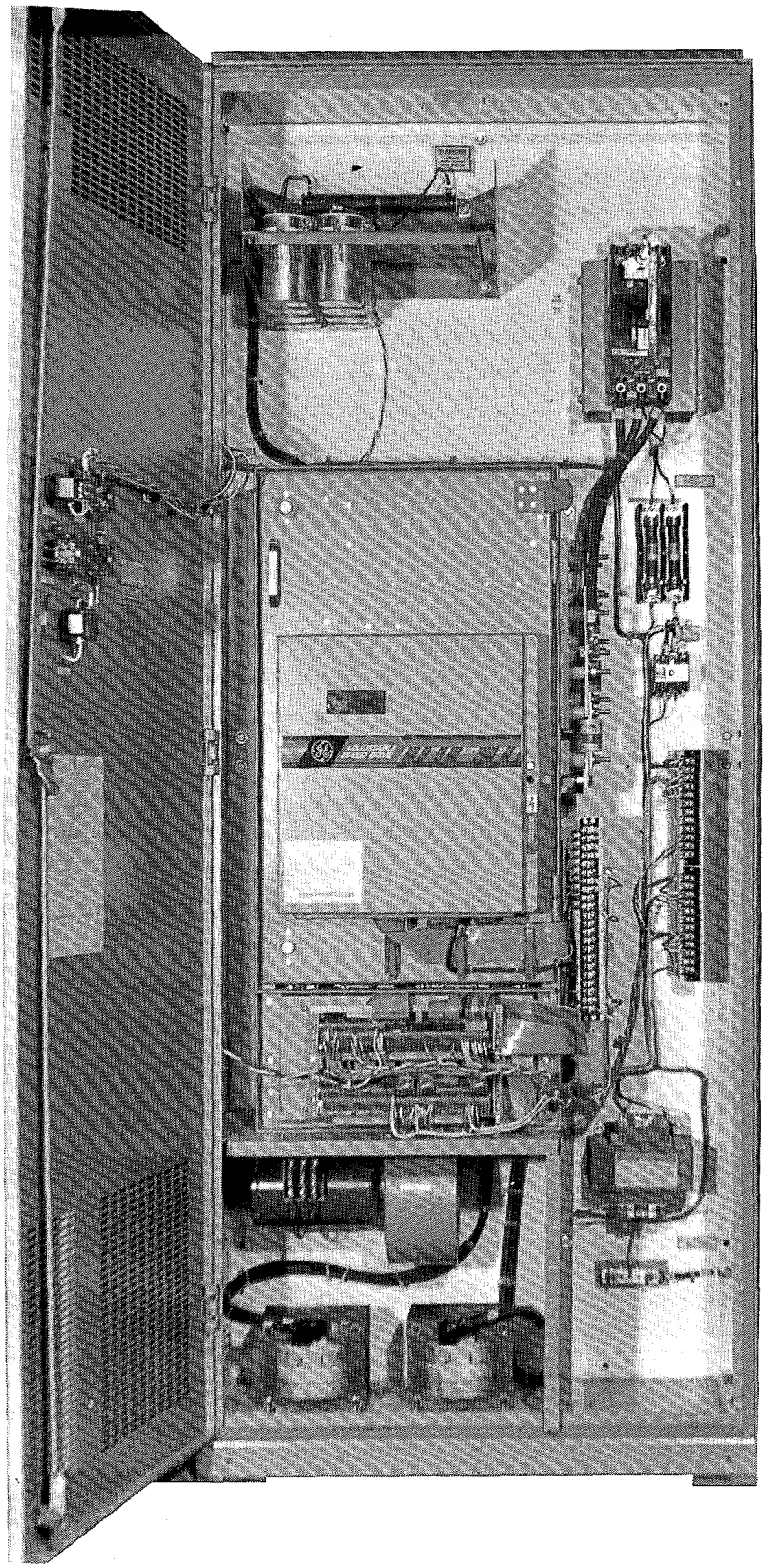


FIGURE 1.2
AC-500 CONTROLLER - 75 TO 200 KVA FRAME

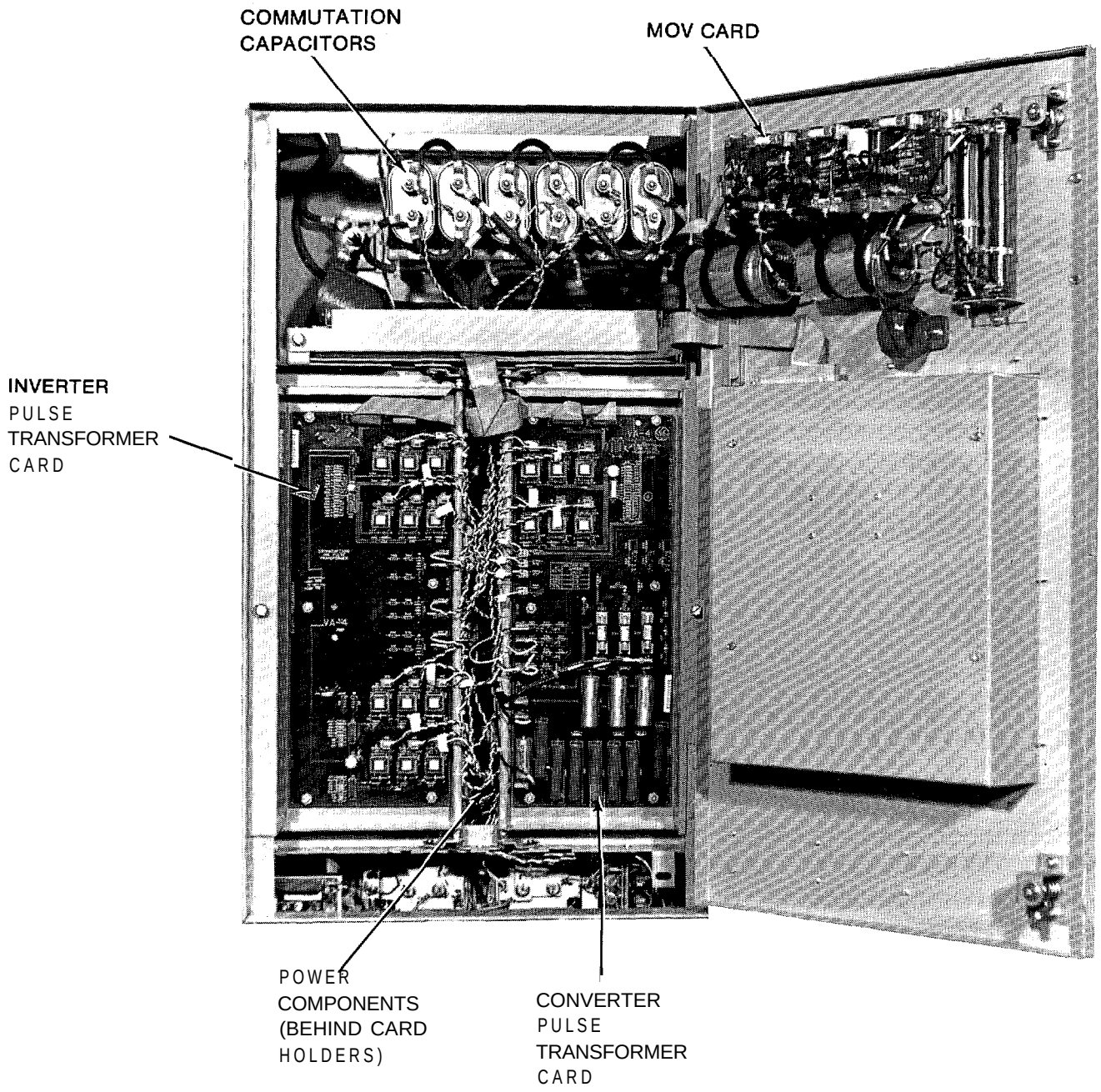


FIGURE 1.3
200 KVA POWER SECTION DOOR OPEN

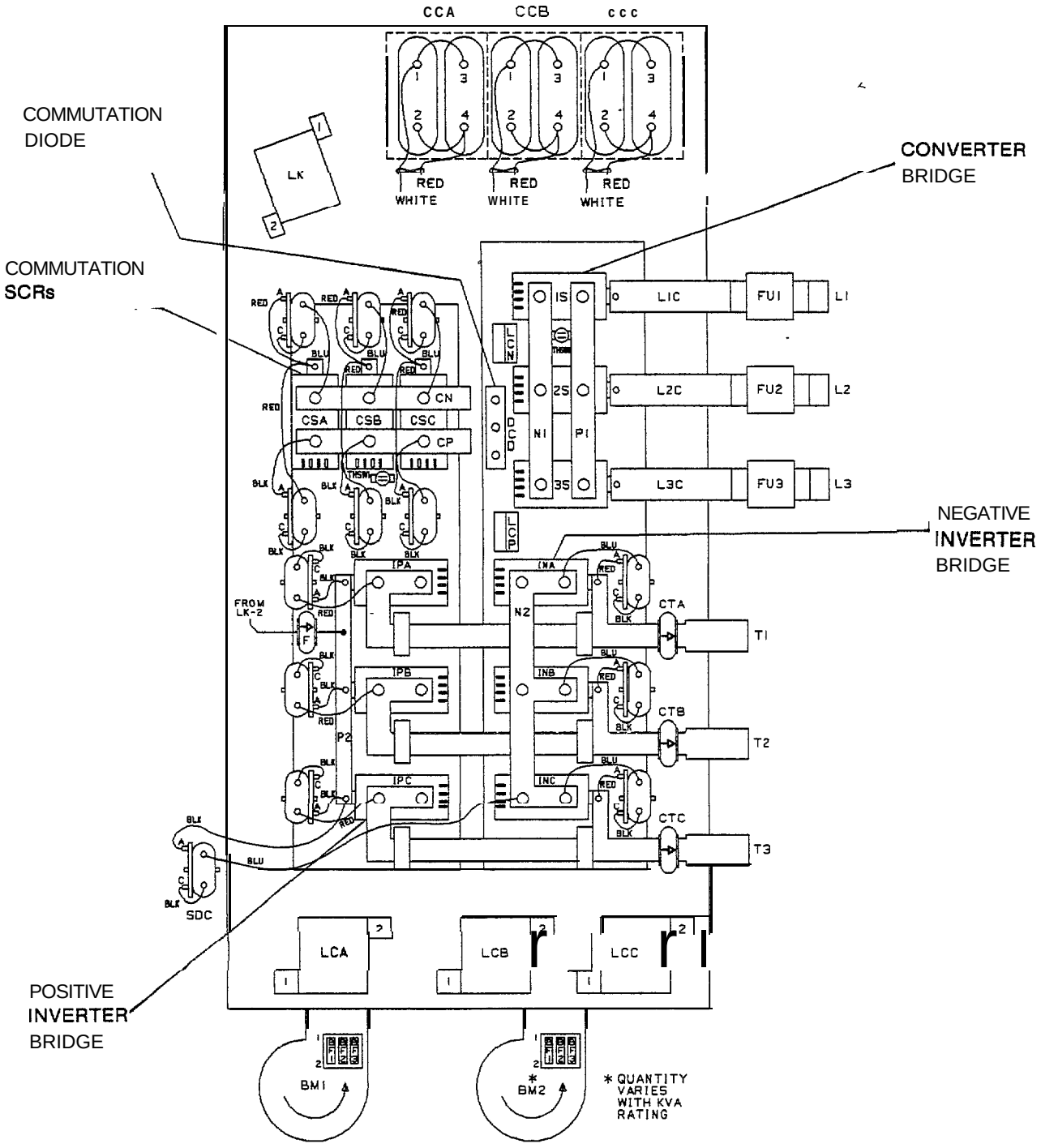


FIGURE 1.4
POWER SECTION LAYOUT (200 KVA FRAME)

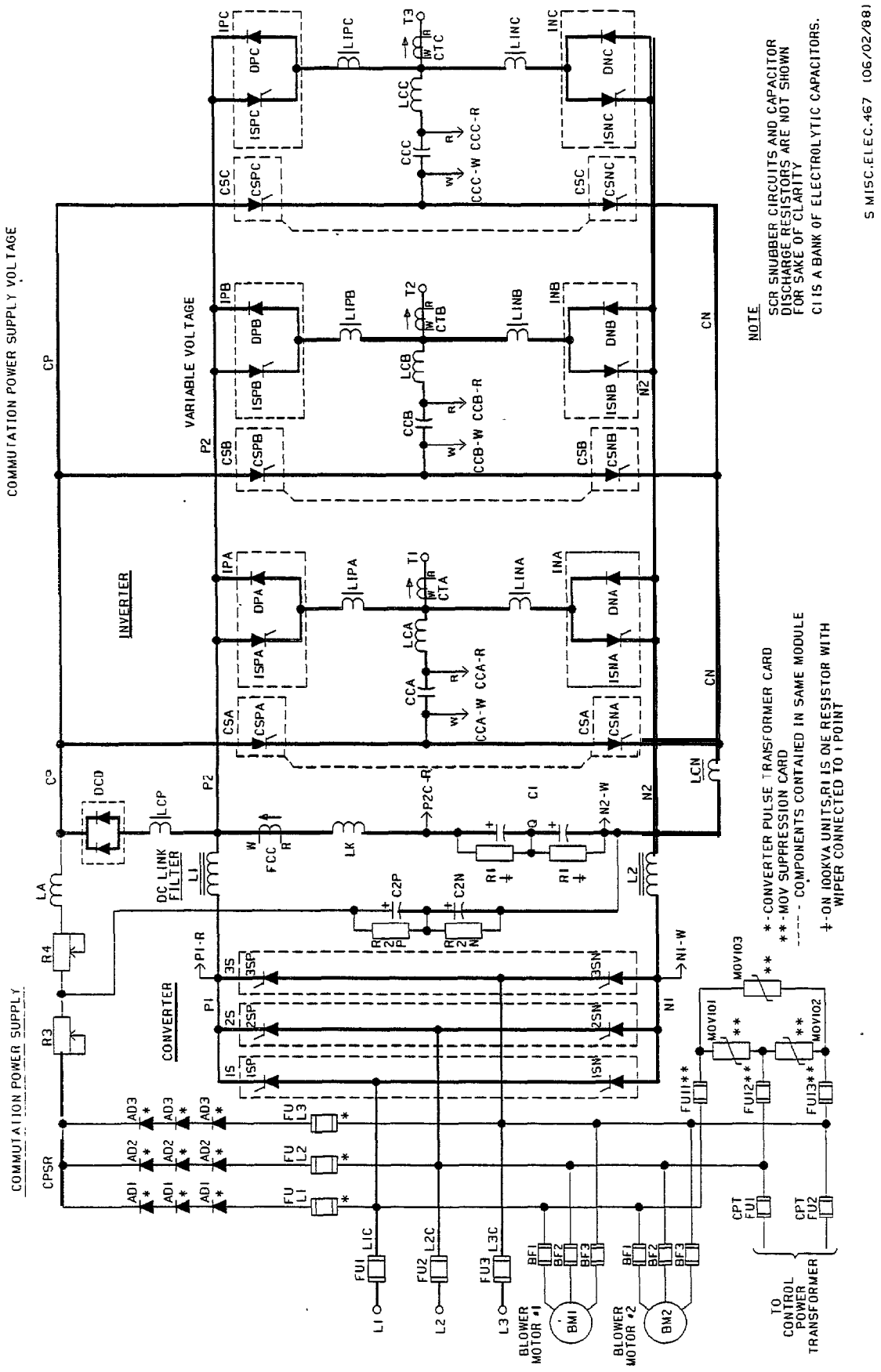


FIGURE 1.5
AC-500 POWER CIRCUIT (200 KVA FRAME)

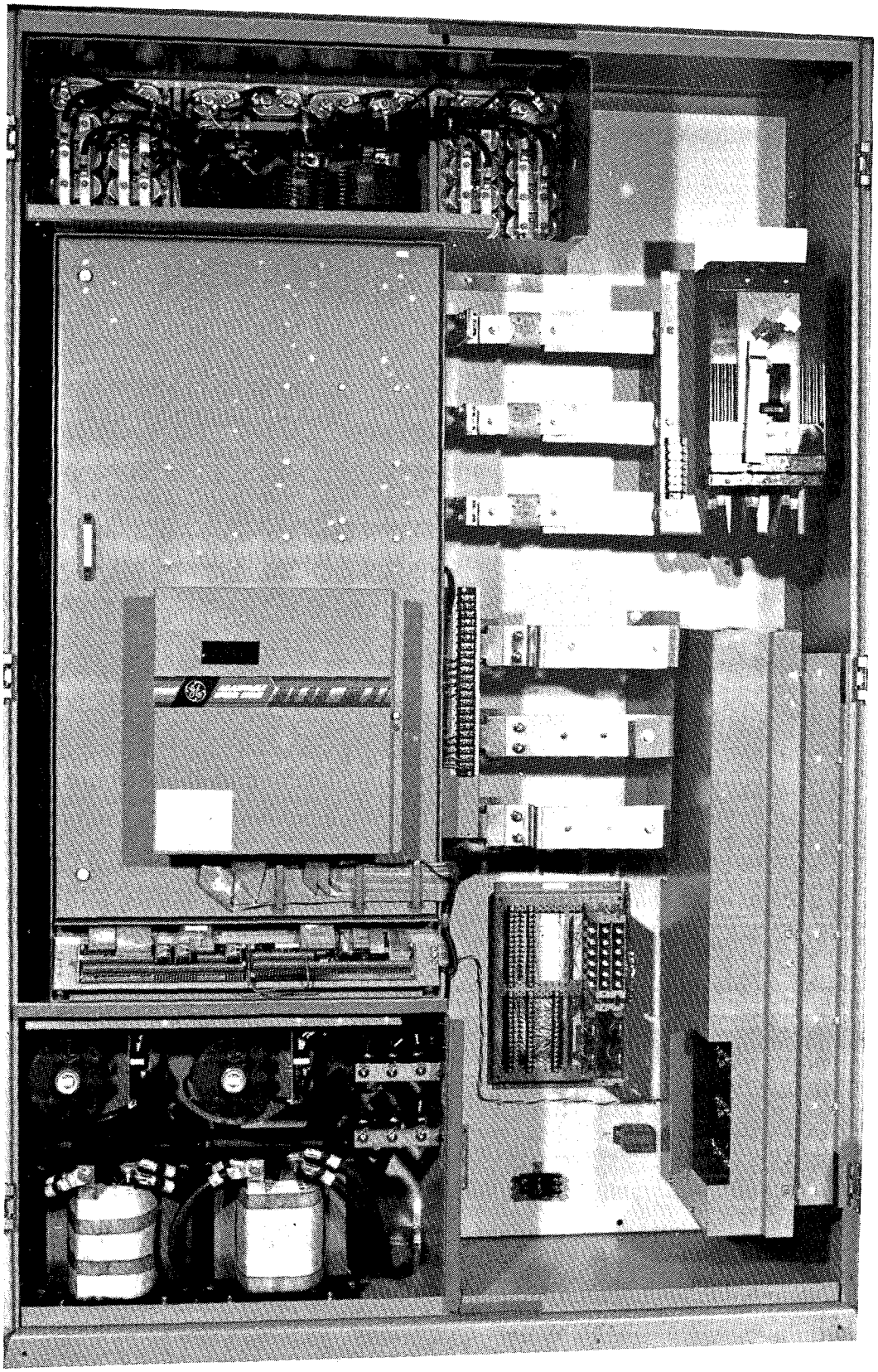


FIGURE 1.6
AC-500 CONTROLLER - 250 TO 500 KVA FRAME

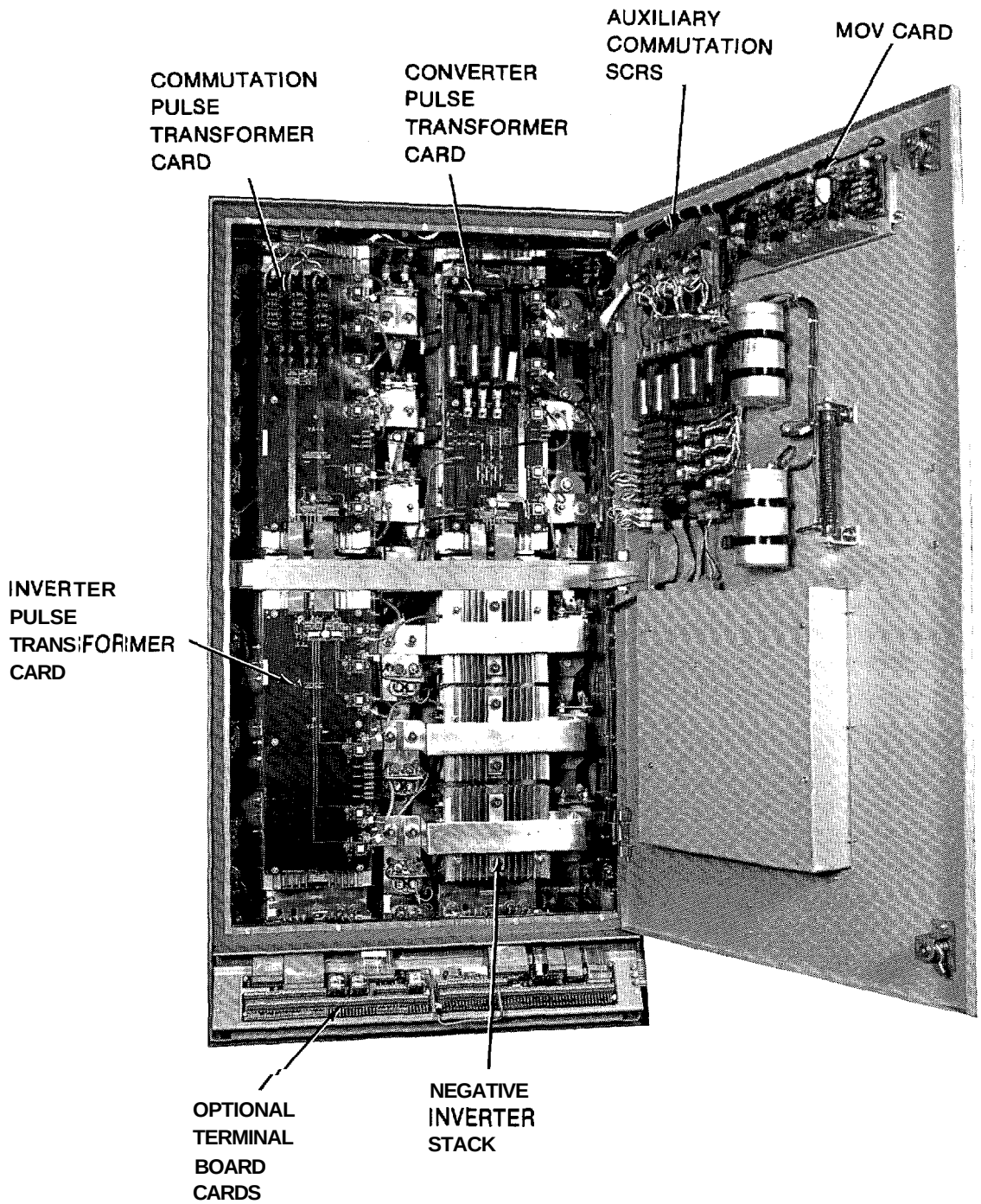


FIGURE 1.7
500 KVA POWER SECTION DOOR OPEN

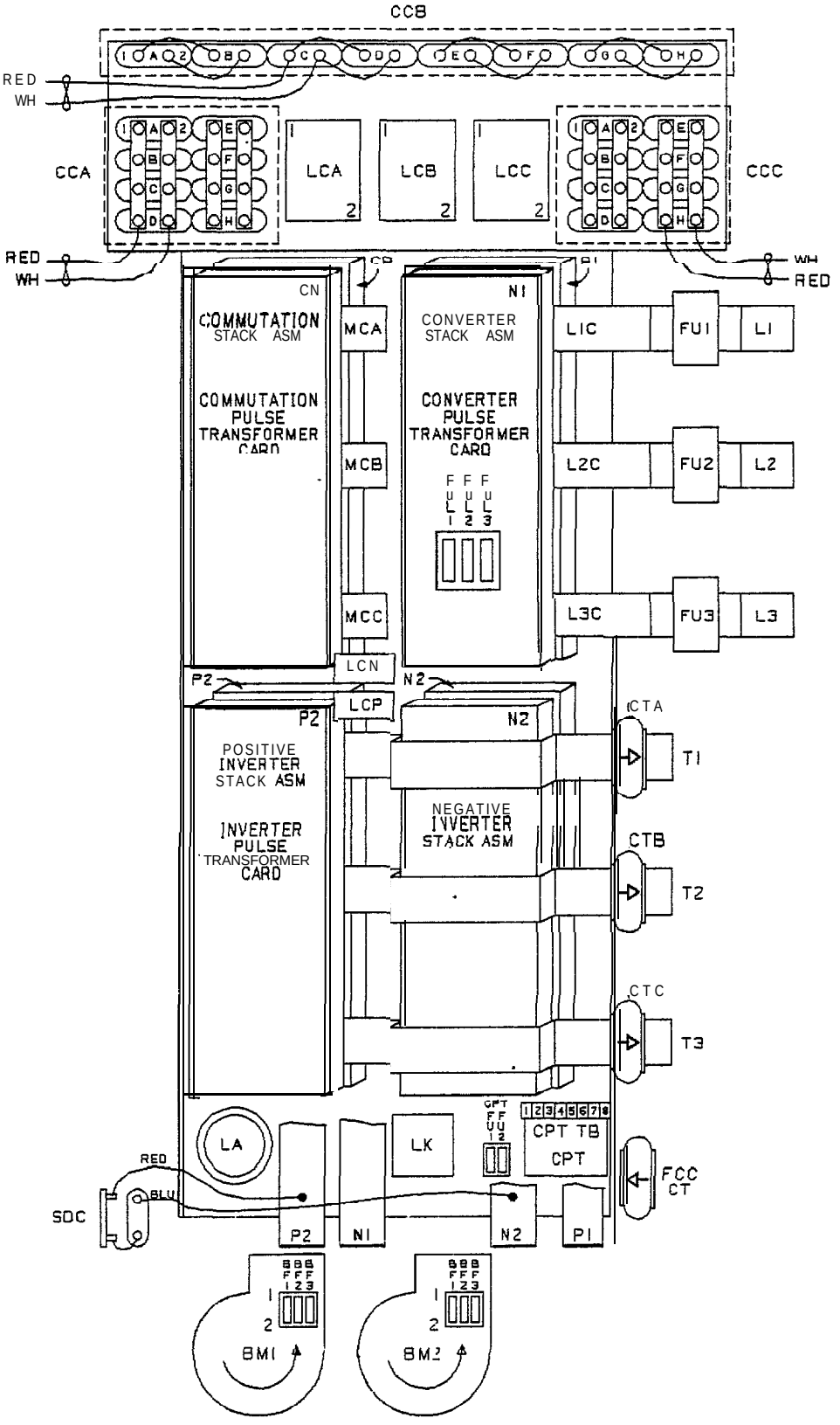


FIGURE 1.8
POWER SECTION LAYOUT (500 KVA FRAME)

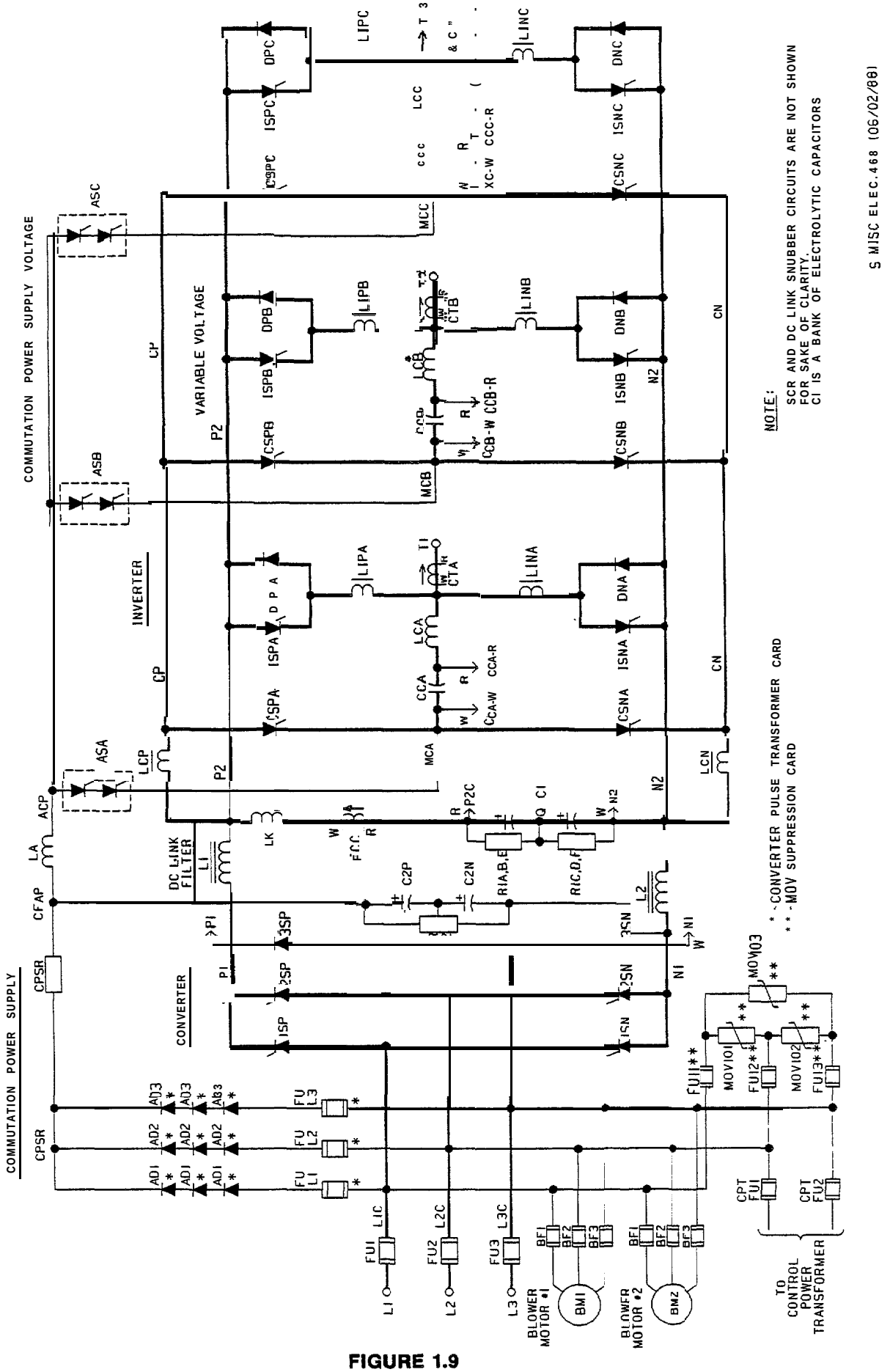


FIGURE 1.9
AC-500 POWER CIRCUIT (500 KVA FRAME)

INVERTER SECTION

The three phase inverter as shown in Figures 1.5 and 1.9 consists of three identical single phase inverter circuits. Each phase consists of two inverting SCRs, two commutation SCRs, two bypass diodes and an LC commutating circuit. Output phase A (T1) of Figure 1.5 will be described, since all three phases operate in an identical manner, except for being displaced by 120 degrees in phase relationship.

The AC motor lead Tl is alternately connected to the positive P2 DC bus and negative N2 DC bus by inverting SCRs ISPA and ISNA respectively. The frequency of this alternation seen at T1 is the fundamental frequency applied to the motor which determines its speed by the following equations:

$$\text{SYNCHRONOUS SPEED (RPM)} = \frac{60 \times 2 \times \text{FREQUENCY}}{\text{NUMBER OF POLES}}$$

Although an SCR can readily be turned on by applying a firing signal to its gate, it must be commutated off by supplying an alternate path for the current which was flowing through the SCR, and by applying a small reverse voltage to the SCR for a short period of time. This is accomplished by means of the commutation SCRs CSPA and CSNA, and by the commutation reactor LCA and commutation capacitor CCA.

At the time when inverter SCR ISPA is to be commutated off, capacitor CCA is charged such that the T1 side is positive. The gate signal is removed from ISPA and commutation SCR CSPA is fired, to create a commutation current pulse that takes the form of a half cycle sine wave due to the interaction of capacitor CCA with reactor LCA. During this pulse the motor current flowing through ISPA is diverted to the alternate path of CSPA, LCA and CCA due to the voltage charge on CCA. When the commutation current in this alternate path exceeds the motor current, no current exists in ISPA. As capacitor CCA discharges further, the excess commutation current (above the motor current level) flows through the LIPA reactor and diode DPA back to CSPA. The voltage drop across DPA produces a small reverse voltage across ISPA to cause it to return to its blocking or off state. Therefore, for successful commutation, the commutation current must exceed the motor current for the amount of turn-off time required for the SCR. In order to minimize this time and the commutation energy required, special inverter grade SCRs are used which have a short turn-off time. Up to 200KVA, asymmetrical SCRs with less than 20 microseconds turn-off time are used. This requires a commutation current base width of about 45-50 microseconds. At higher KVA ratings the SCR turn-off time is less than 40 microseconds and the commutation current base width is about 80-90 microseconds.

When commutating current peaks and starts diminishing, the charge on the capacitor CCA reverses, and the energy stored in reactor LCA charges CCA up in the opposite direction. At the point in time when the commutation current falls below the level of the motor current, the current in diode DPA goes to zero and the potential of the T1 motor lead changes from the inverter positive bus P2 to the negative bus N2 so that diode DNA can furnish motor current. The above action occurs if the oncoming inverter SCR ISNA is not fired before this point in time. If ISNA is fired earlier, the transistion of T1 from positive to negative bus will occur earlier in the commutation interval. In any case, capacitor CCA becomes charged in the opposite direction (T1 side negative) at the end of the ISPA commutation interval. It is now charged correctly to commutate off inverter SCR ISNA when commutating SCR CSNA is fired. This commutation action is the same as the one desribed for ISPA. At the end of each commutating interval, the commutation SCR is commutated off by the charge on capacitor CCA producing a reverse voltage to the commutating SCR which had just been conducting.

The ferrite core reactors, LIPA, LINA and commutation ferrite core reactors LCP and LCN act in conjunction with the SCR snubber circuits (not shown in Figures 1.5 and 1.9) to limit dv/dt and protect the SCRs against voltage transients.

The commutation losses, although small in relation to the total commutation energy, must be replaced in order to keep the commutation capacitor charged up to the proper voltage. These losses are replaced from the DC link P2 and N2 when it is near its maximum value. The amount of energy replaced, and thus the level of the commutation capacitor voltage, is determined by the firing point of the oncoming inverter SCR in the commutation interval. As the DC link voltage is reduced down to zero volts however, the commutation losses are replaced from the commutation power supply.

COMMUTATION POWER SUPPLY

The commutation power supply consists of a three-phase half wave diode bridge (AD1, AD2, and AD3) which in conjunction with the negative converter SCRs (1SN, 2SN, and 3SN), supplies a rectified voltage through CPSR (R3 on the 200 KVA package) to the RC filter section R2 (R2P and R2N on the 200KVA package, Figure 1.5) and C2P, C2N. On application of AC power the three negative converter SCRs will start operating to energize the commutation power supply.

An inductor, LA (and a resistor, R4, on 200KVA units) is connected at the output to control the charging of the commutation capacitor. These devices are relatively small compared to the other power circuit components, since only the commutation circuit losses are furnished by this section.

The two different methods of replenishing these losses are discussed below:

200KVA (and below)

In this circuit, Figure 1.5, the positive commutation SCRs are connected **to the commutation** power supply directly. A rectifier, DCD is connected between the DC link, P2, and the commutation power supply, CP. Each positive commutation current pulse flows through the DCD rectifier. While simpler, this commutation power supply is less efficient and thus only used at lower KVA levels.

500KVA (and down to 250KVA)

Six small auxiliary SCRs (two in series per phase), ASA, ASB and ASC connect the commutation circuit as shown in Figure 1.9. At the end of the positive commutation cycle, when the negative inverter SCR starts firing, gate signals are simultaneously applied to the auxiliary SCRs in that phase.

For both packages the amount of energy added to the commutation capacitors is controlled by regulating the SCR firing point between the peak and the end of the **commutation** current pulse. The commutation capacitor voltage is fed back to the firing logic to regulate the voltage which results in regulating the commutation current pulse.

COMMUTATION FAILURE PROTECTION

For several reasons such as circuit failure or high output current, **an** inverter SCR may not be commutated off prior to the firing of the oncoming inverter SCR. This situation results in a direct short or fault across the DC link from P2 to N2. Reactors L1 and L2 prevent any sudden increase in current from the incoming SCR bridge. Capacitor Cl must supply most of the fault current controlled by the air core reactor LK. The rise in the filter capacitor current is detected by current transformer FCC, which signals the firing logic to fire all main inverter SCRs. Three parallel paths now share the fault current from capacitor Cl, so each SCR carries only one third of the total current.

Cl and reactor LK and other inductances in the discharge path form an oscillating circuit that will circulate and reverse in direction through the diodes DP and DN (A, B, & C). This second half of the cycle reverse biases the inverter SCRs clearing the fault with no damage to the equipment. Current limiting fuses ahead of this incoming SCR bridge provide additional protection.

AC POWER REQUIREMENTS

The AC-500 drive is available in either 230V AC +10%, -5% (40-250HP) or 460V +10%, -5% (75-500HP) three-phase power configurations. Nominal line frequency is 50/60Hz with tolerance of ± 2Hz.

AC LINE FUSES (FU1-3)

AC line fuses provide short circuit protection for the SCRs and the wiring inside the controller. A molded case disconnect switch or circuit breaker is also available as an option. Additional output overcurrent protection is provided by a timed overcurrent function.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuses (CFU1 and 2). The secondary windings provide two isolated voltages

1. 115V AC to operate the coil of the RUN (K1) Relay and controller card cooling fan, capacitor bank cooling fan, and door fan.

2) 38V AC center tapped for the Power Supply card to generate the DC control voltage necessary for controller operation.

CONTROL CARD

This card provides the intelligence for the AC-500 controller. It contains three microprocessors and performs all of the regulating functions, converter SCR burst firing, diagnostic and most protective functions for the drive, as well as many customer and motor interfacing functions. The functions performed on the card are implemented in hardware and/or software.

Six LEDS are provided on this card to display the fault status of the AC-500. The LEDs which are illuminated are interpreted as a binary number indicative of the drive fault. See the troubleshooting section for listing of Fault Codes. Note also that during proper controller operation the LEDs will blink sequentially right to left when the drive is stopped and sequentially right to left in pairs when the drive is running.

POWER SUPPLY/INTERFACE CARD

The Power Supply/Interface card provides the necessary +5V DC, ±15V DC, and ±24V DC power used throughout the controller. Three fault circuits with LEDs for analog Instantaneous Overcurrent (IOC), Ground Fault (GF) and Inverter Fault (IFT) detection are located on this card. Voltage and current feedback isolators and adjustments, as well as VCOs and D/A converters are also on this card. Additionally, three relays: "RUN" (K1), "MAX" (K2) and "FAULT" (K3) are included on this card.

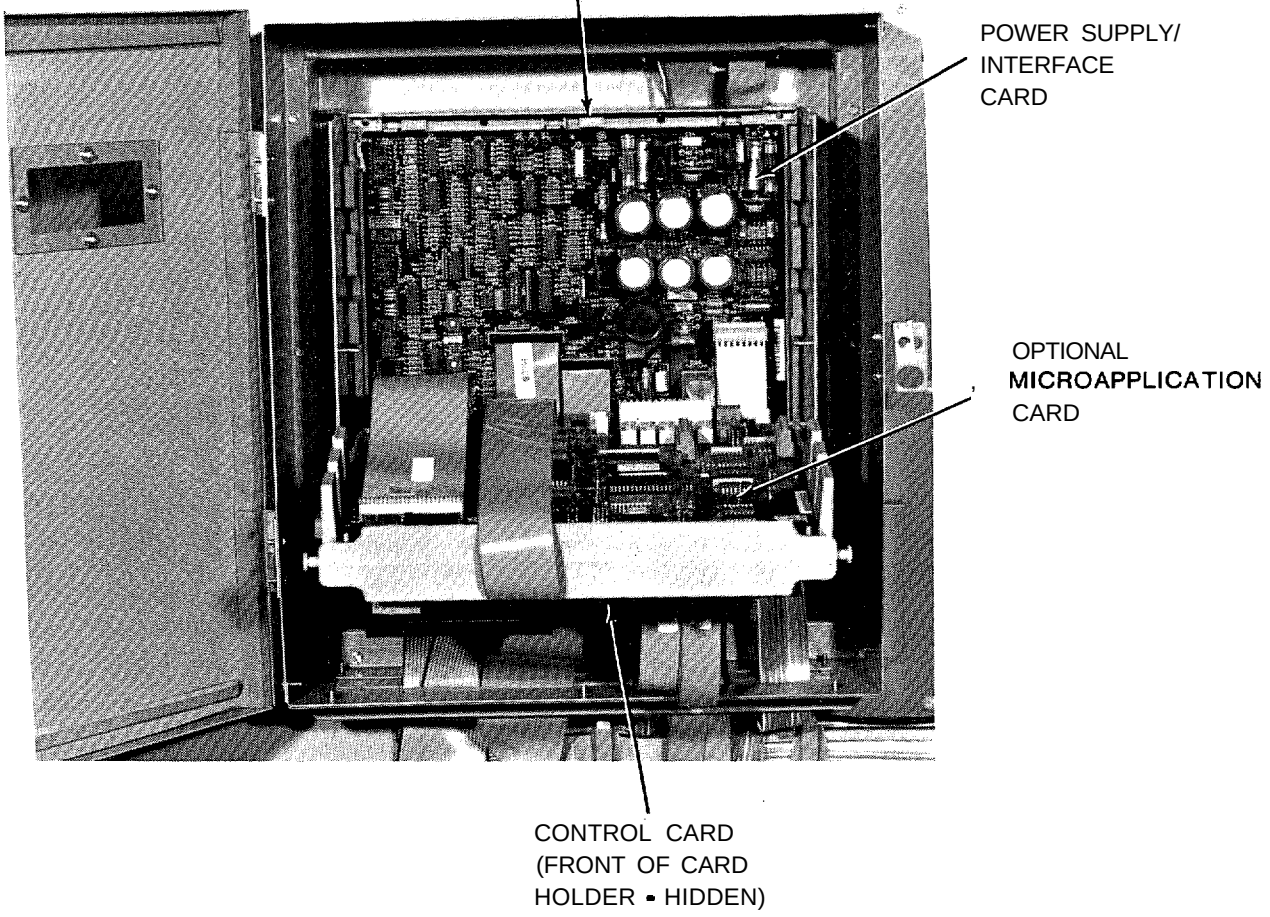
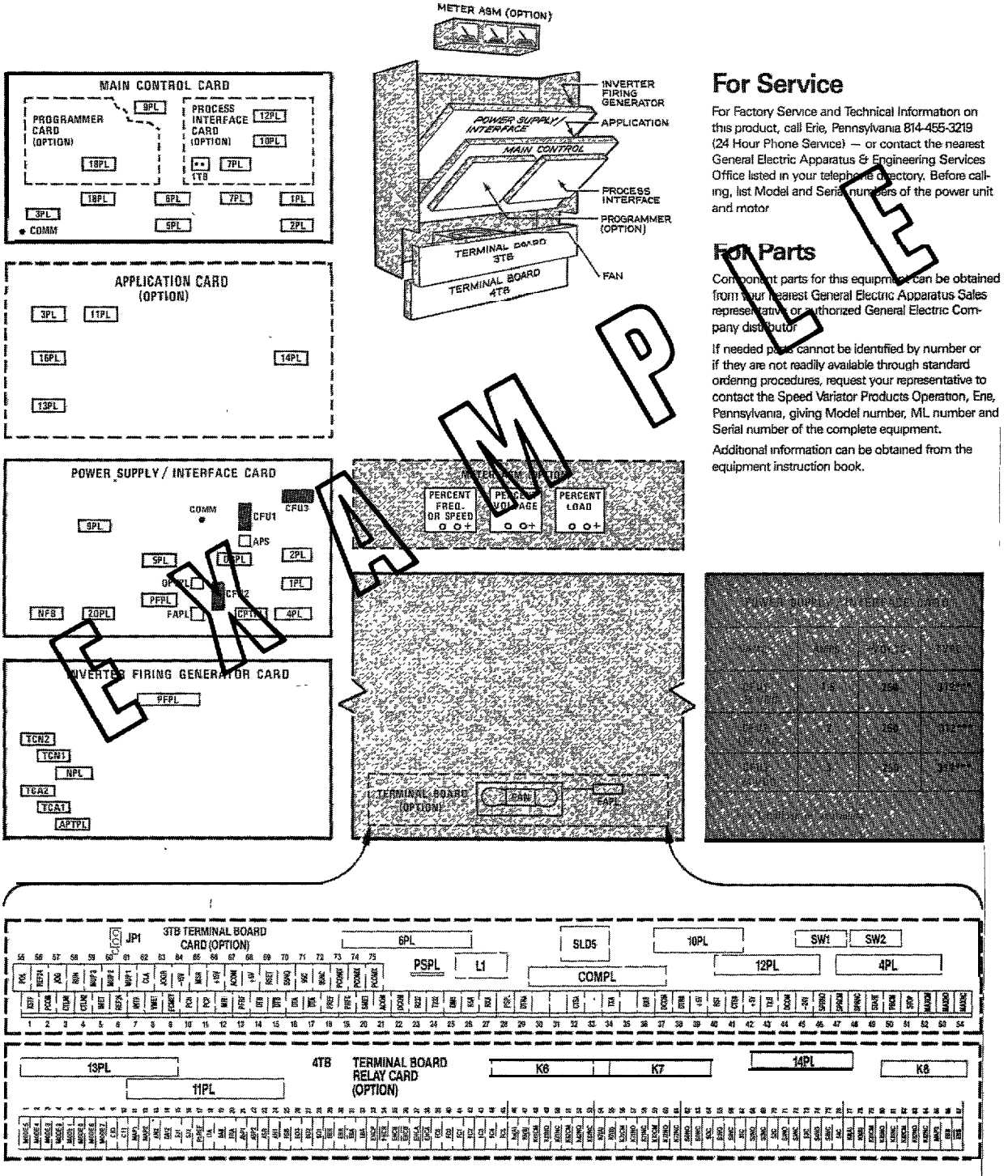


FIGURE 1.10
AC-500 CONTROLLER WITH REGULATOR DOOR OPEN
(FRONT CARD HOLDER DOWN)

Instructions



For Service

For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest General Electric Apparatus & Engineering Services Office listed in your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor.

For Parts

Component parts for this equipment can be obtained from your nearest General Electric Apparatus Sales representative or authorized General Electric Company distributor.

If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Speed Variator Products Operation, Erie, Pennsylvania, giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.

EXAMPLE

FIGURE 1.11
AC-500 INFORMATION LABEL

INVERTER FIRING GENERATOR CARD

This card provides the firing sequence timing and firing signals for the inverter and commutation SCRs. Commutation capacitor voltage regulation is controlled by this card on a per phase basis, and three LEDs for commutation capacitor overvoltage indication for each phase (ACOV, BCOV and CCOV) are also on this card.

CONVERTER PULSE TRANSFORMER CARD (200KVA)

The Converter Pulse Transformer card contains the firing signal amplifiers and pulse transformers as well as feedback and isolation circuits for the AC input, converter and link voltage feedback, as well as the output current feedback and filter current feedback. AC line fuses, AC line to line and converter output DC snubbers, and the commutation power supply rectifiers generate the power to the commutation circuit. The card is located inside the power section in front of the right-hand side of the heatsink assembly.

INVERTER PULSE TRANSFORMER CARD (200 KVA)

The Inverter Pulse Transformer card contains the firing signal amplifiers and pulse transformers for the inverter and commutation SCRs as well as commutation capacitor voltage feedback. It is located inside the power section in front of the left-hand side of the heatsink assembly.

CONVERTER PULSE TRANSFORMER CARD (500KVA)

The Converter Pulse Transformer card contains the firing signal amplifiers, pulse transformers, and AC line voltage feedbacks for the converter SCRs. AC line fuses, AC line to line and converter output DC snubbers, load resistors for LCP and LCN coils, and the commutation power supply rectifiers generate the power to the commutation circuit. The card is located inside the power section on the front of the upper right-hand heatsink assembly.

INVERTER PULSE TRANSFORMER CARD (500KVA)

The Inverter Pulse Transformer card contains the firing signal amplifiers and pulse transformers for the inverter SCRs. This card is located inside the power section on the front of the lower left-hand side heatsink assembly.

COMMUTATION PULSE TRANSFORMER CARD (SOOKVA)

The Commutation Pulse Transformer card contains the firing signal amplifiers and pulse transformers as well as commutation capacitor voltage feedback isolation circuits. This card is also located inside the converter on the front of the upper left-hand heatsink assembly.

COMMUTATION POWER SUPPLY (500KVA)

The Commutation Power Supply card contains firing signal amplifiers and pulse transformers for the commutation power supply SCRs. Also located on this card are SCR snubbers and capacitor sharing resistors. This card is located on the back of the conversion assembly door.

SNUBBER CARD (200 and 500KVA)

The Snubber contains a polarized snubber circuit. Each inverter SCR and commutating SCR has a Snubber card associated with it.

OPTIONAL APPLICATION CARD

This optional card contains a combination of analog interfaces between the Control card and external equipment. These interfaces allow the core drive to be expanded to handle functions required in specific applications such as machine tool spindles. Two versions of the Application card are available. The Application card contains the following optional Functions:

- Eight Analog Inputs
- Eight Digital Inputs
- Sixteen Bit Parallel Word Input
- Five Programmable Signal-Level-Detector Relays (SLDS)
- Seven-Bit Digital Fault Status Output
- Three Programmable Status Bit Outputs
- Two Encoded Interfaces or a Frequency Reference
- Two Analog Outputs
- Two General Purpose Amplifiers

The second type of Application card available is the Microapplication card. In addition to the above features, it also contains the Local Area Network (LAN) feature for communicating over a single pair of wires with up to 30 other drives and a Series Six™ Programmable Control. The Microapplication card and LAN are discussed in the Local Area Network User's Guide, GEK-85789.

OPTIONAL PROCESS INTERFACE CARD

This optional card contains a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions on this card are:

- Analog (AC or DC) tachometer interface.
- Analog speed meter driver (analog tachs only).
- Process Follower (current or voltage reference) interface.
- Digital voltmeter probe.
- RS232C and RS422 serial link interface.

OPTIONAL TERMINAL BOARD CARDS (3TB and 4TB)

These optional cards provide convenient termination points for control interface to the drive. Connections to the Control, Power Supply and optional Process Interface cards are made thru 3TB, which also contains an optional Signal-Level-Detector (SLD5). Connections to the optional Application card are made thru 4TB. In addition, three 115V AC relays are available on this card.

OPTIONAL PROGRAMMER MODULE

The Programmer is available as either a "Local" Programmer, mounted on the Control card, or as a "Hand Held" Programmer which is plugged into 18PL on the Control card. The Programmer contains a 10-digit fluorescent alpha-numeric display and keypad for communicating with the drive.

The Programmer has three modes:

- **Operate** mode provides all necessary functions to run the drive and displays the drive status including fault messages.

- **Parameter** mode allows examination and changes, if necessary, of the system operating parameters.

- **Diagnostic** mode allows the user to monitor drive vaiues while in operation, self test of the controller, and diagnostic RUN and STATIC mode. Diagnostic mode also provides a number of drive tests. Refer to "Use of Optional Programmer Module" section for further information.

SERVICE AND PARTS INSTRUCTION LABELS (Figure 1.11)

Each drive contains, inside the door, a multi-color label similar in appearance to that shown in Figure 1.11. This label provides information on connection point and component loctions essential in connecting or troubleshooting the drive. The label contains the following information in color-coded format:

- Card location
- SCR locations
- Fuse information and nomenclature
- Plug locations
- Incoming control connections at 3TB and 4TB
- Power connections
- Factory "Hot Line" telephone number (1-814-455-3219)

CONNECTION DIAGRAMS (Figures 1.12 and 1.13)

Connection diagrams for the controllers are shown in Figures 1.12 and 1.13. These graphically describe the internal connections for all plugs and cables. These diagrams aid control hook-up as well as troubleshooting when connection integrity of one of the cables is in question. The functions of the connectors are described in the "Electrical Connections" section.

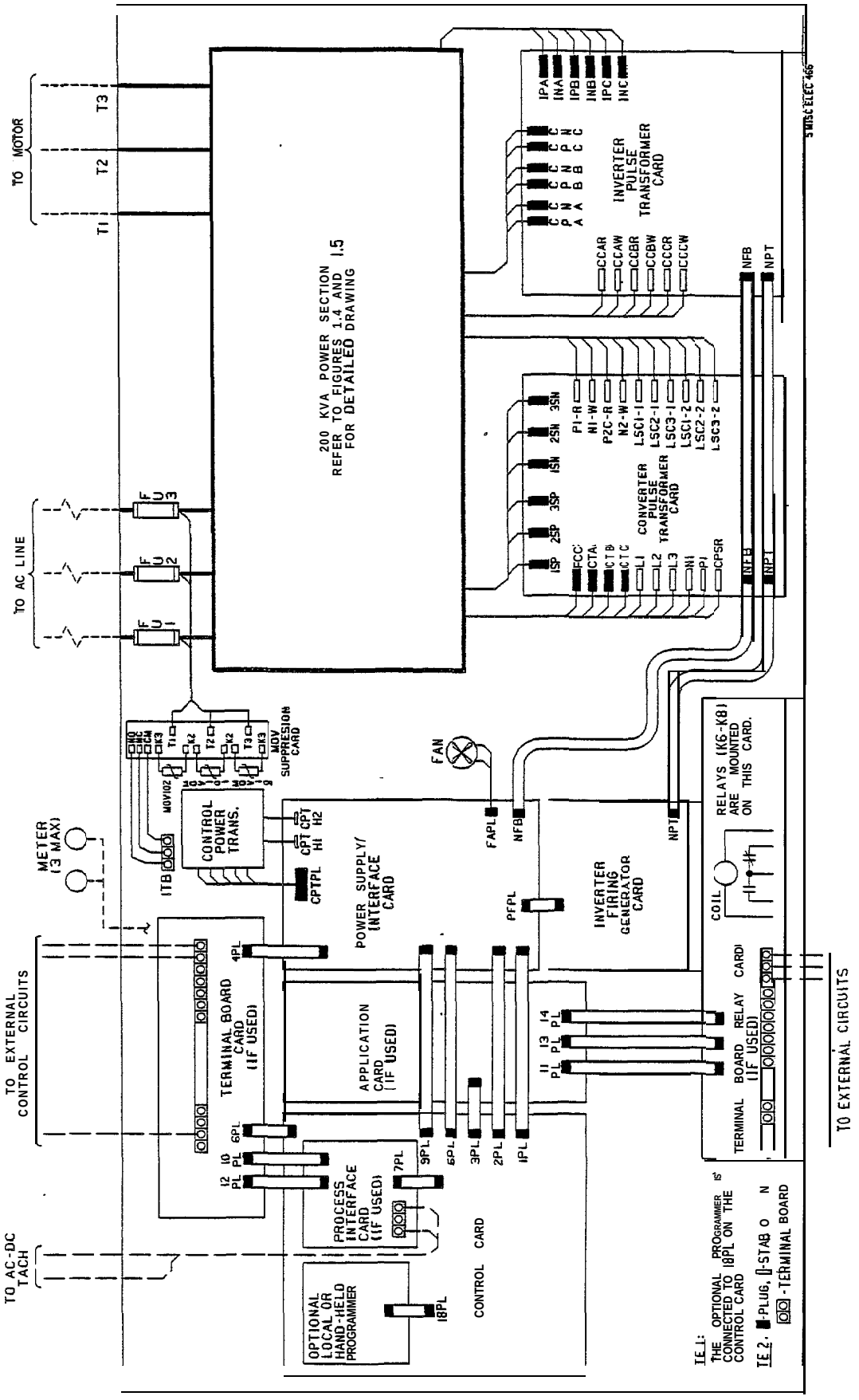


FIGURE 1.12
CONNECTION DIAGRAM - 200 KVA FRAME

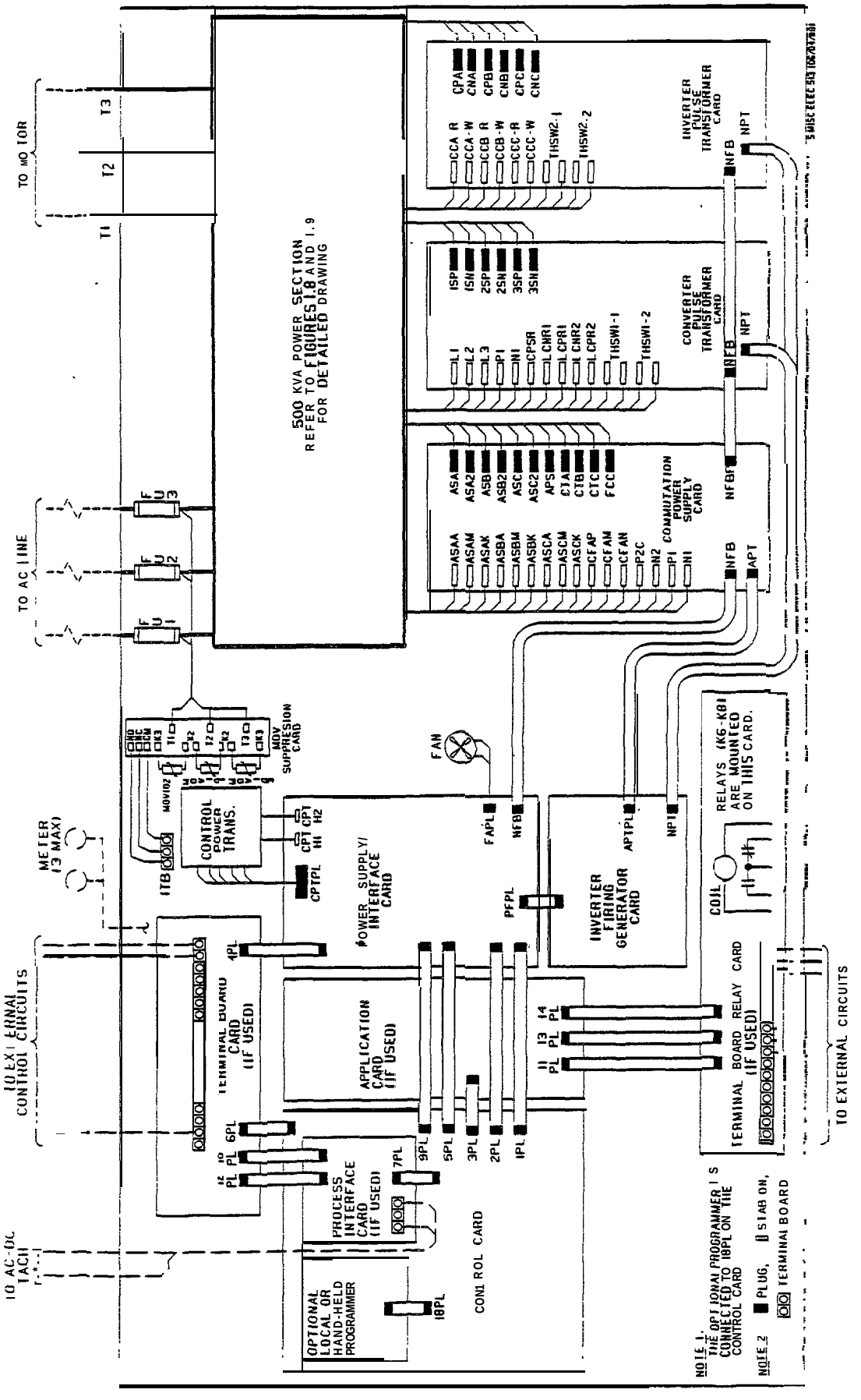


FIGURE 1.13
CONNECTION DIAGRAM - *500 KVA FRAME*

CONNECTOR 1PL — BETWEEN CONTROL CARD AND POWER SUPPLY/INTERFACE CARD (PSC)

PIN	NOMEN-CLATURE	DESCRIPTION
1	DCOM	Digital Common
2	CVRDB	DB Module Control Signal
3	DCOM	
4	FC-1	
5	DCOM	
6	RT1	AC Phase 1 Input from Power Supply/Interface Card (PSC)
7	RT2	AC Phase 2 Input from Power Supply/Interface Card (PSC)
8	RT3	AC Phase 3 Input from Power Supply/Interface Card (PSC)
9	FRC	Fault Relay control drives the 24 volt fault relay on PSC.
10	-24V	Unregulated -24V return path for FRC and RP
11	RP	Run relay driver (Run - K2 on PSC).
12	RUN	Drive run signal from RUN (K1) relay contact on PSC.
13	REF 24	Supplies +/-24V to RUN (K1) relay contact on PSC depending on setting of jumpers JP19 and 20 on Control card.
14	W5V	Watchdog reference provides 24V and 5V reference from the PSC to under voltage detector on the Control card.
15	IFT	Inverter shoot-through fault detection output to Control card; signals FAULT 53.
16	IOC/COV/ D BT/GFT	Instantaneous overcurrent, converter overvoltage, dynamic-braking trip, ground fault trip output to Control card; signals FAULT 03.
17	IFT	
18	IOC/COV/ D BT/GFT	
19	DCOM	
20	CFB	Motor current feedback input from PSC.

CONNECTOR 2PL — BETWEEN CONTROL CARD AND POWER SUPPLY/INTERFACE CARD

PIN	NOMEN-CLATURE	DESCRIPTION
1	ACOM	Analog signal return for +/-15V.
2	-15V	-15V regulated analog power supply input.
3	+15V	+15V regulated analog power supply input.
4	+24V	+24V unregulated power supply input.
5	-24V	-24V unregulated power supply input.
6	+5V	+5V regulated digital power supply input.
7	DCOM	Digital signal return for +5V.
8	+5V	
9	PCOM	Power supply return for +/-24V.

CONNECTOR 3PL — BETWEEN CONTROL CARD AND OPTIONAL APPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	DO	U4 data bus bit 0 (least significant bit).
2	D1	U4 data bus bit 1.
3	DCOM	Digital common.
4	D2	U4 data bus bit 2.
5	D3	U4 data bus bit 3.
6	D4	U4 data bus bit 4.
7	DCOM	
8	D5	U4 data bus bit 5.
9	D6	U4 data bus bit 6.
10	D7	U4 data bus bit 7.
11	DCOM	
12	A8	U4 address bus bit 8.
13	A9	U4 address bus bit 9.
14	A10	U4 address bus bit 10.
15	DCOM	
16	A11	U4 address bus bit 11.
17	SEL	Signal from U3. Used for Microapplication card synchronization.
18	ALE	Address latch from U4.
19	DCOM	
20	/RD	Read control line from U4.
21	/WR	Write control line from U4.
22	A15	U4 address bus bit 15, used for Application card I/O select.
23	RST	Card reset signal to Application card.
24	+5V	+5V regulated digital supply to Application card.
25	+5V	
26	+5V	
27	PCOM	+/-24V power supply return.
28	+24V	+24V non-regulated power supply to Application card.
29	-24V	-24V non-regulated power supply to Application card.
30	DM4	Provides “dummy” connection to Application card.
31	+15V	+15V regulated analog supply to Application card.
32	-15v	-15V regulated analog supply to Application card.
33	ACOM	Analog signal return for +/-15V.
34	OPTA	Input to A-D converter from Application card multiplexer.

CONNECTOR 4PL — BETWEEN POWER SUPPLY/INTERFACE CARD AND OPTIONAL 3TB

PIN	3 TB TERM#	NOMEN-CLATURE	DESCRIPTION
1	46	FLTNO	Fault Relay (K3) Normally Open Contact
2	47	FLTCM	Fault Relay (K3) Common Connection
3	48	FLTNC	Fault Relay (K3) Normally Closed Contact
4	49	START	Connection Point for Start Pushbutton
5	50	PBCM	Common Point for Start and Stop Pushbuttons
6	51	STOP	Connection for Stop Pushbutton
7	52	RLJNCM	RUN Auxiliary Relay (K2) Common Connection
8	53	RUNNO	RUN Auxiliary Relay (K2) Normally Open Contact
9	54	RUNNC	RUN Auxiliary Relay (K2) Normally Closed Contact

NOTE: Contact rating: K2 and K3 relays are 1.0 amp resistive load at 115 volts AC or 29 volts DC.
Pilot duty: 115V AC of 0.2 amp holding and 1.0 amp in-rush inductive load.
Low current: 30mV at 30mA.

CONNECTOR 5PL — BETWEEN CONTROL CARD AND POWER SUPPLY/INTERFACE CARD

PIN	NOMEN-CLATURE	DESCRIPTION
1	A6F	Drives cell 3SN gate pulse transformer.
2		No connection.
3	A5F	Drives cell 2SN gate pulse transformer.
4		No connection
5	A4F	Drives cell 1SN gate pulse transformer.
6		No Connection
7	A3F	Drives cell 3SP gate pulse transformer.
8		No connection.
9	A2F	Drives cell 2SP gate pulse transformer.
10		No connection.
11	A1F	Drives cell 1SP gate pulse transformer.
12 thru 24		No connection
25	LVFD-1	Link voltage signal output to Control card VMET circuit.
26	LFD-COM	Link voltage signal common.

CONNECTOR 6PL — BETWEEN CONTROL CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	69	RSET	A hard reset of drive is initiated when RSET is connected to +15V. Must be open to run. RSET stops firing of all SCRs, removes delayed firing power, drops out the MA contactor and fault relay, and holds all microprocessors in a reset state until RSET is opened. The drive will not automatically restart if reset while running. Do not apply RSET while running.
2	68	+5V	Not for use external to drive.
3	67	ACOM	Not for use external to drive.
4	66	+15V	Signal voltage for System Reference pot, RSET circuit and others.
5	65	MSR	Manual system reference input to drive, ±15V maximum- Positive for forward motoring with POL at zero or open.
6	64	-15v	Signal voltage for system reference pot and others.
7	63	JOGR	Jog reference input (can be selected to bypass linear time), ±14V = Top speed. Programmable to 7V = top speed for fast update rate. Refer to Custom Software Description.
8	62	CLA	Current limit adjust-adds to software setting of current limit. +10V increases current limit by 100% of rated, -lOV decreases it by 100%.
9	61	MUP1	Multiuse point 1 - connects to connection test point MUP1, which may be jumpered to another connection point. Example: DA0 may be jumpered to MUP1 for output on 3TB.
10	60	MUP2	Multiuse point 2 - connects to the connection test point MUP2, which may be jumpered to another connection point.
11	59	MUP3	Multiuse point 3 - connects to connection test point MUP3, which may be jumpered to another connection point.
12	58	RUN	Alternate run command input point in place of "RUN" relay. Do not pull high if run relay is also being used. Drive runs when connected to 3TB56 (6PL-14), REF24. With JP19 and JP20 on the Control card in the l-2 and 3-4 positions respectively, input is active (controller runs) when RUN is pulled down to -24V DC. Input is inactive (controller stops) when RUN is open or driven to +24V DC. With JP19 and JP20 in the l-3 and 2-4 positions respectively, input is active (controller runs) when RUN is driven to +24V DC. Input is inactive (controller stops) when RUN is open or pulled down to -24V DC.
13	57	JOG	Jog command input point. Drive jogs when connected to 3TB56 (6PL-14), REF24. Polarity controlled same as RUN above.
14	56	REF24	Selectable + or -24V supply for RUN, JOG, POL or XTSP input. Polarity set by JP19 and JP20 on Control card.

CONNECTOR 6PL — BETWEEN CONTROL CARD AND OPTIONAL 3TB (continued)

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
15	55	POL	Control input which reverses the direction of the drive for any applied reference; e.g. activating POL low is equivalent to reversing the polarity of the input at MSR. Activated/deactivated same as RUN above. Programmable for multiple functions. Refer to Custom Software Description and elementaries if used.
16	1	XSTP	Must be activated to enable drive. Activated/deactivated same as RUN above. Causes a normal, coast, quick or trip stop when open (see ADR004-STPJP in the Custom Software Description).
17	2	PCOM	Not for use external to drive.
18	3	CTLN1	Control on must be connected to CTLN2 to allow the drive to run. Openingthis connection unconditionally drops MA contactor after a 50 millisec attempt by microprocessor to precondition the drive.
19	4	CTLN2	See CTLNl.
20	5	IMET	Analog load meter output proportional to motor current. 2 to 10V DC output at drive rated current. See P2 and JP15 on the Control card (Table 1) for details. Load not to exceed 1000 ohms/volt.
21	6	REF24	Selectable ±24V supply for RUN, JOG, REV, POL or XSTP, input. Polarity set by JP19 and JP20 on the Control card.
22	7	INTR	24V drive for signal level detector SLD5 on 3TB terminal board.
23	8	VMET	Converter output voltmeter output. This output is adjustable from 4.0 to 10.0 volts at rated voltage output. See page 8 on the Control card (Table 1) for details. Maximum load is 1000 ohms volt.
24	9	FCMET	Inverter analog frequency meter output. Output gain determined by P2 on the Power Supply/Interface card and P5 on the Control card. Refer to Table 1, Control card, for details.

CONNECTOR 7PL — BETWEEN CONTROL CARD AND OPTIONAL PROCESS INTERFACE CARD

PIN	NOMENCLATURE	DESCRIPTION
1	ACOM	Analog signal return.
2	DVM	Output from digital voltmeter.
3	SFB	Output from analog tach circuit.
4	ASFB	Amplified signal from analog tach circuit, for improved resolution at low speed operation.
5	ACOM	
6	-15v	-15V supply to analog circuits on Process Interface (P.I.) card.
7	+15V	+15V supply to analog circuits on P.I. card.
8	+5V	+5V supply to logic circuits on P.I. card.
9	+5V	
10	TRS	RS422 driver tri-state command to the P.I. card.
11	DCOM	Digital signal return.
12	DIR	Signal from digital tach interface signalling direction of motor rotation when a a quadrature track tach is used.
13	TPN	Net pulse train signal from digital tach or AC analog tach interface used to determine tach speed.
14	PRF	Not used.
15	DCOM	
16	DCOM	
17	RXD	Input from either the RS422 or RS232C serial link receiver from the P.I. card.
18	TXD	Output to the RS422 and RS232C serial link drivers from the P.I. card.
19	/DTR	Data terminal ready—output indicating the P.I. card is ready to receive a byte on the serial link. Low when ready, high when not.
20	/CTS	Clear to send-Input from serial link signalling when the receiver on the other end of the link is ready to receive a byte. When low, drive may transmit. When high drive must not send anything out on TXD.
21	RST	Reset line from Control card to P.I. card, active high.
22	+5V	
23	+5V	
24	DCOM	
25	DCOM	
26	-24v	-24V raw power to P.I. card.

CONNECTOR 9PL -- BETWEEN CONTROL CARD AND POWER SUPPLY/INTERFACE CARD (PSF)

PIN	NOMENCLATURE	DESCRIPTION
1	/EN	TUP's inverter SCR firing enable.
2	SIGN	Sign of the analog signal being converted by A/D on PSF.
3	UDAD	Inverter Phase A up down control signal.
4	UDBD	Inverter Phase B up down control signal.
5	UDCD	Inverter Phase C up down control signal.
6	NC	No connection
7	NC	No connection
8	NC	No connection
9	A2D0	Bit 0 of the bi-directional data bus.
10	A2D1	Bit 1 of the bi-directional data bus.
11	A2D2	Bit 2 of the bi-directional data bus.
12	A2D3	Bit 3 of the bi-directional data bus.
13	A2D4	Bit 4 of the b&directional data bus.
14	A2D5	Bit 5 of the bi-directional data bus.
15	A2D6	Bit 6 of the bi-directional data bus.
16	A2D7	Bit 7 of the b&directional data bus.
17	FF	Force fire command to the inverter from the Control card.
18	ENA	RUP's back-up inverter SCR firing enable override on disable from the Control card
19	/INH	Inhibit analog type trips on the PSF board from the Control card.
20	NC	No connection
21	NC	No connection
22	NC	No connection
23	NC	No connection
24	NC	No connection
25	NC	No connection
26	cvco	Converter voltage VCO feedback from PSF.
27	RST	Reset control from Control card.
28	LVCO	Inverter voltage VCO feedback from PSF.
29	LE/RD	Dual function latch and read control line for the analog to digital conversion on the PSF.
30	NC	No connection
31	NC	No connection
32	NC	No connection
33	NC	No connection
34	NC	No connection

CONNECTOR 10PL — BETWEEN OPTIONAL PROCESS INTERFACE CARD OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	10	PCN	Optional negative input for process follower current (l-5, 4-20, 5-50, or 5-65mA.) or voltage (0-10V DC) reference.
2	11	PCP	Optional positive input for process follower reference.
3	12	MR1	Optional reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card reference input via auto/manual selection.
4	13	PFRF	Optional process follower interface output (0-10V) which connects to a speed reference or other analog process input.
5	14	DTB	Optional square wave tach quadrature input (track B high). Operating Voltage: -0.5V to +0.8V is logic 0; +2.4V to +15V is logic 1.
6	15	/DTB	Optional inverted square wave differential tach quadrature input (track B low).
7	16	DTA	Optional square wave tach main track input (track A high). Operating voltages: -0.5V to +0.8V is logic 0, +2.4V to +15V is logic 1.
8	17	/DTA	Optional inverted square wave differential tach quadrature input (track A low).
9	18	PREF	Optional pulse reference frequency positive input, 0 to 6KHz maximum.
10	19	PRFC	Optional high impedance common for pulse reference input.
11	20	SMET	Analog speed meter output, 4V DC. See P2 on the P.I. card (Table 1).
12	21	ACOM	Common for optional RS232C communications.
13	22	DCOM	Not for use external to drive.
14	23	R232	Optional RS232C receive line.
15	24	T232	Optional RS232C transmit line.
16	25	DM5	Spare connection tied *to* undedicated point DM5.

NOTE: /or = NOT: Example /C or C = Not C

TERMINAL BOARD 1TB LOCATED ON OPTIONAL PROCESS INTERFACE CARD

1	TKN	Negative tachometer feedback signal. Maximum input is 310V DC to 344V AC.
3	TKP	Positive tachometer feedback signal.

CONNECTOR 11PL — BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	30	EBR	No connection (used with Microapplication card).
2	29	/EBR	No connection (used with Microapplication card).
3	85	MAP3	No connection (used with Microapplication card).
4	25	PSB	Strobe for optional I6 bit parallel input on 16PL. See Application card JP7 (Table 1). Inputs must be set 50 micro-set before strobe and strobe held for 100 micro-sec.
5	26	B03	Optional programmable 24V bit output #3.
6	27	B02	Optional programmable 24V bit output #2.
7	28	B01	Optional programmable 24V bit output #1.
8	32	EBA	No connection (used with Microapplication card).
9	31	/EBA	No connection (used with Microapplication card).
10	86	EBB	No connection (used with Microapplication card).
11	87	/EBB	No connection (used with Microapplication card).
12	33	ENCR	Optional non-inverting differential or single-ended encoder marker pulse input.
13	34	/ENCR	Optional inverting differential encoder marker pulse input.
14	35	ENCB	Optional non-inverting differential or single-ended encoder Track B input.
15	36	/ENCB	Optional inverting differential encoder Track B input.
16	37	ENCA	Optional non-inverting differential or single-ended encoder Track A input.
17	38	/ENCA	Optional inverting differential encoder Track A input.
18	39	FC6	Bit 6 of optional fault code 24V output.
19	40	FC0	Bit 0 of optional fault code 24V output.
20	41	FC1	Bit 1 of optional fault code 24V output.
21	42	FC2	Bit 2 of optional fault code 24V output.
22	43	FC3	Bit 3 of optional fault code 24V output.
23	44	FC4	Bit 4 of optional fault code 24V output.
24	45	FC5	Bit 5 of optional fault code 24V output.

CONNECTOR 12PL — BETWEEN OPTIONAL PROCESS CARD AND OPTIONAL 3TB

PIN	3TB TERM #	NOMENCLATURE	DESCRIPTION
1	—	—	No connection.
2	—	—	No connection.
3	—	—	No connection.
4	—	—	No connection.
5	—	—	No connection.
6	—	—	No connection.
7	32	CTSA	Optional inverted RS422 clear-to-send input from serial link signalling when the Control card may transmit a byte (active high). (EIA standard nomenclature is CTSA).
8	29	DTRA	Optional inverted RS422 data-terminal-ready output from Control card signalling when it is ready to receive a byte on the serial link (active high). (EIA standard nomenclature is DTRA) .
9	34	TXA	Optional inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA) .
10	27	RXA	Optional inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXA) .
11	36	RXB	Optional non-inverted RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB).
12	43	TXB	Optional non-inverted RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB).
13	38	DTRB	Optional non-inverted RS422 data-terminal-ready output from Control card. (EIA standard nomenclature is DTRB).
14	41	CTSB	Optional non-inverted RS422 clear-to-send input from serial link. (EIA) standard nomenclature is CTSB).
15	40	RST	Reset signal from Control card, active high, not for use external to drive.
16	39	+5V	Not for use external to drive.
17	42	+5V	Not for use external to drive.
18	37	DCOM	Not for use external to drive.
19	44	DCOM	Not for use external to drive.
20	45	-24V	Not for use external to drive.

CONNECTOR 13PL — BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

(NOTE: Refer to the Custom Instruction Book, software jumper description, for definitions of I/O through 13PL for particular application).

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	6	MODE 0	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
2	5	MODE 1	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
3	4	MODE 2	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
4	3	MODE 3	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
5	2	MODE 4	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
6	1	MODE 5	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
7	7	MODE 6	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
8	8	MODE 7	Programmable mode select input. Refer to elementaries and Custom Software Description* for information.
9	9	CI0--	No connection (used with Microapplication card).
10	10	CI1	No connection (used with Microapplication card).
11	11	MAP1	Microapplication multiuse point 1 - connects to connection point MAP1, which may he jumpered to another connection point.
12	12	MAP2	Microapplication multiuse point 2 - connects to connection point MAP2, which may he jumpered to another connection point.
13	13	AN2	No connection (used with Microapplication card).
14	14	DA12	Optional analog output from 12 bit D-A converter. Range = -lOV to +10V, 1MA with 450 ohms output impedance. Update rate is 30 to 40 times/second.
15	15	SJ1	Optional analog input to summing junction #1. Operating range = ± 10V; maximum input = ±15V.
16	16	SJ2	Optional analog input to summing junction #2. Operating range = ±10V; maximum input = ±15V.
17	17	PSREF	Optional analog position or process reference input (±10V).
18	18	LTA	Optional analog linear time adjust. Positive voltages increase timing, negative voltages decrease timing. ±10 volts maximum. 10 volts is equal to 66 second adjust.
19	19	DA8	Optional analog output from 8 bit D-A converter. See DA12 for specifications. Normal update rate of 180 times/sec. Reduced to 30 to 40 times/sec. if fast update selected for DAO.

CONNECTOR 13PL — BETWEEN OPTIONAL APPLICATION CARD OPTIONAL 4TB (continued)

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
20	20	FRA	Optional analog field reference adjust. Positive voltages up to +10V increase field current. Negative voltages down to -lOV decrease field current. See Application card JP32 and P2 (Table 1).
21	21	ASP1	Optional spare analog input #1 (programmable).
22	22	ASP2	Optional spare analog input #2 (programmable).
23	23	ASR	Optional auto speed reference input to drive, ±15V maximum, positive for forward cell firing while motoring. ASR is the active speed reference in the auto mode when analog auto reference is selected.
24	24	AN1	No connection (used with Microapplication card).

***NOTE:** Functions of MODE0 - MODE7 are determined by Addresses 000-015 (ADR000 - ADR015) in the software description.

CONNECTOR 14PL — BETWEEN OPTIONAL APPLICATION CARD AND OPTIONAL 4TB

PIN	4TB TERM #	NOMENCLATURE	DESCRIPTION
1	62	SON0	Optional SLD #0 normally open contact.
2	63	SONC	Optional SLD #0 normally closed contact.
3	64	S0C	Optional SLD #0 common contact.
4	65	S1NO	Optional SLD #1 normally open contact.
5	66	S1NC	Optional SLD #1 normally closed contact.
6	67	S1C	Optional SLD #1 common contact.
7	68	S2NO	Optional SLD #2 normally open contact.
a .	69	S2NC	Optional SLD #2 normally closed contact.
9	70	S2C	Optional SLD #2 common contact.
10	71	S3NO	Optional SLD #3 normally open contact.
11	72	S3NC	Optional SLD #3 normally closed contact.
12	73	S3C	Optional SLD #3 common contact.
13	74	S4NO	Optional SLD #4 normally open contact.
14	75	S4NC	Optional SLD #4 normally closed contact.
15	76	S4C	Optional SLD #4 common contact.

CONNECTOR 16PL — LOCATED ON OPTIONAL APPLICATION CARD

PIN	NOMENCLATURE	DESCRIPTION
1	DCOM	Digital common.
2	+5V	Not used.
3	PI0	Optional parallel word input least significant binary input. For all inputs refer to Table 1, JP8 description for voltage level select. Input must be set for 50 micro-see prior to strobe.
4	PI1	Optional parallel binary bit input 1.
5	PI2	Optional parallel binary bit input 2.
6	PI3	Optional parallel binary bit input 3 (or PIO-PI3 = BCD units digit).
7	PI4	Optional parallel binary bit input 4.
8	PI5	Optional parallel binary bit input 5.
9	PI6	Optional parallel binary bit input 6.
10	PI7	Optional parallel binary bit input 7 (PI4—PI7 = BCD tens digit).
11	PI8	Optional parallel binary bit input 8.
12	PI9	Optional parallel binary bit input 9.
13	PI10	Optional parallel binary bit input 10.
14	PI11	Optional parallel binary bit input 11 (PI8—PI11 = BCD hundreds digit).
15	PI12	Optional parallel binary bit input 12.
16	PI13	Optional parallel binary bit input 13.
17	PI14	Optional parallel binary bit input 14 (PI12—PI14 = BCD thousands digit).
18	PI15	Optional sign bit input.
19	DCOM	Digital common.
20	DCOM	Digital common

CONNECTOR 18PL — BETWEEM CONTROL CARD AND OPTIONAL PROGRAMMER CARD

PIN	NOMENCLATURE	DESCRIPTION
1		No connection
2		No connection
3		No connection.
4		No connection.
5		No connection.
6		No connection.
7		No connection.
8		No connection.
9		No connection
10		No connection
11	RXT	Input from the Programmer card serial link driver to Control card.
12	TXT	Output from the Control card to the Programmer card serial link receiver.
13	/DTRD	Data terminal ready-output from Control card indicating that it is ready to receive a byte from the Programmer card. Signal goes low when ready, high when not.
14	/CTSD	Clear to send-input from Programmer card signalling when it is ready to receive a byte. Signal low when ready, high when not.
15	RST	Reset line from Control card to the Programmer card, active high.
16	+5V	+5V regulated power supply to the Programmer card.
17	+5V	
18	DCOM	Signal and power return for the Programmer card.
19	DCOM	
20	-24V	-24V unregulated power supply to Programmer card.

CONNECTOR NFB (200KVA) - BETWEEN POWER SUPPLY/INTERFACE, CONVERTER PULSE TRANSFORMER AND INVERTER PULSE TRANSFORMER CARDS

PIN	NOMENCLATURE	DESCRIPTION	TERMINATED AT
		Fused AC line 1 voltage feedback sense	Converter (531X195CPT_)
			Converter
			erter
			erter
			verter
			verter
			verter
			nverter
			mverter
			onverter
			onverter
			Converter
			Converter
			Both
			Both

15-07-40

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CONNECTOR NFB (SOOKVA) — BETWEEN POWER SUPPLY/INTERFACE, COMMUTATION POWER SUPPLY (CPS), CONVERTER PULSE TRANSFORMER, INVERTER PULSE TRANSFORMER AND COMMUTATION PULSE TRANSFORMER CARDS

PIN	NOMENCLATURE	DESCRIPTION	TERMINATED AT
1	RT1	Fused AC line 1 voltage feedback sense	Converter (531X185CPT_)
2	RT2	Fused AC line 2 voltage feedback sense	Converter.
3	RT3	Fused AC line 3 voltage feedback sense	Converter
4	CTA	Current Feedback, phase A	CPS (531X186CPS_)
5	CTB	Current Feedback, phase B	CPS
6	CTC	Current Feedback, phase C	CPS
7	AY	Current Feedback common	CPS
8	LVP	Positive line voltage feedback	CPS
9	LVN	Negative link voltage feedback	CPS
10	FCC	Filter capacitor current feedback	CPS
11	AY	Filter capacitor current feedback	CPS
12	CVP	Positive converter voltage feedback	CPS
13	CVN	Negative converter voltage feedback	CPS
14	PCOM	Unregulated power signal return for +24V	All except CPS
15	PCOM		
16	+24V	+24V unregulated power supply	All except CPS
17	+24V		
18	NC	No connection	
19	NC	No connection	
20	NC	No connection	

CONNECTOR NPT (200KVA) • BETWEEN INVERTER FIRING GENERATOR, CONVERTER PULSE TRANSFORMER AND INVERTER PULSE TRANSFORMER CARDS

PIN	NOMENCLATURE	DESCRIPTION	TERMINATED AT
1	ISPA	Phase A positive inverter SCR firing signal	Inverter (531X183XPT_)
2	ISNA	Phase A negative inverter SCR firing signal	Inverter
3	ISPB	Phase B positive inverter SCR firing signal	Inverter
4	ISNB	Phase B negative inverter SCR firing signal	Inverter
5	ISPC	Phase C positive inverter SCR firing signal	Inverter
6	ISNC	Phase C negative inverter SCR firing signal	Inverter
7	CSPA	Phase A positive commutation SCR firing signal	Inverter
8	CSNA	Phase A negative commutation SCR firing signal	Inverter
9	CSPB	Phase B positivexommutation SCR firing signal	Inverter
10	CSNB	Phase B negative commutation SCR firing signal	Inverter
11	CSPC	Phase C positive commutation SCR firing signal	Inverter
12	CSNC	Phase C negative commutation SCR firing signal	Inverter
13	AP	Phase 1 positive converter SCR (1SP) firing signal	Converter (531X195CPT_)
14	AN	Phase 1 negative converter SCR (1SN) firing signal	Converter
15	BP	Phase 2 positive converter SCR (2SP) firing signal	Converter
16	BN	Phase 2 negative converter SCR (2SN) firing signal	Converter
17	CP	Phase 3 positive converter SCR (3SP) firing signal	Converter
18	CN	Phase 3 negative converter SCR (3SN) firing signal	Converter
19	CVIA	Phase A commutation capacitor voltage feedback (+)	Inverter (531X183XPT_)
20	CVCA	Phase A commutation capacitor voltage feedback (-)	Inverter
21	CVIB	Phase B commutation capacitor voltage feedback (+)	Inverter
22	CVCB	Phase B commutation capacitor voltage feedback (-)	Inverter
23	CVIC	Phase C commutation capacitor voltage feedback (+)	Inverter
24	cvcc	Phase C commutation capacitor voltage feedback (-)	Inverter
25	CLS	Commutation voltage limit SCR firing signal	Inverter
26	NC	No connection	

CONNECTOR NPT (SOOKVA) - BETWEEN INVERTER FIRING GENERATOR, CONVERTER INVERTER, AND COMMUTATION PULSE TRANSFORMER CARDS

PIN	NOMENCLATURE	DESCRIPTION	TERMINATED AT
1	ISPA	Phase A positive inverter SCR firing signal	Inverter (531X184IPT_)
2	ISNA	Phase A negative inverter SCR firing signal	Inverter
3	ISPB	Phase B positive inverter SCR firing signal	Inverter
4	ISNB	Phase B negative inverter SCR firing signal	Inverter
5	ISPC	Phase C positive inverter SCR firing signal	Inverter
6	ISNC	Phase C negative inverter SCR firing signal	Inverter
7	CSPA	Phase A positive commutation SCR firing signal	Commutation (531X199SPT_)
8	CSNA	Phase A negative commutation SCR firing signal	Commutation
9	CSPB	Phase B positive commutation SCR firing signal	Commutation
10	CSNB	Phase B negative commutation SCR firing signal	Commutation
11	CSPC	Phase C positive commutation SCR firing signal	Commutation
12	CSNC	Phase C negative commutation SCR firing signal	Commutation
13	AP	Phase 1 positive converter SCR (1SP) firing signal	Converter (531X185CPT_)
14	AN	Phase 1 negative converter SCR (1SN) firing signal	Converter
15	BP	Phase 2 positive converter SCR (2SP) firing signal	Converter
16	BN	Phase 2 negative converter SCR (2SN) firing signal	Converter
17	CP	Phase 3 positive converter SCR (3SP) firing signal	Converter
18	CN	Phase 3 negative converter SCR (3SN) firing signal	Converter
19	CVIA	Phase A commutation capacitor voltage feedback (+)	Commutation (531X199SPT_)
20	CVCA	Phase A commutation capacitor voltage feedback (-)	Commutation
21	CVIB	Phase B commutation capacitor voltage feedback (+)	Commutation
22	CVCB	Phase B commutation capacitor voltage feedback (-)	Commutation
23	CVIC	Phase C commutation capacitor voltage feedback (+)	Commutation
24	cvcc	Phase C commutation capacitor voltage feedback (-)	Commutation
25	CLS	Commutation voltage limit SCR firing signal	Commutation
26	NC	No connection	

APTPL - BETWEEN INVERTER FIRING GENERATOR AND COMMUTATOR POWER SUPPLY

PIN	MNEMONIC	DESCRIPTION
1	ASA	SCRASA1 Firing Signal
2	ASA	SCRASA2 Firing Signal
3	ASB	SCRASB1 Firing Signal
4	ASB	SCRASB2 Firing Signal
5	ASC	SCRASC1 Firing Signal
6	ASC	SCRASC2 Firing Signal

CONNECTOR CPTPL - BETWEEN POWER SUPPLY/INTERFACE AND CPT TRANSFORMER

PIN	NOMENCLATURE	DESCRIPTION
1	—	19V AC, controller power supplies
2	—	19V AC, controller power supplies
3	—	Center tap point, controller power supplies common
4	—	Not used.
5	—	115V AC for controller control and cooling fan. (Hot side)
6	—	115V AC for controller and cooling fan. (Gnd side)

Connector FAPL - Between Supply/Interface card & Controller Cooling Fan

Pin	Nomenclature	Description
1	X1	115V AC for controller fan. (Hot side)
2	x2	115V AC for controller cooling fan. (Gnd side)

Connector OPTPL - Located OR Power Supply/Interface card

Pin	Nomenclature	Description
1	FX1	Fused 115V AC from control power transformer.
2	x2	Grounded 115V AC return from control power transformer.

Connector APSPL - Located on Power Supply/Interface card

Pin	Nomenclature	Description
1	+24V	24V supply for commutation power supply (500KVA only) (not for use external to drive)
2	PCOM	Common Return for +24V supply (not for used external to drive)

COMPL is a 25 pin D-connector provided for use with the RS232C communication option. Refer to GEK-85790, Serial Line User's Guide, for details on the converter configuration.

NOTES

AC-500 ADJUSTMENTS

The following section details adjustments available in hardware and software. The hardware portion describes the location and function of jumpers and potentiometers on the AC-500 cards. The software portion describes key software "parameter" adjustments available.

Hardware Jumper and Pot Adjustments

Most of the jumper selections and pot adjustments have been factory set. When one of the cards has to be changed, make sure the replacement card switches, jumpers and pots are set to be the same as the replaced card. For full description of the jumper and pot adjustments, see Table 1.

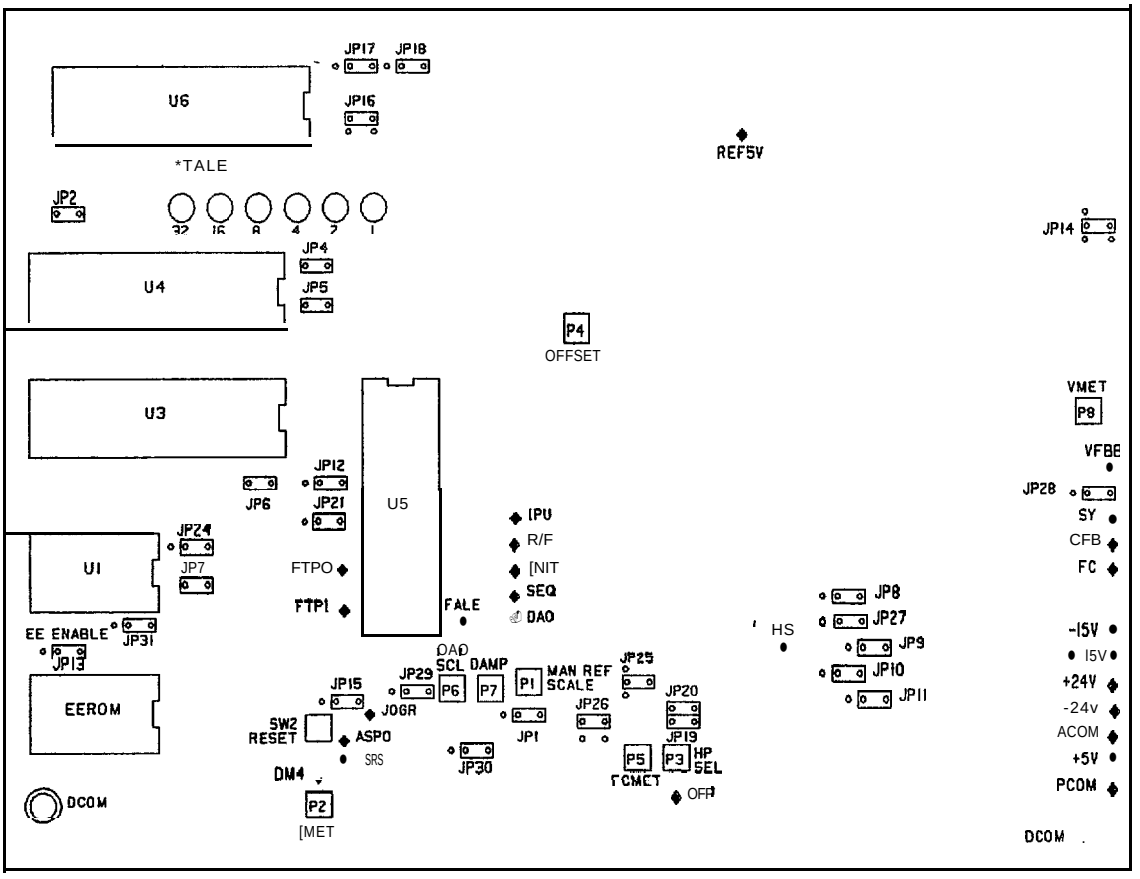
NOTE: If the Control card is replaced, the EEROM (U12) from the original card should be put into the replacement card. If the same failure symptoms still exist, the new EEROM, shipped with the replacement card, can be re-installed but must be reprogrammed per the custom software jumper and adjustment values supplied in the Custom Software Description of parameter list in the drive door pocket.

Hardware jumpers and pots are located on the Control, Power Supply/Interface, Inverter Firing Generator, Inverter Pulse Transformer and Converter Pulse Transformer cards, optional Process Interface card, optional Application card, and optional Programmer card. Figures 3.1 thru 3.12 show the positions of pots, jumpers, test points and other key components.

Most of the pots and jumpers are used in the interface circuits of the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use. The test data sheets supplied with each controller (in the drive door pocket) indicate the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by following the procedures in this book.

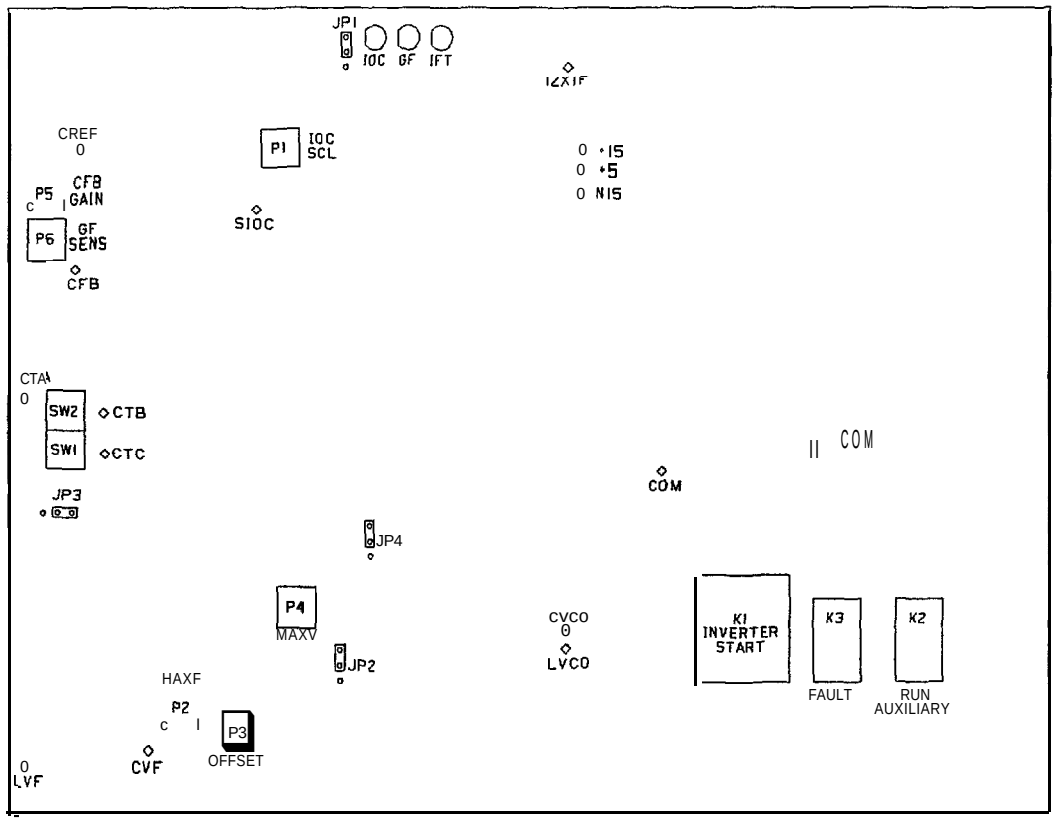
NOTE: Only high impedance digital voltmeters or the optional "on-board" DVM (Test 3) should be used to make the indicated measurements.

NOTE: Before changing any jumper positions, compare actual jumper positions with "as shipped" jumper positions through use of the Custom Software Description.



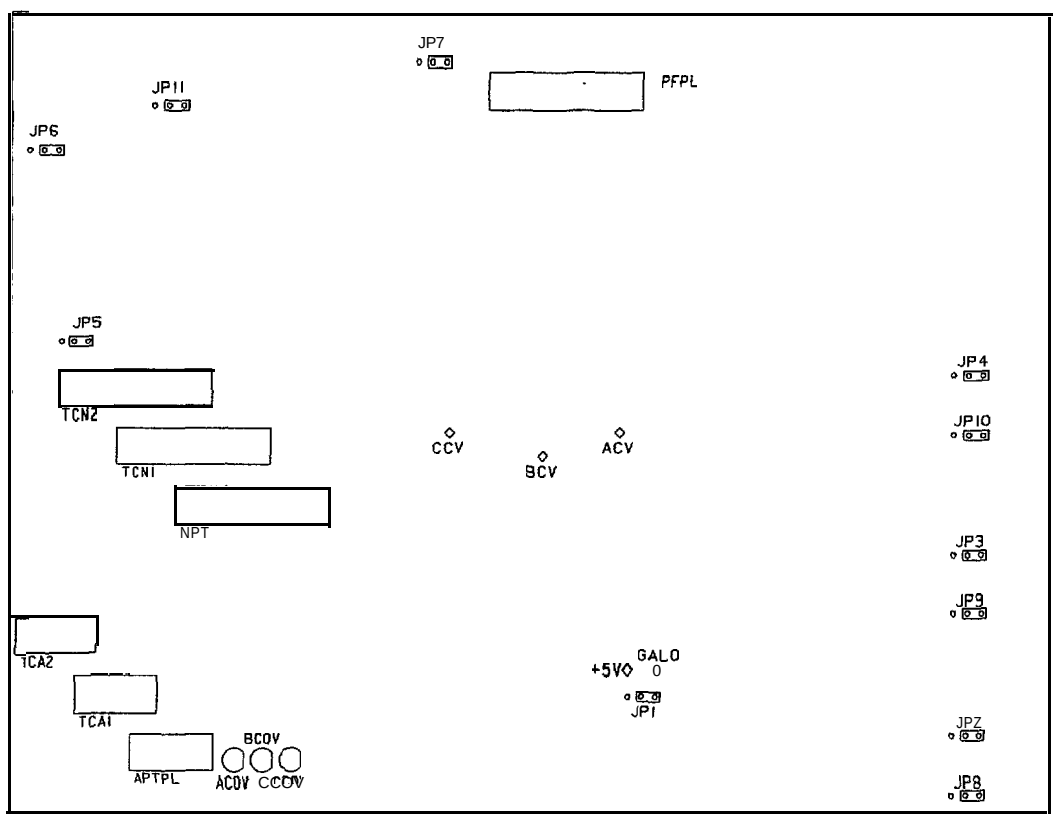
MISC ELEC 442

FIGURE 3.1
CONTROL CARD



S MISC.ELEC 475

FIGURE 3.2
POWER SUPPLY/INTERFACE CARD



S.MISC ELEC.474

FIGURE 3.3
INVERTER FIRING GENERATOR CARD

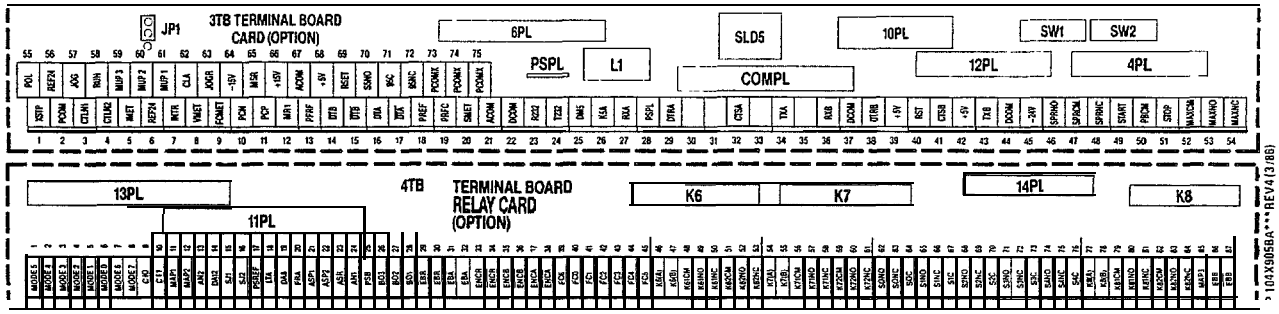
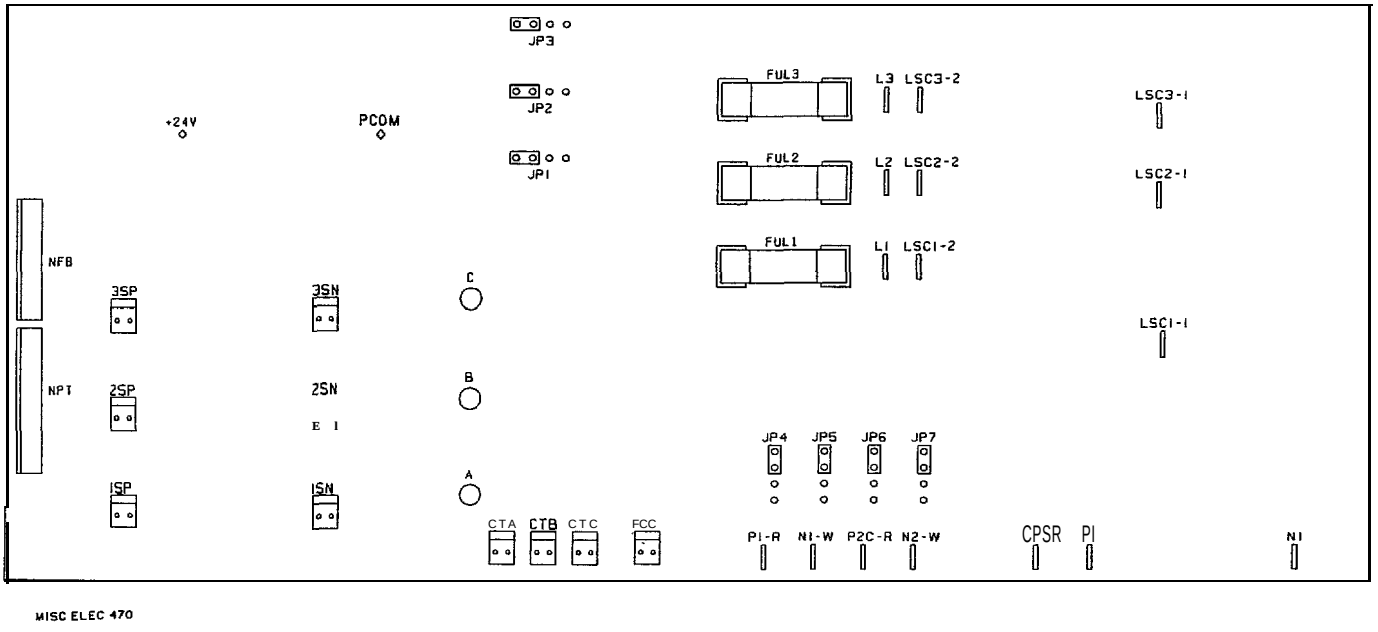
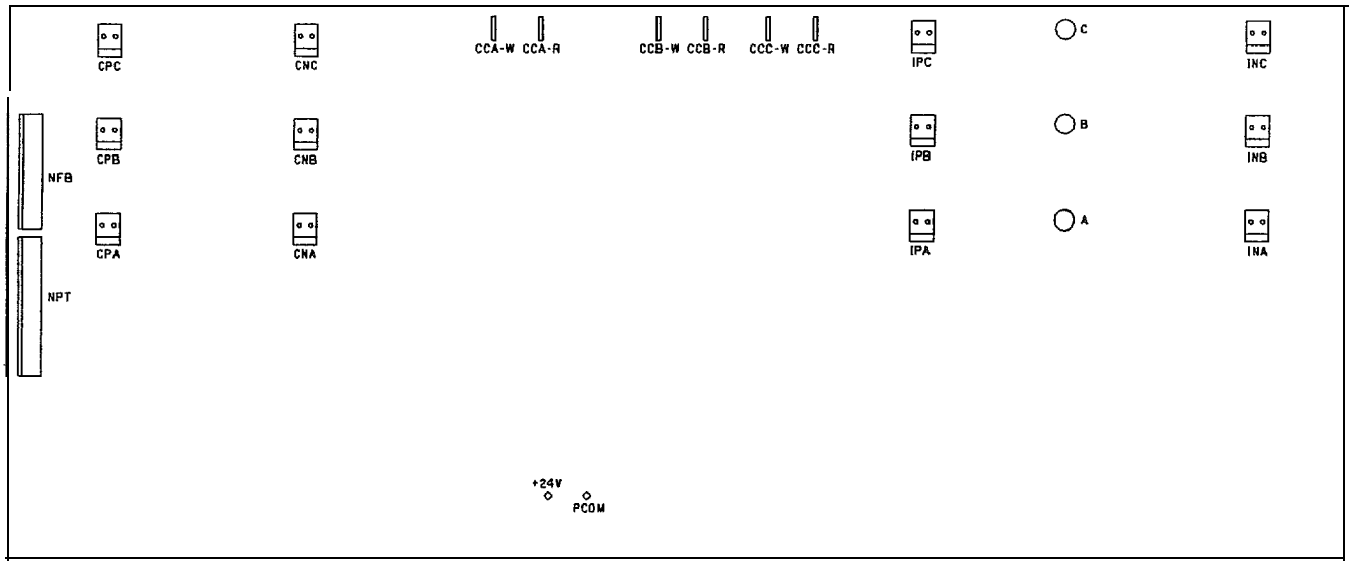


FIGURE 3.4
OPTIONAL TERMINAL BOARDS (3TB AND 4TB)



MISC ELEC 470

FIGURE 3.5
CONVERTER PULSE TRANSFORMER CARD (200 KVA)



5 MISC.ELEC.473

FIGURE 3.6
INVERTER PULSE TRANSFORMER CARD (200 KVA)

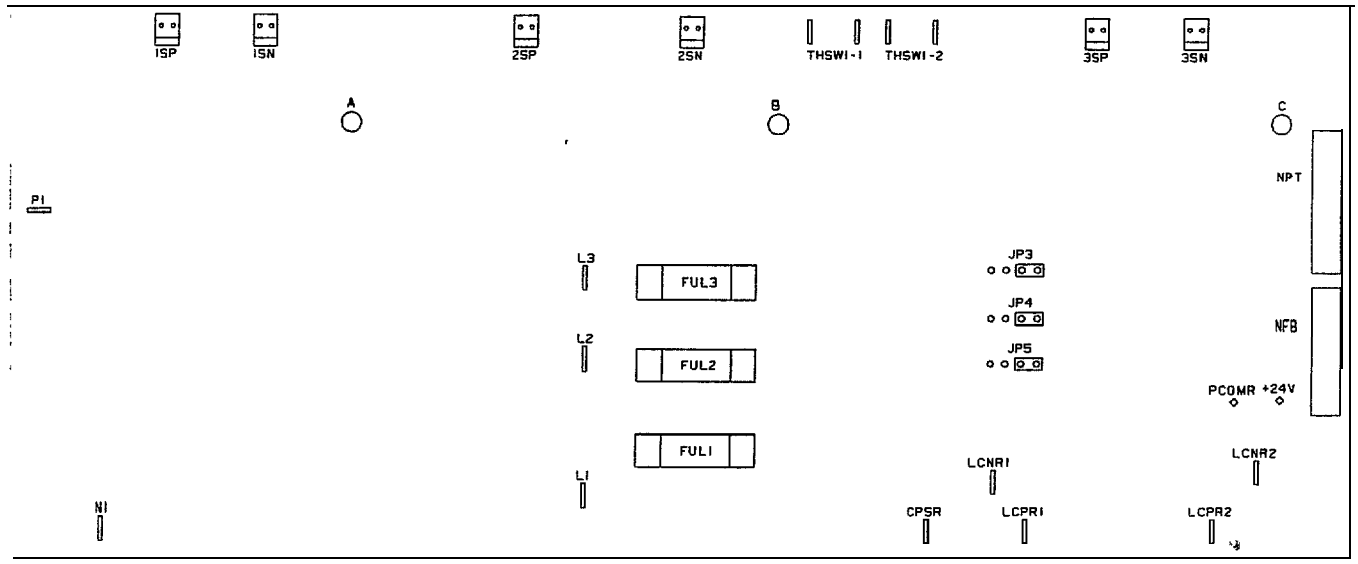
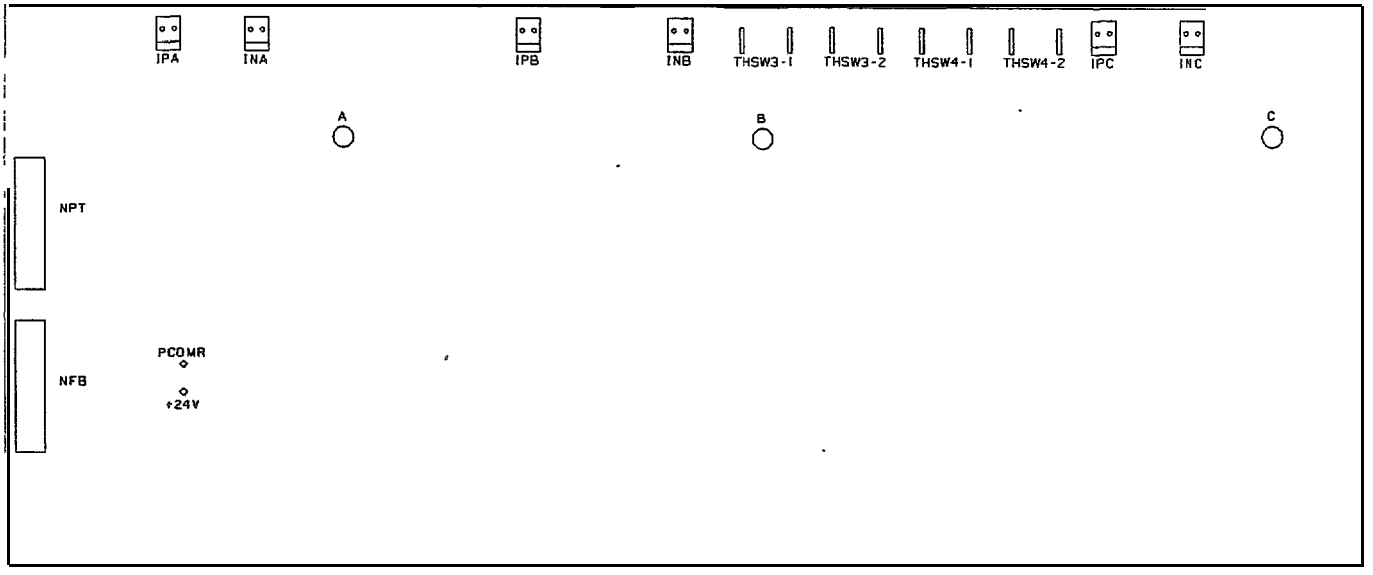
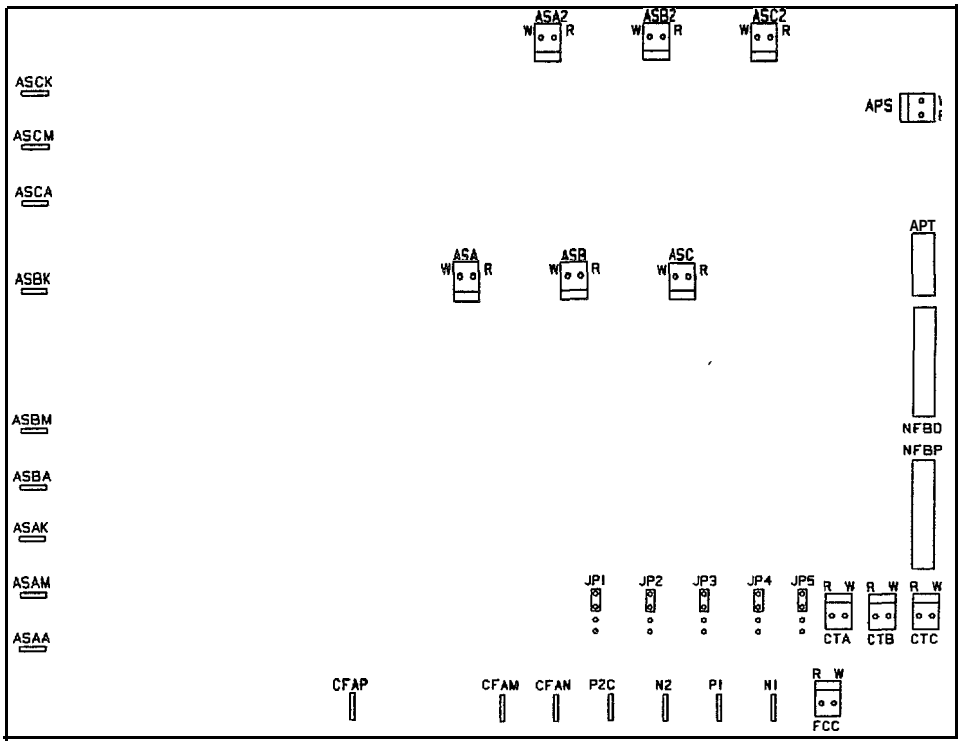


FIGURE 3.7
CONVERTER PULSE TRANSFORMER CARD (500 KVA)



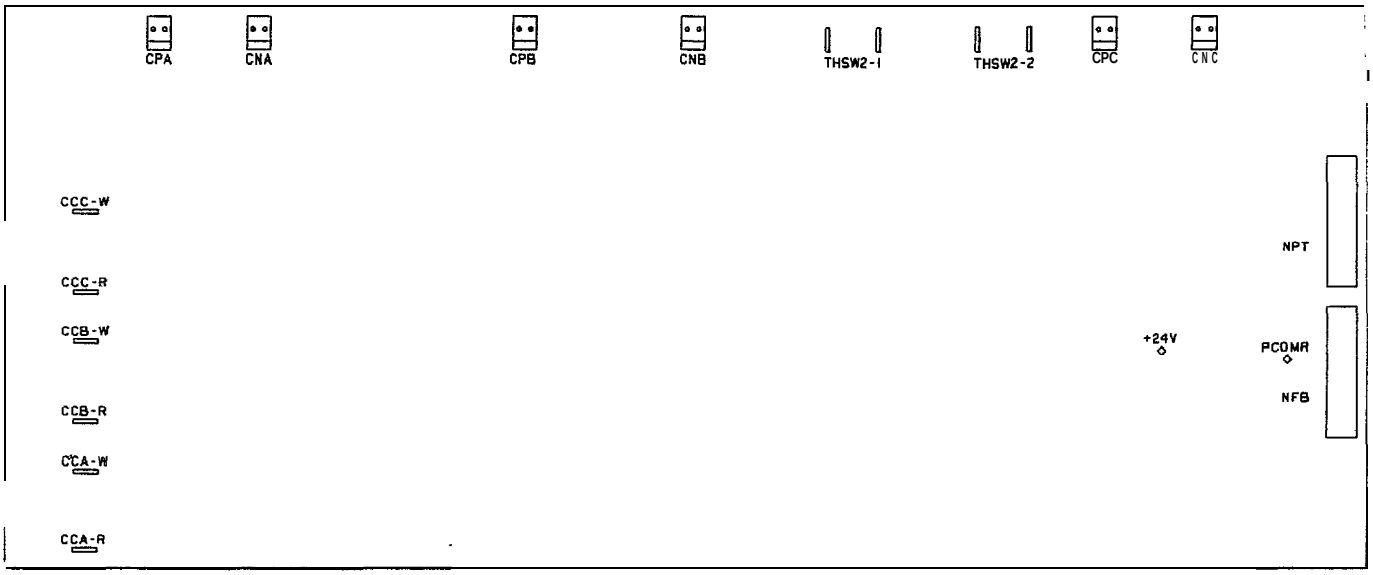
MISC ELEC.472

'FIGURE 3.8
INVERTER PULSE TRANSFORMER CARD (500 KVA)



S.MISC ELEC.478

FIGURE 3.9
COMMUTATION POWER SUPPLY CARD (500 KVA)



S.MISC.ELEC.469

FIGURE 3.10
COMMUTATION PULSE TRANSFORMER CARD (500 KVA)

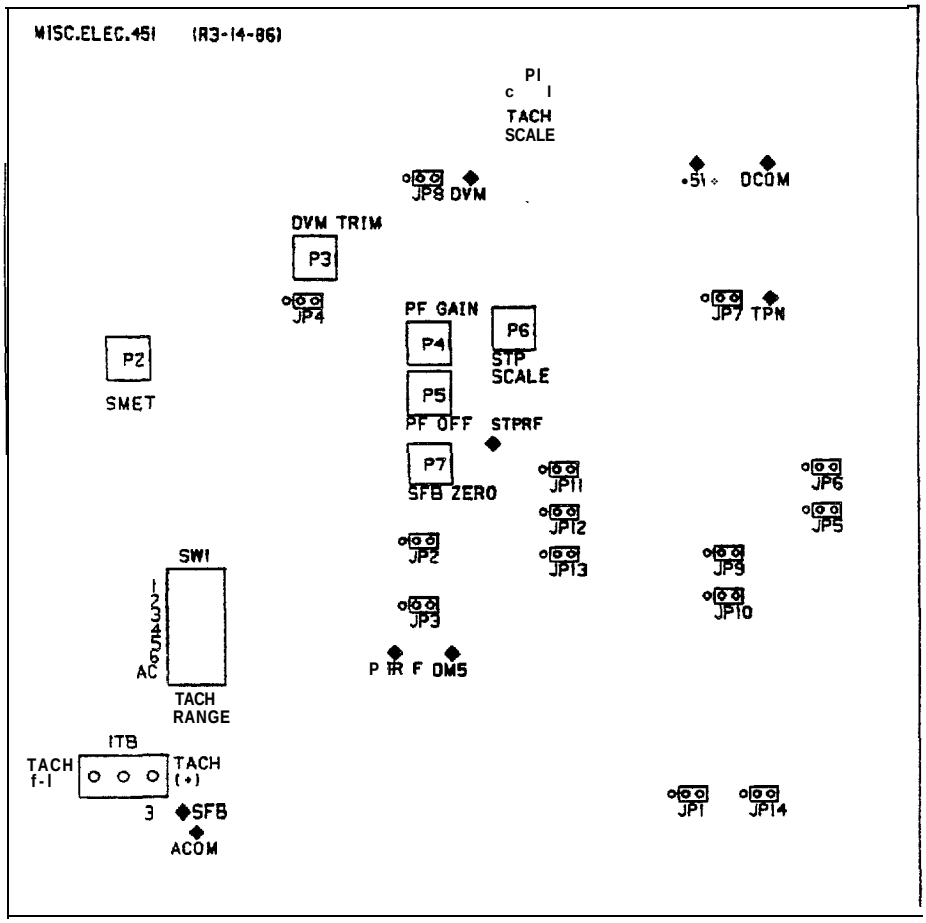


FIGURE 3.11
OPTIONAL PROCESS INTERFACE CARD

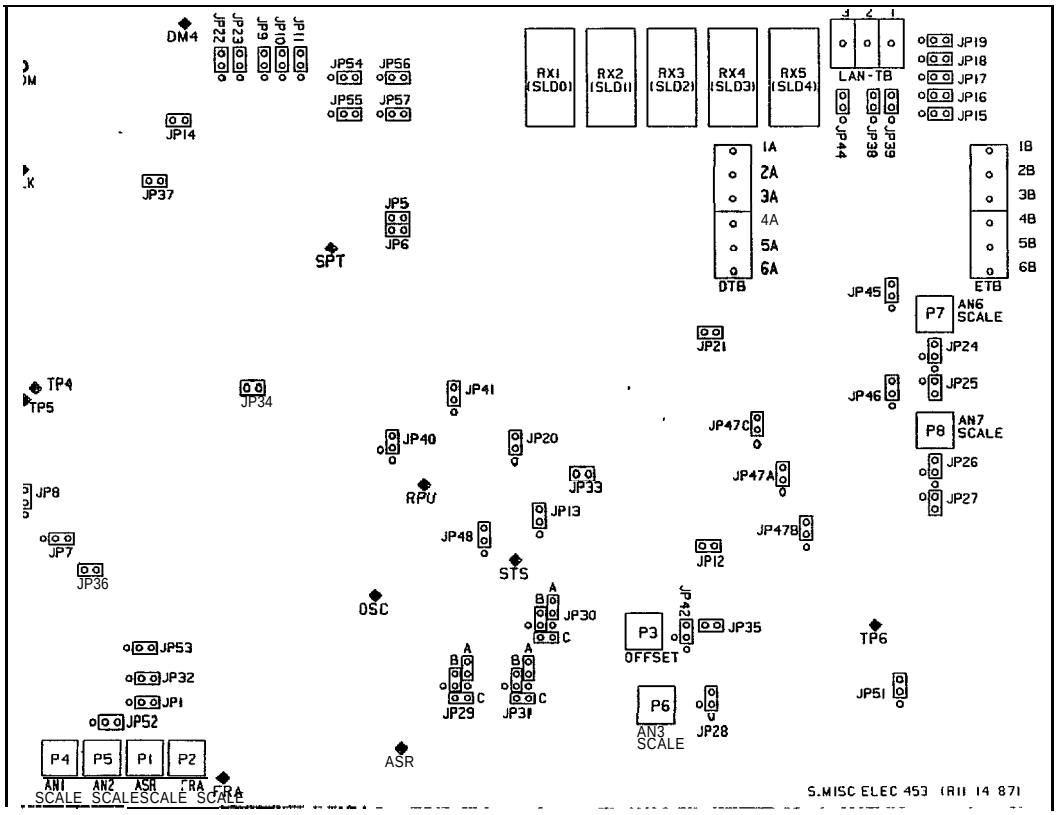


FIGURE 3.12
OPTIONAL **APPLICATION** CARD

TABLE 1

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Pot	P1	MAN REF	To adjust test point SRS to four (4) volts when top speed -reference-is at MSR (6PL-5; connector 6PL, pin number 5). Used in conjunction with jumper JP1.
		P2	IMET	This pot is used to adjust the output of the analog load voltmeter circuit. Adjust the output voltage to be from 2 to 10 volts at rated load. Maximum current load on this signal is 1mA.
		P4	OFFSET	Adjusted by the manufacturer, no need to be adjusted in the field unless directed to do so by fault codes. Refer to adjustment procedure in "Use of the Optional Programmer Module" section if a Programmer is available.
		P5	FCMET	This pot is used to adjust the output of the analog frequency voltmeter circuit. Adjusts the output gain dependent on setting of JP2 on the Power Supply/Interface card. Maximum current load on this signal is 1mA.
		P6	DAO SCL	This pot is used to adjust the gain of the 12 bit digital to analog converter (DAO) located on the Control card. Full CCW sets the full scale gain at 0.25, full CW sets the full scale gain at 1.25 at test point DAO. Refer to Table 3 for DAC output levels.
		P7	DAMP	Adjusted by the manufacturer. To adjust in the field, set to make it visually the same on replacement cards.
		P8	VMET	This pot is used to adjust the output of the analog motor voltage meter circuit. Adjust the output voltage to be 3.5 to 10 volts at rated voltage. Maximum current load on this signal is 1mA.
	Jumper	JP1	MAN REF	This jumper selects the maximum voltage range for the manual speed reference input. 1-2: 8 to 15 volts 2-3: 5 to 8 volts
		JP2 JP5 JP6 JP7		JP2 - JP7 Set as follows: 1-2: should be in this position for proper operation. 2-3: only for manufacturer's use.

TABLE 1 (continued

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP12		SCR Test without Programmer. 1-2: Disables SCR test (normal operation). 2-3: Place in this position before power up for SCR test. An SCR test will be executed the first time a RUN is commanded after power-on or a "hard" reset. Subsequent RUN commands will execute normally.
		JP13		This jumper provides a hardware write protection for the software jumpers and adjustments. 1-2: Protected (should be in this position for maximum security and reliability of the drive) 2-3: Enables reprogramming. Do not perform a "hard" reset or turn power on or off in this position.
		JP14		This jumper selects Phase Loss Trip sensitivity as follows: l-2: Trip below 80% of nominal input voltage. 2-3: Trip below 67% of nominal input voltage. Other settings only per manufacturer instruction.
		JP15		This jumper selects whether the load meter driver output is to be bipolar or absoluted. 1-2: Absoluted (always positive) 2-3: Bipolar (zero center)
		JP16		This jumper selects the line sync reference: 1-2: Only for manufacturer's use. 2-3: Enable line sync. 1-4: Enable Pulse reference input for line sync.
		JP17 JP18		JP17 - JP18 set as follows: 1-2: Only for manufacturer's use. 2-3: Should be in this position for proper operation.
		JP19 JP20		JP19 - JP20 set the polarity of RUN, JOG, POL and XSTOP. 1-2 & 3-4: Inputs are active when pulled down between -24 to 0.4 VDC. Inputs are inactive when open or tied to +3.5 to +24 VDC. 1-3 & 2-4: Inputs are active when tied between +3.5 to +24 VDC. Inputs are inactive when open or held at -24 to 1.5 VDC.
		JP21		Allows interactive card test. 1-2: Disables card test (normal operation). 2-3: Initiates card test if drive is not running. To exit card test, move JP21 back to the 1-2 position and hard reset drive (push SW2).

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control card	Jumper	JP24 JP25 JP26 JP27 JP28		JP24-28 set by manufacturer. Refer to jumper listing in Custom Instruction Book or drive door pocket for proper setting.
		JP29		This jumper selects whether the DAO output has a fixed gain or adjustable gain. 1-2: Fixed gain (See Table 3, RAM Variables) 2-3: Gain adjustable by P6-DAO SCL.
		JP30 JP31		These jumpers set by manufacturer. Refer to jumper listing in Custom Instruction Book or drive door pocket for proper setting.
		JP32		This jumper selects line voltage feedback for line transfer operation. l-2: Standard operation 2-3: AC-500 Line Transfer
	Switch	SW2	Hardware	This switch causes a hardware reset ("hard" reset) of the controller when pressed. Used to clear some drive faults (refer to Table 5, Specific Fault Troubleshooting). The power supplies are toggled "OFF" and "ON" and the drive will return to the initial power up condition.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Power Supply/ Interface card	Pot	P1	IOC SCL	This pot is used to adjust the analog Instantaneous Over-current (IOC) trip setpoint. Factory set for 200% controller rated current (3V @ SIOC)
		P2	MAXF	Adjusted by the manufacturer. Sets gain of the Link Voltage Controlled Oscillator (LVCO) to 53.4Khz/rated link voltage. Do not adjust in field (See * below)
		P3	OFFSET	Adjusted by the manufacturer. Sets VCO offset to 500Hz at zero link volts. Do not adjust in field. (See * below)
		P4	MAXV	Adjusted by the manufacturer. Sets VCO gain to 70.6KHz at rated converter output voltage. Do not adjust in field. (See * below)
		P5	CFB GAIN	This pot is used to adjust the output current feedback (CFB) gain dependent on KVA rating. Check for proper setting of SW1 and SW2 before adjusting (See * below)
		P6	GF SENS	Sets the Ground Fault trip level. (See * below) **NOTE:** The AC-500 controller input power **must** be supplied by an isolation transformer with a grounded neutral WYE output, to allow the ground Fault Function to operate.
	Jumper	JP1		This jumper enables ground fault detection: 1-2: Ground fault detection enabled. 2-3: Ground fault detection disabled.
		JP2		This jumper selects the maximum output DC voltage at FCMET on the Control card. **Frequency (Hz)** 30 / 60 / 75 / 90 1-2: 2-11V / 4-12V / 5-12V / 6-12V 2-3: 1-3V / 1.5-6V / 2-8V / 2-9V
		JP3		This jumper allows for voltage imbalance trip. 1-2: Standard Operation 2-3: Parallel AC-500 (Master drive only)
		JP4		This lumper allows for current imbalance trip. 1-2: Standard Operation 2-3: Parallel AC-500 (Master drive only)
	Switch	SW1 SW2		These switches set the CT load resistor for various KVA ratings. (See * below)

***Detailed information for these adjustments is provided in the "Power Supply/Interface Card Adjustment" procedure following Table 3 in this section.**

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
nvert er Firing Gener- rtor	Jumper	JP1		This jumper sets the minimum timed delay between commutation and inverter SCR firings, dependent on frame size. l-2: 250-500KVA (500KVA frame) 2-3: Up to 200KVA (200KVA frame)
		JP2 JP3 JP4		These jumpers set the commutation pulse width, dependent on frame size. l-2: 250-500KVA (500KVA frame) 2-3: Up to 200KVA (200KVA frame)
		JP5 JP6		These jumpers set the synchronization of inverter to commutation firings, dependent on KVA. 1-2: 250-500KVA (500KVA frame) 2-3: Up to 200KVA (200KVA frame)
		JP7		JP7 set as follows: 1-2: Should be in this position for proper operation. 2-3: Only for manufacturer's use.
		JP8 JP9 JP10		These jumpers set the commutation overvoltage trip. l-2: Low setting (Default) 2-3: High setting
3TB Terminal Board	Jumper	JP1		JP1 selects whether optional SLD5 mounted on 3TB is controlled by the Control card or by an external contact closure. 1-2: Control card operation of SLD5, programmed by ADR085, 086 and 087. 2-3: External control of SLD5 by connecting 3TB-26, K5A to 3TB-56, REF24.
	Switch	SW1 SW2		SW1 and SW2 select the mode of serial link communication through COMPL on 3TB. Refer to the Serial Link Users Guide, GEK-85790, for information on configuring COMPL using SW1 and SW2.

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Converter Pulse Transfmr card [200KVA]	Jumper	JP1 JP2 JP3		Selects AC voltage feedback resistance: 1-3: 360460 VAC input voltage 2-3: 180-230 VAC input voltage 3-4: 520-575 VAC input voltage
		JP4 JP5 JP6 JP7		Selects DC voltage feedback resistance: 1-2: 360-460 VAC input voltage 2-3: 180-230 VAC input voltage 3-4: 520-575 VAC input voltage
Converter Pulse Transfmr card [SOOKVA]		JP3 JP4 JP5		Selects AC voltage feedback isolation resistance: 1-2: 360-460 VAC input voltage 2-3: 180-230 VAC input voltage 3-4: 520-575 VAC input voltage
Comm. Power Supply card [500KVA]		JP1 JP2 JP3 JP4		Selects DC voltage feedback isolation resistance: 1-2: 360-460 VAC input voltage 2-3: 180-230 VAC input voltage 3-4: 520-575 VAC input voltage
		JP5		Inverter fault trip 1-2: 500 KVA Hardware 3-3: 300 KVA Hardware

TABLE 1 (continued)

<table>
<tr><th>.OCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td>'rocess nterface :ard optional)</td><td>Switch</td><td>SW1</td><td>Tach Range Select</td><td>These switches provide coarse scaling of the tach circuit as follows:
DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend.

<table>
<tr><th>SWI POSITION NO. "CLOSED" OR "ON"</th><th>TACH VOLT AT MAX SPEED DC TACH</th><th>AC TACH</th></tr>
<tr><td>All "Open" or "Off"</td><td>10 - 12.5V</td><td>11.1 - 13.9V</td></tr>
<tr><td>1</td><td>12.5 - 15.5V</td><td>13.9 - 17.2V</td></tr>
<tr><td>2</td><td>15.5 - 19V</td><td>17.2 - 21.1V</td></tr>
<tr><td>1,2</td><td>19 - 22v</td><td>21.1 - 24.4V</td></tr>
<tr><td>3</td><td>22 - 25v</td><td>24.4 - 27.8V</td></tr>
<tr><td>1,3</td><td>25 - 3ov</td><td>27.8 - 33.3V</td></tr>
<tr><td>1,2,3</td><td>30 - 37.5V</td><td>33.3 - 41.7V</td></tr>
<tr><td>4</td><td>37.5 - 46V</td><td>41.7 - 51.1V</td></tr>
<tr><td>1,2,4</td><td>46 - 56V</td><td>51.1 - 62.2V</td></tr>
<tr><td>1,2,3,4</td><td>56 - 70V</td><td>62.2 - 77.8V</td></tr>
<tr><td>5</td><td>70 - 87V</td><td>77.8 - 96.7V</td></tr>
<tr><td>2,3,5</td><td>87 - 105V</td><td>96.7 - 117V</td></tr>
<tr><td>1,2,4,5</td><td>105 - 120v</td><td>117- 133v</td></tr>
<tr><td>1,2,3,4,5</td><td>120 - 144V</td><td>133 - 160V</td></tr>
<tr><td>6</td><td>144 - 180V</td><td>160 - 2oov</td></tr>
<tr><td>1,2,4,6</td><td>180 - 205V</td><td>200 - 228V</td></tr>
<tr><td>5,6</td><td>205 - 250V</td><td>228 - 278V</td></tr>
<tr><td>1,2,3,4,5,6</td><td>250 - 310V</td><td>278 - 344V</td></tr>
</table>

NOTE: SW1 position 7 must be off (or open) for DC tachs, on (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.</td></tr>
<tr><td></td><td>Pot</td><td>P1</td><td>TACH SCALE</td><td>Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.</td></tr>
<tr><td></td><td></td><td>P2</td><td>SMET</td><td>Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.</td></tr>
<tr><td></td><td></td><td>P3</td><td>DVM TRIM</td><td>With DVM test point connected to +5V and optional Programmer set to DVM operation, adjust P3 until the Programmer displays 5.00.</td></tr>
</table>

TABLE 1 (continued)

OCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
'rocess nterface ard optional)	Pot	P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, 5 to 65 mA transducer inputs. To adjust, start with PF gain at the full CCW position, and perform the following procedure: — Apply the current reference corresponding to minimum speed. — Adjust PF OFF for 0 volts at test point PFRF. — Adjust the current reference corresponding to maximum speed. — Adjust PF GAIN for 10.0 volts at PFRF. Repeat as necessary. **NOTE:** To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.
		P6	STP SCALE	Set point scale, used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JPI selects circuitry for the serial link communication mode. 1-2: Enables RS422 Reception 2-3: Enables RS232C Reception
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2 JP3** l-2 l-2 5-65 mA current loop follower l-2 2-3 5-50 mA current loop follower **2-3 I-2 4-20 mA** current loop follower **2-3 2-3** 1-5 mA current loop follower or 10V voltage follower
		JP4		JP4 selects the speed meter (SMET) circuit mode. 1-2: Unipolar speed meter driver mode. 2-3: Bipolar meter drive mode (DC tach only).

TABLE 1 (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Process Interface card (optional)	Jumper	JP5		JP5 selects digital tach quadrature mode. 1-2: quadruples frequency of digital tach input (only when JP6 is doubler mode). 2-3: disables quadrature tach mode.
		JP6		JP6 selects digital tach doubler mode. 1-2: doubles frequency of digital tach input at 3TB16. 2-3: disables doubler mode.
		JP7		JP7 selects type of digital tach . 1-2: AN tach linearization mode or reluctance digital tach. 2-3: square wave digital tach.
		JP8		JP8 selects analog tach filter mode. 1-2: enables 4MS filter in analog tach feedback; required for AC tach operation. 2-3: disables 4MS filter: required for speed regulator performance above 20 radians (DC or Digital Tach only)
		JP9 JP10		JP9 and JP10 select the digital tach mode. 1-2: Differential RS422 compatible with terminating resistor. 2-3: Single-ended input at 3TB14 and 3TB16. Removed: Differential operation with no terminating resistor.
		JP11 JP12 JP13 JP14		JP11 JP12 JP13 JP14 1-2 l-2 l-2 l-2 RS422 Serial Link- Standard 2-3 2-3 2-3 2-3 RS232C Serial Line - Standard (DTR low) 2-3 2-3 l-2 2-3 RS232C Serial Link L(DTR hi) l-2 2-3 2-3 2-3 RS422 Xmit/RS232C Hand-shake (Series 6 Programmable controller compatible)
Application card (optional)	Pot	P1	ASR SCALE	P1 is used in conjunction with JP1 to trim test point ASR to four (4) volts for top speed reference applied to 13PL-23 (4TB23).
		P2	FRA SCALE	P2 is used to scale the field reference adjust input at 13PL-20 (4TB20). One volt amplitude at test point FRA increases/decreases the FLDMX parameter setting by 204 counts.
		P7	AN6 SCALE	P7 is used to provide a variable input lead for general purpose amplifier #1. Refer to elementaries and operating notes for further information if used.

TABLE (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Application card (optional)	Pot	P8	AN7 SCALE	P8 is used to provide a variable input lead for general purpose amplifier #2. Refer to elementaries and operating notes for further information, if used.
	Jumper	JP1		JP1 selects the voltage range for the auto speed reference input (ASR) at 4TB23 (13P:-23). l-2: 8 to 15 volts 2-3: 5 to 8 volts
		JP2		JP2 selects whether the encoder Track A input at 11PL-16 (4TB37) is single ended or differential. l-2: RS422 differential 2-3: TTL compatible single-ended.
		JP3		JP3 selects whether the encoder track B input at ll-PL-14 (4TB35) is single-ended or differential. l-2: RS422 differential 2-3: TTL compatible single-ended.
		JP4		JP4 selects whether the encoder marker track input at 11PL-12 (4TB33) is single-ended or differential. l-2: RS422 differential 2-3: TTL compatible single-ended.
		JP5 JP6		JP5 and JP6 select polarity of the eight logic inputs. MODE O-MODE 7, at 13PL-1 thru 8 (4TB1 thru 8). 1-2 & 3-4 The. control is active low. Inputs are active when pulled below 0.8 volts. l-3 & 2-4: The control is active high. Inputs areactive when pulled above 3.5 volts or driven by TTL high output.
		JP7		JP7 selects whether the sixteen bit parallel input is latched on the high or low level of the strobe. l-2: Data latched on high level of PSB. 2-3: Data latched on low level of PSB.
		JP8		JP8 selects whether the sixteen bit parallel input is strobe signals are driven by an active high or low driver. l-2: Lines internally pulled up to +5V. Must be driven by active pull-down to common driver. 2-3: Lines internally pulled down to DCOM. Must be driven by an active pull-up (above 3.5 volt) drive.
		JP14		JP14 set as follows: l-2: Should be in this position for proper operation Open: Only for manufacturer's use.

TABLE I

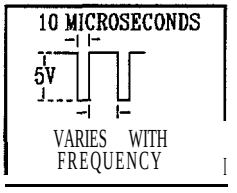
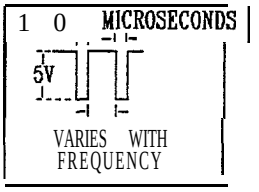
LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Application Card (optional)	Jumper	JP24 JP25 JP26 JP27 JP28 JP29 JP30 JP31		JP24 thru JP31 define functions for the on-board general purpose amplifiers (GP AMP #1 and GP AMP #2). Refer to drive elementaries, operating notes and Custom Software Description for more information (if used).
		JP32		JP32 selects the scaling range of P2-FRA SCALE: 1-2: Normal range 2-3: Extended range
		JP36 JP37		JP36 & JP37 set as follows: 1-2: Should be in this position for proper operation. Open: Only for manufacturer's use.
		JP40 JP41		JP40 & JP41 select whether the encoder input is used as a feedback or a pulse reference: **JP40 JP41** 1-2 1-2 Normal encoder operation. 2-3 1-2 Enables encoder track A as a pulse reference frequency input at 3TB37 (11PL-16, ENCA), 40K Hz maximum. **2-4** — Not used.
		JP45		JP45 selects whether P7-AN6 SCALE is in or out of the GMP AMP #1 circuit: 1-2:. Connects P7 into amplifier circuit. 2-3: Disconnects P7 from amplifier circuit.
		JP46		JP46 selects whether P8-AN7 SCALE is in or out of the GMP AMP #2 circuit: 1-2: Connects P8 into amplifier circuit. 2-3: Disconnects P8 from amplifier circuit.
Programmer Card (optional)	Jumper	JP1 JP2		JP1 and JP2 set as follows: 1-2: Should be in this position for proper operation. 2-3: Only for manufacturer's use.
		JP3 JP4		JP3 and JP4 select whether the Programmer is Local or Hand-held. 1-2: Selects Local or Hand-held Programmer 2-3: Selects Remote Operators Station.
		JP5		JP5 is set at factory. Set so same position on replacement cards.

As a further aid to troubleshooting the AC-500 drive controller, a series of test points have been provided to allow for specific signal observation and/or measurements. Test points not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common for analog circuit power supplies.	
CFB	Current feedback voltage signal developed from the three AC output currents.	Approximately 1.7 VDC at drive rated current.
DAO	Scaled output from 12 bit digital-to-analog converter (DAC).	
DCOM	Common for digital circuit power supply.	
DFP	Delayed firing power for the SCR gate circuit. Maximum external loading is 25 mA. Enabled only when SCR firing is allowed.	+24V when enabled or OV when disabled
JOGR	Jog reference input at 6PL-7. May also be used as an alternate reference. Refer to Elementaries and Custom Software Description.	+/-7 to 14 VDC
MUP1	Multiuse point 1, connects to 6PL-9. Multiuse points may be used to jumper signals from other test points to the associated connector for custom applications, Refer to elementaries supplied with drive.	
MUP2	Multiuse point 2, connects to 6PL-10.	
MUP3	Multiuse point 3, connects to 6PL-11.	
SRS	Scaled speed reference signal	4 volts at top speed reference.
VFBB	Voltage feedback signal. Maximum external load is 1mA. Not for external use.	4 volts at rated link voltage.
-24V	-24 VDC power supply	-22.8 to -25.2 VDC
+24V	+24 VDC power supply	+22.8 to +25.2 VDC
-15v	-15 VDC power supply	-14.25 to -15.75 VDC
+15V	+15 VDC power supply	+14.25 to +15.75 VDC
+5V	+5 VDC power supply	+4.95 to +5.05 VDC

POWER SUPPLY/INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
CTA	Current transformer signal for phase A(T1) of the inverter output. The gain in volts per amps is adjustable by switches 1 and 2 on the Power Supply/Interface card. (See * below)	0.65 to 1.1 VRMS at rated drive current
CTB	Current transformer signal for phase B(T2) of the inverter output. The gain in volts per amps is adjustable by switches 1 and 2 on the Power Supply/Interface card. (See * below)	0.65 to 1.1 VRMS at rated drive current
CTC	Current transformer signal for phase C(T3) of the inverter output. The gain in volts per amps is adjustable by switches 1 and 2 on the Power Supply/Interface card. (See * below)	0.65 to 1.1 VRMS at rated drive current
SIOC	Analog IOC setting test point. Used to set P1-IOC SCALE on the Power Supply/Interface card. (See * below)	1.5 VDC at 100% drive rated current.
12XIF	12 times inverter frequency pulse. A pulse is generated at 12 times the fundamental inverter output operating frequency.	+15 VDC high going pulse, 120 microseconds wide.
CFB	Current feedback voltage signal developed from the three AC output output currents	Approximately 1.7 VDC at drive rated current.
CREF	Current feedback scale adjustment test point. (See * below)	
CVF	Converter voltage feedback.	5.4 VDC at rated converter voltage.
CVCO	Converter Voltage Controlled Oscillator (VCO). 0 to 590 VDC = 29.1 to 70.6 KHz	
LVF	Link capacitor voltage feedback.	-7.5 VDC at rated output voltage.
LVCO	Link capacitor voltage VCO 0 to 590 VDC equals 500 HZ to 53.4 KHz	

*** Detailed Information for these adjustments is provided In the "Power Supply/Interface Card Adjustments" procedure following Table 3 in this section.**

INVERTER FIRING GENERATOR TEST POINTS

Test Point Name	Description	Nominal Value
ACV	Phase A commutation capacitor voltage feedback. Biased from common by one half of the +15 VDC power supply voltage.	4 to 6 volts, peak to peak
BCV	Phase B commutation capacitor voltage feedback. Biased from common by one half of the +15 VDC power supply voltage.	4 to 6 volts, peak to peak
CCV	Phase C commutation capacitor voltage feedback. Biased from common by one half of the +15 VDC power supply voltage.	4 to 6 volts, peak to peak

OPTIONAL PROCESS INTERFACE CARD TEST POINTS

Test Point Name	Description	Nominal Value
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5 VDC power supply	+4.95 to +5.05 VDC
DVM	Input probe connection for digital voltmeter.	
SFB	Analog speed feedback signal. No value when no tach is being used in drive system.	4 volts at top speed.
PFRF	Output of process follower interface circuitry.	10 volts at top speed.
DM5	Dummy point 5, connects to 10PL-16.	

OPTIONAL APPLICATION CARD TEST POINTS

Test Point Name	Description	Nominal Value
ASR	Auto System Reference signal.	4 volts at top speed reference.

SOFTWARE ADJUSTMENTS

Table 2 is listing of "standard" AC-500 software adjustments (parameters) common in most AC-500 applications. Included is the "AC-500 Parameter Map" (equivalent analog representation of parameters), the "Key Parameter Adjustments", and the "RAM Variable Scaling" table, which defines the scaling of RAM addresses available for interrogation of the drive control "circuit". The parameters can be examined and adjusted with the optional Programmer (refer to "Use of Optional Programmer Module"} or via the serial link.

Additionally, a Custom Software Description is shipped with every drive. The Custom Software Description lists the setting and/or function of various programmable jumpers and adjustments (hardware and software) that are identified with the specific model number of a drive. Included in that book is a listing of specific programmed options, their description and a set of software adjustment definitions pertinent to the specified drive.

PARAMETER MAP (Figure 3.13)

The Parameter Map is a diagnostic aid designed to reference specific software adjustments and correlate them to their specific regulating functions. For example, ADR015 • BSLPG is represented as a potentiometer feedback across "the base speed regulating amplifier, giving an effective linear gain adjustment of input to output.

Software adjustments are shown next to their symbol with the mnemonic and the parameter address in a box beside it. Hardware adjustments are represented by a symbol, the adjustment name and the card it is located on (e.g., VMET is a potentiometer. P8 on the Control card). RAM addresses that can be used to examine outputs of certain stages of regulation through TEST 11, TEST 21, TEST 22 or D-A converters, are shown as a three digit number in a circle, connected to the regulating section it is associated with (e.g.: Timed reference can be examined by using RAM address 065 in TEST 11, 21, 22 or the D-A converters).

The Programmer display scaling adjustments are shown in the lower section of the map as well as their relation to feedback.

TABLE 2
KEY PARAMETER ADJUSTMENTS

The following parameter adjustments, listing Programmer mnenomic, address number, function name, and description, will prove helpful in setup and troubleshooting of the AC-500 drive. This is a partial list. For more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive. The following is a description of column headers:

NAME	Mnemonic displayed by Programmer when interrogating parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example: RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.

NOTE: Parameters in Table 2 are grouped functionally per the Parameter Map.

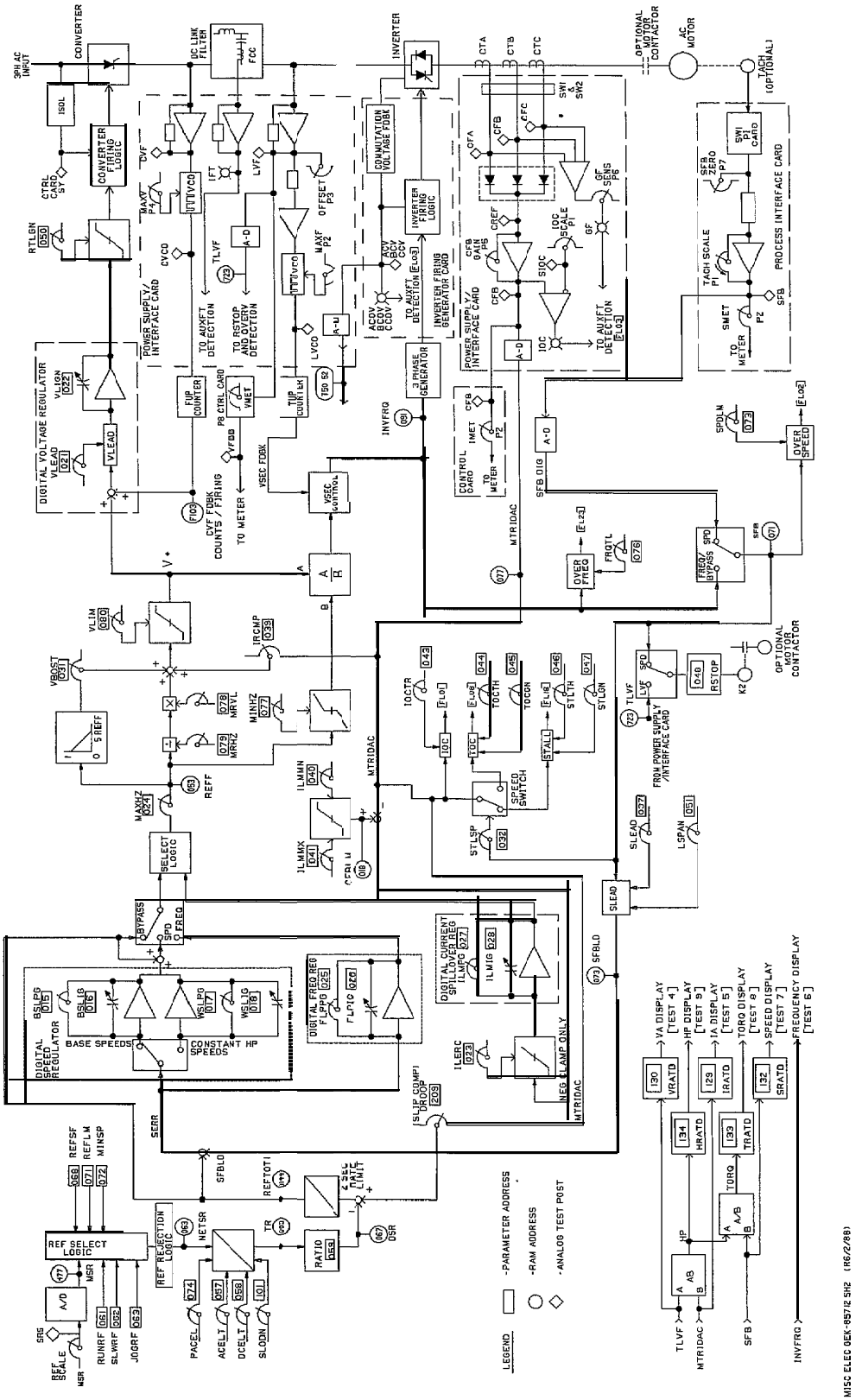


FIGURE 3.13
AC-500 PARAMETER MAP
(ANALOG REPRESENTATION)

TABLE 2
KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
VOLTAGE ADJUSTMENTS						
MRVL	078	VRMS	180	805	1	480

Motor Rated Voltage - - Sets the RMS voltage at motor rated speed or frequency. One unit = 1 VRMS.
NOTE: Output voltage is limited by the value entered in the ADR080-VLIM.

VLIM	080	VRMS	0	805	1	460*

Output Voltage Limit - - Sets the maximum level of RMS output voltage at which the controller will clamp. One unit = 1 VRMS.

CAUTION

If the controller is configured for line transfer, VLIM sets the difference between inverter output voltage and in line voltage. Maximum VLIM setting is 20 in this mode. Refer to Custom Software Description for details.

VBOST	031	Bits/V	0	200	1	0

Voltage Boost - - Sets the voltage boost scale (4.45 units = 1 AC motor volt). Used to provide maximum breakway torque based on load. The additional voltage is automatically tapered out at 50% of rated output voltage as set by ADR078-MRVL.

IRCMP	039	None	0	255	1	0

IR Compensation - - Sets positive or negative IR compensation for the voltage reference calculation.

$$\text{VAC (boost for IRCOMP)} = \frac{460 \times 0.076 \times \text{IRCMP} \times (\text{motor rated current})}{100 \times (\text{drive rated current})}$$

FREQUENCY ADJUSTMENTS						
MINHZ	077	Hz	1	12	1	3

Minimum Inverter Frequency - - Sets the minimum operating frequency of the controller. One unit = 1 Hz.

MRHZ	079	HZ	30	120	1	80

Motor Rated Frequency - - Sets the frequency at base speed. One unit = 1 Hz.

MAXHZ	024	HZ	30	120	1	60

Maximum Full Scale Frequency - - Sets the maximum frequency at full-scale reference. One unit = 1 Hz.

NOTE: By adjustment of this parameter and ADR080 VLIM the controller can run the motor in constant horsepower operation above the frequency set at ADR079-MRHZ.

FRQTL	078	HZ	30	150	1	70

Output Frequency Trip Level - - Frequency at which drive will trip on over frequency (FL23-OVERF) if exceeded. Maximum value should not exceed 1.2 x MAXHZ.

TABLE 2
KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
LINEAR TIME ADJUSTMENTS						
PACEL	074	Declsec	20	9999	1	600
		Pre-Acceleration Time - Sets the rate at which the motor speed will increase in response to a commanded speed change up to 17% rated output voltage. The response above 17% is controlled by ACELT(057). One unit = 0.1 sec.				
ACELT	057	Decisec	20	65535	1	200
		Acceleration Time - Sets the rate at which motor speed will increase in response to commanded speed changes. For linear time, enter time in tenths of a second for accel from 0 to top speed. One unit = 0.1 sec.				
DCELT	058	Decisec	20	65535	1	200
		Deceleration Time - Sets the rate at which motor speed will decrease in response to commanded speed changes. For linear time, enter time in tenths of a second for decel from top speed to 0. One unit = 0.1 sec.				
SLODN	101	None	0	255	1	51
		Slow down ratio - Sets ratio by which speed reference is rescaled when RECAL-REF command is active. 51 corresponds to 20%.				
SPEED REFERENCE, SCALING AND TRIP ADJUSTMENTS						
RSTOP	048	None	40	511	1	60
		Stop Speed - Sets the speed at which the drive is disabled during a stop. One unit = 0.03% speed (for Speed Regulator) or 0.12% of DC link volts. Drive must be below 8% or rated speed/voltage before a restart is allowed, unless the catch mode option is enabled.				
REFSF	068	None	0	6553	1	6553
		Reference Scale Factor - Sets the factor by which references entered from the serial link and Programmer will be multiplied and by which speeds displayed on the Programmer and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). Important - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, parallel input scaling, and speed and reference display. With Tach, REFSF = Top Speed RPM; for frequency regulator, REFSF = Base Speed RPM at rated motor volts and frequency.				
RPMSF	070	RPM	0	6553	1	6553
		RPM Scale Factor - Sets the speed in RPM which corresponds to 4 volts at SFB.				
		NOTE: Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective. Important - affects setting of REFLM, MINSP, SPDLM, tapered current limit speed setting and digital tach scaling. Set the same as REFSF.				

TABLE 2

KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT'
RUNRF	061	RPM	0	6653	1	0

"Run" Key Reference • Sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. Also serves as the top speed reference for motor operated rheostat (MHR) mode. May be a speed, torque or position reference depending on jumper ADR006-RUNJP. Normally RUNRF = Top Speed (RPM).

SLWRF	062	RPM	0	6663	1	0

"Slow" Key Reference • • Sets the preset reference for SLOW commands from the serial link. SLWRF = Slow Speed (RPM).

JOGRF	063	RPM	0	6663	1	655

"Jog" Key Reference • • Sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. JOGRF = Jog Speed (RPM).

REFLM	071	RPM	0	6666	1	6660

Maximum Speed • • Sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF. Typical, REFLM = RUNRF + 100.

MINSP	072	RPM	50	6663	1	666

Minimum Speed • • Sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. MINSP = desired Minimum Speed (RPM). Maximum value should not exceed RPMSF.

SPDLM	073	RPM	0	7936	1	726%

Speed Limit Trip Set Point • • Sets the speed at which the drive will trip. SPDLM = RPM at drive overspeed trip. Should not exceed 1.2 x RPMSF.

RATIO	059	%x100	0	19996	1	10000

Ratio • • Sets the ratio (draw) between the actual speed and the speed reference (NETSR). A ratio greater than 100% (RATIO set at 10,000) causes the drive to run faster than the speed reference; below 100%, slower.

TABLE 2
KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
VOLTAGE REGULATOR ADJUSTMENTS						
VLEAD	021	None	0	255	1	10
Voltage Loop Lead -- Sets the lead time constant of the voltage loop.						
VLIGN	022	None	0	255	1	80
Voltage Loop Integral Gain -- Sets the integral gain of the voltage loop.						
RTLGN	050	None	0	18	1	10
Retard Limit Gain -- Sets the maximum retard limit angle based on voltage reference.						
SPEED REGULATOR ADJUSTMENTS						
BSLPG	015	None	0	255	1	7
Base Speed Loop Proportional Gain --Sets the proportional gain of the outer speed regulator below base speed when tachometer feedback is used.						
BSLIG	018	None	0	255	1	8
Base Speed Loop Integral Gain -- Sets the integral gain of the outer speed regulator below base speed when tachometer feedback is used.						
WSLPG	017	None	0	255	1	7
Weak Field Speed Loop Proportional Gain -- Sets the proportional gain of the outer speed regulator above base speed.						
WSLIG	018	None	0	255	1	8
Weak Field Speed Loop Integral Gain -- Sets the integral gain of the outer speed regulator above base speed.						
FLPPG	025	None	0	255	1	7
(Tachless) Frequency Loop Proportional Gain -- Sets the proportional gain of the outer frequency regulator when frequency is used as a feedback instead of a tachometer.						
FLIPG	028	None	0	255	1	8
(Tachless) Frequency Loop Integral Gain -- Sets the gain of the outer frequency regulator when a tach is not used.						
SLEAD	037	None	0	255	1	10
Speed Feedback Lead -- Sets the lead time constant on tachometer feedback.						
LSPAN	051	None	0	15	1	0
Speed Feedback Lead/Lag Span -- Sets the span of the tachometer lead/lag network. A setting of 0 is treated as a two-to-one span.						
DROOP	209	None	0	255	1	0
Speed Droop Gain -- Sets the gain of the speed droop function which causes speed to droop positive or negative as a function of current.						

TABLE 2
KEY PARAMETER ADJUSTMENT

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
CURRENT LIMIT REGULATOR AND TRIP ADJUSTMENTS						
ILMMN	040	% Rated	10	225	2	10
		Minimum Current Limit . . Sets the minimum allowable current limit level; must be set below the desired maximum curent limit setting. Also acts as the third current limit setting when tapered current limit is used. One unit = 1% current.				
ILLMX	041	% Rated	0	225	2	150
		Maximum Current Limit . . Sets the maximum current level allowed by the drive. One unit = 1% current.				
ILMPG	027	None	0	255	1	10
		Current Limit Proportional Gain . . Sets the proportional gain of the spillover current limit regulator.				
ILMIG	028	None	0	255	1	30
		Current Limit Integral Gain . . Sets the integral gain of the spillover current limit regulator.				
ILERC	023	None	10	255	1	12
		Current Limit Error Rate Clamp . . Sets the maximum spillover curent error presented to the current regulator.				
IOCTR	043	% Rated	50	225	2	200
		Instantaneous Overcurrent Trip Setpoint . . Sets the current level at which the drive will trip. One unit = 1% current.				
TOCTH	044	% Rated	10	85535	2	120
		Timed Overcurrent Sense Threshold . . Sets the current level at which the TOC integrator begins to time-up. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.				
TOCGN	045	None	0	225	1	38
		Timed Overcurrent Response. .Sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used.				

$$\text{TOCGN} = \frac{66{,}666}{(\text{Trip Time})\ (\text{X.TOCTH})}$$

where x = % current corresponding to desired trip time.

NOTE: The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 60 seconds.

STLTH	048	% Rated	10	85535	2	30
		Stalled Overcurrent Sense Threshold . . Sets the current level at which the stalled motor TOC integrator begins to time up when motor speed is below 1%. One unit = 1% current. Refer to the Custom Software Description if multi-motor function is used.				
STLGN	047	None	0	65535	1	55
		Stalled Overcurrent Response . . Sets the time constant of the stalled motor TOC integrator. The defining equation is the same as for ADR045-TOCGN. Refer to the Custom Software Description if multi-motor function is used.				

NOTE: The default values for STLTH and STLGN allow the drive to operate at 150% current for 10 seconds.

TABLE 2
KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
STLSP	032	**% Rated**	0	100	1	10

Stall setpoint - - sets the lower speed range, as a % of RPMSF, that stall TOC gains/limits are used.

DIAGNOSTIC TEST ADJUSTMENTS

DACO	090	**None**	0	65535	1	0

D-A #0 Output Control - - Controls what internal variable is sent to the 12 bit D-A converter (DACO) on the Control card. Enter the desired RAM Variable number into ADR090 to program DACO.

Refer to "RAM Variables", Table 3 at the end of this listing for a list of variables and their relative values.

DAC8	066	**None**	0	65535	1	0

D-A #1 Output Control - - Controls what internal variable is sent to the 8 bit D-A converter on the Application card. See DACO for programming information.

DAC12	097	**None**	0	65535	1	0

D-A #2 Output Control - - Controls what internal variable is sent to the 12 bit converter (DAC12) on the Appliction card. See DACO for programming information.

BIT#1	096	**None**	0	66535	1	0

Bit #1 Output Control - - Controls what internal bit is sent to bit output #1 (B01) on the Application card. Possible outputs are shown in Table 3, "RAM Variables", as Logic Signals.

In addition, the sign of all of the Variables given in Table 3 may be used as outputs.

BIT#2	099	**None**	0	65535	1	0

Bit #2 Output Control - - Controls what internal bit is sent to bit output #2 (B02) on the Application card. See Bit#l for programming information.

BIT#3	100	**None**	0	66535	1	0

Bit #3 Output Control - - Controls what internal bit is sent to bit output #3 (B03) on the Application card. See Bit#l for programming information.

DIAG1	122	**None**	0	65635	1	0

Diagnostic Variable #1 - - Selects 1 of 2 variables which can be displayed on the serial link or Programmer using "display diagnostics". See RAM variables, Table 3.

DIAG2	123	**None**	0	65535	1	0

Diagnostic Variable #2 - - Selects second variable which can be displayed on the serial link or Programmer using "display diagnostics". See RAM variables, Table 3.

SCHRL	127	**None**	1	26	1	16

Left Scaled Unit Character - - Determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B"; 26, "Z".

SCHRR	126	**None**	1	26	1	13

Right Scaled Unit Character - - See ADR127-SCHRL for programming information.

TABLE 2
KEY PARAMETER ADJUSTMENTS (cont'd)

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT
IRATD	129	None	0	3276	1	100
		Displayed motor current scaling - - Controls the factor by which per unit current is multiplied for display on the Programmer (Test 5).				
VRATD	130	None	0	2621	1	100
		Displayed Output Voltage scaling - - Controls the factor by which per unit output voltage is multiplied for display on the Programmer (Test 4).				
SRATD	132	None	0	6200	1	100
		Displayed Speed scaling - - Controls the factor by which per unit speed is multiplied for display on the Programmer (Test 7).				
TRATD	133	None	0	3276	1	100
		Displayed Torque scaling - - Controls the factor by which per unit torque is multiplied for display on the Programmer (Test 8).				
HRATD	134	None	0	3276	1	100
		Displayed HP scaling - - Controls the factor by which per unit horsepower is multiplied for display on the Programmer (Test 9).				

TABLE 3
RAM VARIABLES FOR PROGRAMMER AND DAC OUTPUTS

RAM ADR	VARIABLE NAME	VALUE AT 100% SIGNAL DECIMAL	VALUE AT 100% SIGNAL HEX	VOLTS ON DAC AT 100% SIGNAL
053	REFF-Frequency Reference Output (60 Hz. equals 100%)	2047	0800	—
063	NETSR-Net Speed Reference	6553	1999	8.0
065	TR-Timed Reference	6553	1999	8.0
067	DSR-Draw Speed Reference	6553	1999	8.0
069	JR-Jog Reference	6553	1999	8.0
071	SFB-Speed Feedback	6553	1999	8.0
073	SFBLD-Speed Feedback Lead	6553	1999	8.0
077	MTRIDAC-Motor Current	100	0064	8.0
091	INVFRQ-Inverter Frequency	6553	1999	8.0
445	Filtered Voltage Feedback	1638	0666	8.0
448	Filtered Current Feedback	1638	0666	8.0
477	MSR-Manual Speed Reference	6553	1999	8.0
501	TOCT-Timed Overcurrent Timer (Trips at 1024)	1024	0400	0.3
723	TLVF-Link Voltage Feedback (590 VDC = 100% signal)	177	00B1	—
F103	CVF-Converter Voltage Feedback	120	0078	—

NOTE: F103 is a Firing Microprocessor (FUP) RAM variable. To use this variable for any diagnostic or device control, program ADR125-FUPDI as 103 (this sets the FUP Diagnostic variable at 103 - CVF). Then program the desired diagnostic or control parameter (DAC, SLD, etc.) with a value of 93. This loads the FUPDI variable into the desired function.

TABLE 3

RAM VARIABLES FOR PROGRAMMER AND DAC OUTPUTS (cont'd)

LOGIC SIGNALS

RAM ADR	LOGIC SIGNAL NAME
137	Trip Fault
140	Current Limit Flag
147	Ready to Start Flag
148	Reference Enable
152	Precondition
156	Aux Fault (Analog FL13-AUXFT) Flag
175	Start/Stop Control

When used with Bit outputs 0 VDC is logic 0 (off) and ±24VDC is logic 1 (on).
When used on DIAG1 or DIAG2 variable, 0000 is logic 0 and 8080 is logic 1.

POWER SUPPLY/INTERFACE CARD ADJUSTMENTS

On the Power Supply/Interface card there are three settings that require adjustment when the card is replaced. They are:

P1-IOC SCL	Analog IOC Trip Level
P5-CFB GAIN	Current Feedback Gain
P6-GF SENS	Ground Fault Sensitivity

Follow the procedures outlined below to set these adjustments.

NOTE: All adjustments and test points referred to in this section are located on the Power Supply/Interface card unless specifically stated as otherwise.

P1-IOC SCALE (Instantaneous overcurrent)

Monitor the voltage from the SIOC test point to the COM test point. Adjust Pl-IOC SCL for the desired SIOC voltage as determined from the equation:

$$\text{SIOC VDC} = \frac{1.5 \times \text{IOC Amps}}{\text{Drive Rated Amps}}$$

NOTE: Normally set at 3.0 VDC for an IOC trip level of 200% of rated drive current.

P5-CFB GAIN (Current Feedback Gain)

1. With controller power off, disconnect the motor from the controller by removing the output leads (T1, T2, T3) or disabling the output starter. Connect a Programmer to the drive, if not already provided.

2. Set SW1 and SW2 switches per Table A.

3. Apply power and start the drive. Jumper the +5V test point to the CREF test point on the Power Supply/Interface card.

4. By reading the DC voltage from the CFB test point to the COM test point or using the % LOAD readout on the Programmer; adjust P5-CFB GAIN for the appropriate "Static Settings" from Table A.

NOTE: Depending on the settings of stall or running TOC and IOC, the drive may trip while attempting the adjustments. If this occurs use the voltage reading at the CFB test point.

5. Reconnect and run the motor at greater than 75% speed and greater that 75% load if possible to verify and fine tune the P5 adjustment with the % LOAD readout.

$$\text{\% LOAD readout} = \frac{\text{Actual output amps} \times 100\%}{\text{Rated Drive-Amps}}$$

TABLE A

	460V DRIVE RATED		POWER SUPPLY CAR SWITCH SETTINGS		TEST POINT CTA, CTB AND CTC	STATIC SETTINGS VALUES	
	KVA	AMP	SW1	SW2	VAC AT RATED AMPS	% LOAD/	CFB (VDC)
500 KVA	500	650	Closed	Closed	1.0V	117%	2.32V
Frame Size	400	520	Closed	Closed	0.83V	151%	2.96V
	350	455	Closed	Open	0.91V	131%	2.57V
(5000:1 CT)	300	390	Open	Closed	0.99V	115%	2.27V
	250	310	Open	Closed	0.82V	145%	2.85V
	200	260	Open	Open	1.1V	99%	1.95V
200 KVA	200	260	Closed	Closed	1.0V	117%	2.32V
Frame Size	150	195	Closed	Open	0.97V	122%	2.40V
(2000: 1 CT)	125	160	Closed	Open	0.80V	142%	2.78V
	100	130	Open	Closed	0.87V	141%	2.77V
	75	98	Open	Open	1.0V	107%	2.11V
	60	77	Open	Open	0.79V	143%	2.76V
	50	70	Open	Open	0.66V	161%	3.08V

NOTE: The AC-500 controller input power must be supplied by an isolation transformer with a grounded neutral WYE output to allow the ground fault function to operate.

P6-GF SENSE (Ground Fault Sensitivity)
If the ground fault feature is enabled (JP1 in the 1-2 position) ; set P6 to the same position (visually) as on the original card. The adjustable range is from 27% drive rated amps at full CCW end and 11% drive rated amps at full CW.

WARNING

THE REMAINING ADJUSTMENTS ON THE POWER SUPPLY/INTERFACE CARD ARE SET AND SEALED AT THE FACTORY. IF THE PROPER OUTPUT VOLTAGE OR FREQUENCY CAN NOT BE OBTAINED (AS SET IN ADR024-MAXHZ AND ADR078-MRVL) BY USING THE NORMAL START-UP PROCEDURES, CONTACT **GE DRIVES DEPARTMENT PRODUCT SERVICE.** THE REMAINING POTENTIOMETERS MAY REQUIRE ADJUSTMENT. IF INSTRUCTEDTO DO SO, THE FOLLOWING PROCEDURE DETAILS ADJUSTMENT OF THESE POTENTIOMETERS.

The remaining Power Supply/Interface card adjustments are:

P4-MAXV	Converter voltage feedback gain
P2-MAXF	Link Voltage feedback gain
P3-OFFSET	Link Voltage feedback offset

P4-MAXV (Converter Voltage Feedback Gain)
The converter voltage feedback consists of a resistance isolated differential amplifier, a gain pot P4 and the Voltage Controlled Oscillator (VCO) circuit. The circuit should produce approximately 29.4 KHZ at OVDC and 70.6 KHZ at the rated voltage of 295,590 or 738 VDC for the 230, 460 or 575V AC jumper settings, respectively.

1. Verify that the isolation resistor jumpers are set for the correct input voltage range on the:

   500 KVA Frames . Commutation Power Supply card JP1, 2, 3 &4
   Converter Pulse Transformer card JP3, 4 & 5

   200 KVA Frames . Converter Pulse Transformer card JP1 through JP7

2. With the controller power off, disconnect the motor from the controller by removing the output leads (T1, T2, & T3) or disable the output starter.

3. Apply power.

4. Using the Programmer verify that ADR130 . VRATD has the value of the rated input voltage from the controller nameplate or 100 for a percent rated voltage display.

5. Start the drive and increase reference to full speed. (4.0 VDC at test point SRS on the Control card for an analog reference at MSR).

6. Using the Programmer in the Diagnostic Mode, Test 4 (ARMV), adjust P4-MAXV for a reading equal to the value in ADR080-VLIM or 100%.

NOTE: The PI . N1 converter output voltage should be (ADR080-VLIM X 1.283) Volts DC.

7. Stop the drive, proceed to the next section to adjust the link voltage VCO circuit.

P2-MAXF & P3-OFFSET

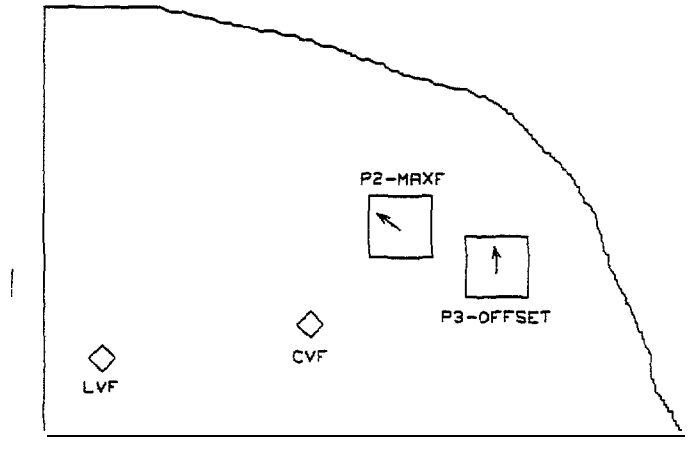
NOTE: This should be factory set; only adjust if maximum frequency cannot be obtained. P4-MAXV must be set first.

The link voltage feedback has similar circuitry as the converter voltage feedback except an offset pot is added to the VCO circuit. The circuit will produce approximately 500HZ at OVDC and 53.4HZ at the rated voltage of 295, 590 and 738VDC at P2C to N2.

1. With controller power off, verify the resistor jumpers are set for the correct input voltage range as done in Step 1 of the P4-MAXV procedure.

2. Disconnect the motor from the controller by removing the output leads T1, T2, T3 or disable the output starter.

3. Adjust position of P3 and P2 as shown in sketch below:



4. Apply power and note the following parameter values.

ADR031-VBOOST _____
ADR024-MAXHZ _____
ADR072-MINSP _____
ADR077-MINHZ _____
ADR078-MRVL _____
ADR079-MRHZ _____
ADR080-VLIM _____

5. Set ADR031-VBOOST equal to 0 and ADR072-MINSP, ADR077-MINHZ equal to 1.
Calculate the following from the EEROM parameters.-,

OFFSET FREQUENCY =
0.172 x ADRO79-MRHZ = _____

OFFSET VOLTAGE =
(ADR024-MAXHZxADR078-MRVL)÷(ADR079-MRHZx6)= _____

6. While monitoring Diagnostic Mode Test 4 (ARMV) ; start the drive and increase reference until the OFFSET VOLTAGE is obtained.

7. Switch to Diagnostic Mode Test 6 (FREQUENCY) and adjust P3-OFFSET until the OFFSET FREQUENCY is obtained.

9. Repeat 6, 7, and 8 several times until the readings are repeatable without any adjustment.

10. Change ADR031-VBOOST, ADR072-MINSP and ADR077-MINHZ back to the original values.

NOTES

START UP PROCEDURE

This startup procedure is meant as a general guide for the initial checkout and startup of an AC-500 Controller, It references some general hardware and software checks relative to most basic drive applications. Refer to the sections on "Hardware Jumpers and Pot Adjustments" and "Key Parameter Adjustments" for more complete information on adjustments available on the AC-500 Controller. Refer to the Custom Software Description and Elementaries for information on wiring and adjustments specific to your drive application.

NOTE: Prior to beginning actual drive start up, read this entire manual and be aware of all pertinent information. If an optional Programmer was supplied with the control, review the "Use of Optional Programmer Module" section. Special diagnostic functions are available which will greatly aid drive start up.

Before Applying Power

Every AC-500 controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are performed:

WARNING

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT. FOLLOW THE START-UP PROCEDURES STEP BY STEP.

READ AND HEED ALL **WARNING, CAUTION** AND **NOTE** LABELS POSTED ON THE EQUIPMENT.

A. Verify that all incoming wiring is in agreement with the elementary drawings supplied with the drive, and conforms to approved wiring practices (wire size, separation of power from control, etc.)

B. Verify that incoming three-phase voltage is in accordance with nameplate requirements (+10%, -5%), unless otherwise specified.

C. Confirm that all jumpers (JP) and switch positions (SW) are in accordance with the jumper selection chart supplied in the controller pocket and Custom Software Description. Refer to the card outlines, Figures 3.1 thru 3.12 for the location of jumper and switches and to Table 1 for jumper/switch functions.

D. Verify that all connections are tight.

E. Verify that the motor has been properly installed and that the shaft is free to rotate.

F. Using an ohmmeter set to the xl ohms scale, check the AC power fuses, all fuses on the Converter Pulse Transformer card, and all control power fuses.

G. If specific information on the motor and application have **not** been submitted prior to shipping, complete the following AC-500 Parameter Quick Calculation Chart; otherwise, skip to the Apply AC Power section of this procedure.

AC-500 PARAMETER QUICK CALCULATION CHART

The following is a method of calculating the basic parameters to satisfactorily run an AC-500 controller for most simple applications. If detailed motor and application information was available prior to factory testing of the controller, then all parameters have been precalculated and entered.

WARNING

IF THE DRIVE IS TO BE OPERATED ABOVE BASE FREQUENCY (60 Hz), THE MOTOR DESIGN MUST BE REVIEWED WITH THE MOTOR MANUFACTURER TO VERIFY THAT OPERATION AT THESE SPEEDS IS SAFE.

Record the following Motor Nameplate information:

Motor HP _____ Motor Amps _____
Motor Volts _____ Motor Hz _____ (Rated Hz)
Motor RPM _____ (at Rated Frequency)

NOTE: For the synchronous RPM value, enter the next highest hundred RPM over Motor RPM.

Example: Use 1800 Synchronous RPM for a 1750 RPM Motor.
Synchronous RPM _____

Record the necessary load information:

Maximum RPM required _____ Minimum RPM _____
Accel Time (Sec) _____ Decel Time (Sec) _____
Drive Amps (from controller nameplate output rating) _____
If possible, estimate the ratio of motor to load inertia (Motor Inertia/Load Inertia) _____

Calculate the following parameter values as shown from the information recorded above.

NOTE: Round the numbers to the nearest whole number.

VOLTAGE SETTINGS:

ADR#	NAME	FORMULA		MIN	MAX	DEFAULT
078	MRVL	= Motor Volts RMS	= _____	180	605	460
080	VLIM	= Motor Volts RMS	= _____	180	605	460
	For line transfer drives see the Custom Instruction Book for ADR080-VLIM settings:					
031	VBOST	= AC Output Voltage Boost = 4.28 x Required voltage boost.	= _____	0	200	0
	NOTE: Normally set to zero, maximum setting = 46 VAC.					
039	IRCMP	= Output circuit resistive drop compensation = Output voltage boost at 100% load current ÷ 0.35 = _____		0	65535	0

FREQUENCY SETTINGS:

ADR#	NAME	FORMULA		MIN	MAX	DEFAULT
077	MINHZ	= normally 3 Hz	= 3	1	12	3
079	MRHZ	= Motor Hz	= _____	30	120*	60
024	MAXHZ	= Max. RPM x Motor Hz/ Sync. RPM	= _____	30	120*	60
076	FRQTL	= 1.17 x ADR024-MAXHZ	= _____	30	150*	70

*Standard AC-500 packages should not exceed 90 Hz. Consult factory for special applications.

SPEED SETTINGS:

ADR#	NAME	FORMULA		MIN	MAX	DEFAULT
068	REFSF	= Maximum RPM	= _____	0	6553	6553
070	RPMSF	= Maximum RPM	= _____	0	6553	6553
071	REFLM	= 1.015 x Max RPM .	= _____	0	6656	6650
072	MINSP	= Minimum RPM	= _____	0	6553	655
073	SPDLM	= 1 . 2 x Max RPM	= _____	0	7936	7208

CURRENT SETTINGS:

ADR#	NAME	FORMULA		MIN	MAX	DEFAULT
032	STLSP	= % maximum speed to switch from stall to TOC function	= _____	0	100	10
041	ILMMX	= X (Motor Amps/Drive Amps) (Where X = % desired for current limit, normally not higher than 150)	= _____	0	225	150
043	IOCTR	= X (Motor Amps/Drive Amps) (Where X = %desired for IOC trip, normally no higher than 200)	= _____	50	225	200
044	TOCTH	= X (Motor Amps/Drive Amps) (Where X = % desired for TOC threshold, normally not higher than 120)	= _____	10	225	120
046	STLTH	= X (Motor Amps/Drive Amps) (Where X = % desired for Stall threshold, normally not higher than 30)	= _____	10	65535	30

TIME SETTINGS:

ADR#	NAME	FORMULA		MIN	MAX	DEFAULT
057	ACELT	(Acceleration Time) = (20 to 9999) for (2.0 to 999.9 sec.) For example 10 = 1 second	= _____	20	65535	200
058	DCELT	(Deceleration Time) = (20 to 9999) for (2.0 to 999.9 sec.) For example 10 = 1 second	= _____	20	65535	200
074	PACEL	= 3 x (ADR057-ACELT)	= _____	20	9999	600
028	CLMIG	= 60 x (ADRO79-MRHZ) / (ADR024-MAXHZ) 60 x (Motor Inertia/Load Inertia)	= _____	0	255	30

NOTE: Estimate if possible or default to a value of 30.

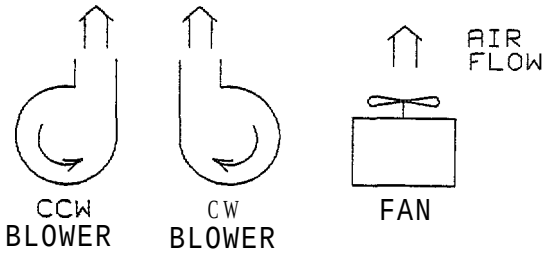
027	CLMPG	= (ADR028-CLIMG) /3	= _____	0	255	10

APPLY AC POWER

A. Verify that the Diagnostic LEDs are blinking one at a time from right to left. Confirm that the programmer displays the word "OPERATE".

NOTE: If at any time the LEDs remain locked, indicating a fault message, the Programmer displays a fault code, or either display is blank, refer to the "Troubleshooting" section for corrective procedures.

B. Check for proper direction of rotation of all three-phase blower motor fans in the power unit(s) and on the AC motor(s).



C. Verify that the following power supply voltages appear at the appropriate Control card test points by measuring from the supply test point to PCOM test point or by using the built-in DVM (requires optional Programmer and Process Interface card, refer to "Use of the Optional Programmer Module" section) .

Test Point	Value		
+24V	23.8	to	29.0V DC
-24V	-23.8	to	29.0V DC
+15V	14.25	to	15.75V DC
-15v	-14.25	to	-15.75V DC
+ 5v	4.95	to	5.05V DC

NOTE: A Programmer is. required for the following steps. For complete information on the Programmer operation refer to the "Use of the Optional Progammer Module" section.

D. Place the EEROM write protect device (JP13) on the Control card in the "ENABLE" position (2-3) to allow programming. (The 1-2 position is the "PROTECT" position.)

CAUTION

DO NOT PERFORM A "HARD RESET" OR TURN POWER "OFF" OR "ON" WITH JP13 IN THE "ENABLE" (2-3) POSITION.

E. Access the Parameter mode by pressing mode [SET] [DRV#] [7] [7] and [ENTER] on the keypad (where the number or function encased in brackets [ ] indicates the Programmer key to be pressed).

The word "PARAMETER" should be displayed. The BLACK keypad functions are now usable. To access a parameter address, press all three numbers on the keypad corresponding to the three-digit number of the desired address. The name of the address and its present value will be displayed.

To access a new parameter, press [CLEAR/MODE] and the three digits corresponding to the next parameter desired.

F. In order to allow programming of new parameter values, first verify that the parameter address #003, DGNJP (ADR003-DGNJP) contains an odd number (1, 3, 9, etc.) by pressing [0] [0] [3]. If this address contains an even number, add a "1" to this number and enter the new value

Example: ADR003 contains the number 16. Add I to obtain 17. Press [1] [7] and [ENTER].

Press [CLEAR/MODE] and the three digits for the next desired parameter.

NOTE: Confirm that the drive parameters agree with the "as shipped" parameter list inside the drive door or in the Custom Software Description prior to making any adjustments. Document any changes.

If parameters were not pre-calculated for the specific application, the Quick Calculation Chart in the "Before Applying Power" part of this section should be filled in, and those parameters entered into the controller before proceeding. Return JP13 to the l-2 position after programming is finished.

G. Enter the DIAGNOSTIC mode from the PARAMETER mode by pressing [CLEAR/MODE]. "DIAGNOSTIC" should appear. The BLUE and BLACK keypad functions will now control the drive.

NOTE: The motor must be connected for the next step.

WARNING

THE OUTPUT STARTER (IF SUPPLIED) WILL BE ENERGIZED AND POWER APPLIED TO THE MOTOR IN THE NEXT STEP, BUT THE MOTOR SHOULD NOT ROTATE.

H. Check the power section integrity by performing an SCR test (Diagnostic Test 12). While in the DIAGNOSTIC Mode, press [TEST] [1] [2] (the display will show the same). Then press [ENTER]. The output starter (if supplied) will be energized and power applied to the motor. Refer to the "Use of the Optional Programmer Module" section for complete information on the SCR test.

After the test is completed (approximately 3 to 4 minutes) the display will show either "PASSED" or a fault. If "PASSED" is displayed, proceed to the next step, if a fault, refer to the SCR Test Instructions and Troubleshooting section.

I. Reference Adjustment - Refer to the Custom Software Description of ADR000-REFJP and the controller elementaries to determine the source of the reference(s) to the controller. Proceed to the appropriate step(s) to adjust the reference.

NOTE: If a Programmer and Process Interface card are included in the drive, an on-board digital voltmeter (DVM) and electronic wire check are available to verify wiring and make the following adjustments. Refer to the "Use of Optional Programmer Module" section for instructions.

1. Programmed Reference - Verify that the settings of the reference parameters (if any) indicated by ADR000-REFJP are set correctly. They may be changed by entering PARAMETER mode, entering the parameter address and then correct reference.

Example: Run reference set by ADR061-RUNRF
ADR061 as shipped set at 3000 RPM
Desired reference = 2000 RPM

- Enter PARAMETER mode and press [0] [6] [1].
- Programmer displays "RUNRF. 3000"
- Press [2] [0] [0] [0] [ENTER]
- The reference is now set at 2000 RPM

2. Analog Reference - There are three commonly used analog reference inputs to the drive: Manual System Reference (MSR) and Jog Reference (JOGR) through 3TB (and 6PL) to the Control card and Auto System Reference (ASR) through 4TB (and 13PL) to the optional Application card. Additonally, an analog current reference (process follower) may be brought into the optional Process Interface card. Adjustment of the process follower reference is described in the "Adjustments" section, Table 1 under Process Interface card.

- Adjustment of MSR (3TB65/6PL-5) and ASR (4TB23/13PL-23)

With the drive stopped, set the reference to MAXIMUM. Adjust Pl-REFSCALE on the Control card to obtain 4.0 VDC at test point SRS (Manual System Reference only. Jumper JP1 on the Control card selects the manual system reference input voltage range). Measure with a volt-ohmmeter (or the DVM diagnostic test as mentioned above) at the appropriate test point and PCOMX. If Auto System Reference (ASR) is used on the optional Application card, adjust Pl-ASR SCALE on the Application card for 4.0 VDC at test point ASR, also located on the optional Application card. (Jumper JP1 on this card selects the reference input voltage range). Reset the reference to 0% speed.

- Adjustment of JOGR (3TB63/6PL-7)

If an analog jog reference is supplied through JOGR, adjust the external source for the desired voltage between Control card test point JOGR and PCOMX on 3TB.

NOTE: ±14 VDC corresponds to full reference. Programmable to 7V = top speed for fast update rate; refer to Custom Software Description.

$$\frac{\text{Desired Jog Speed x 14 VDC}}{\text{Top Speed}} = \text{Volts at JO\&}$$

Example: Top speed = 1800 RPM
Desired JOG speed = 100 RPM

$$\frac{100 \times 14 \text{ VDC}}{1800} = \text{Volts at JOGR}$$

Volts at JOGR = 0.78 VDC
Adjust the external reference source to 0.78 VDC.

J. The following steps will use Diagnostic Run (Test 2-DGR) on the Programmer card to form the capacitors and set-up the feedback(s) and load adjustment. Refer to the "Use of the Optional Programmer Module" section, Diagnostic Run before proceeding.

NOTE: If an output contactor or motor starter is used, a jumper or switch must be connected between 3TB52 (4PL-7) and 3TB53 (4PL-8) to energize the contactor when required.

K. Capacitor Forming - This step is required to prepare the DC Link capacitor bank for normal operation and must be done with the motor disconnected. Remove power from the controller and disconnect the T1, T2 and T3 cables at the controller output. Reapply power.

WARNING

THE FILTER CAPACITORS MUST BE FORMED ON START-UP REGARDLESS OF THE SHIPMENT DATE OR LAST ENERGIZED DATE FROM THE FACTORY. THE CAPACITORS SHOULD ALSO BE FORMED IF THE CONTROLLER IS NOT RUN FOR MORE THAN SIX MONTHS.

1. Connect a DC voltmeter (or the Programmer DVM) between test points LVF on the Power Supply/Interface card and PCOM on the Control card.

2. Enter the DIAGNOSTIC mode as previously described. Verify that the Diagnostic Run reference is set to zero by pressing (DGS), then adjusting the MSR input or DEC key so that the display shows "REF. 0". Press [STOP].

WARNING

THE NEXT STEP WILL APPLY POWER TO THE CONTROLLER OUTPUT.

3. Press [DGR] to start Diagnostic Run, (the LEDs will blink in pairs). Slowly increase the Diagnostic Run reference until the LVF voltage reaches the first level in the following table. Pause at this level for five minutes. Increase to the next level and pause for five minutes. Finally, increase to the top level and pause for five minutes.

4. Reduce the reference to zero and press [STOP] on the Programmer. The DC voltage will decrease to zero volts.

L. Motor Operation and Feedback Adjustment (if used) - Remove power from the controller and reconnect the motor to T1, T2, and T3. Also, if a motor starter or output contactor is used, connect the jumper or switch as discussed in Step J to activate this device.

WARNING

THE NEXT STEP WILL APPLY POWER TO THE MOTOR AND CAUSE IT TO ROTATE. VERIFY THAT THE SHAFT IS FREE TO ROTATE. DO NOT BYPASS ANY PROTECTIVE CONTACTS FROM THE MOTOR STARTER CIRCUIT (IF USED).

1. With zero reference applied, press [DGR]. Slowly increase the Diagnostic Run reference until the motor starts to turn at a steady, smooth speed.

2. Verify that all output currents are balanced to within ±10% of the average of these currents with a clamp on ammeter or by measuring the AC voltage at test points CFA, CFB, and CFC on the Power Supply/Interface card.

3. Slowly increase reference to full speed and verify stable motor and load operation. Refer to the Troubleshooting section if problems are experienced.

4. Reduce the reference to zero and press [STOP] on the Programmer.

M. Analog Tach Feedback Setting - The motor speed feedback is to be calibrated using Diagnostic Run. If no tach or other feedback device is used, skip to Step N.

1. Check that JP8 and SW1, on the Process Interface card, are set properly for the type of tach and maximum tach volts at top speed. Refer to the "Adjustments" section, Table I for the proper setting.

Voltage at TP LVF	Test 4 % Rated	Capacitor Voltage - P2C to N2 230 VAC Controller	480 VAC Controller	Time
5.1 VDC	68	200VDC	400 VDC	5 minutes
6.4 VDC	85	250 VDC	500 VDC	5 minutes
7.7 VDC	101*	300 vDc*	600 VDC*	5 minutes

*Top level voltage may be limited by the maximum reference setting. In this case, use maximum reference as top voltage .

2. With drive stopped, adjust P7-SFB ZERO on the Protess Interface card for 0 VDC at test point SFB also on the Process Interface card.

3. Start the drive by pressing [DGR] on the Programmer and increase the reference to maximum. Check for 4 VDC at test point SFB and adjust, if necessary, with P1-TACH SCALE. Check actual motor speed with a hand tach and verify the proper tach output volts.

4. After top speed has been calibrated, move to step N with the motor still running at top speed.

N. Load Current Scaling - Run the motor in Diagnostic Run at top speed and at as close to rated drive current as possible (if necessary, stop the drive and connect the load if not already connected).

Verify that the % Load readout on the Programmer is:

$$\% \text{ Load} = \frac{\text{Actual RMS Current}}{\text{Controller Nameplate Rated Current}}$$

If there is more than a 5% difference between the reading and the calculated value refer to the Power Supply/Interface card adjustments.

Reduce the Diagnostic Run reference to zero and press[STOP].

NOTE: The speed reference referred to in the following steps is the normal operating reference, not the Diagnostic Run reference.

0. Set the speed reference (speed pot, N/C reference, etc.) at its minimum setting. Start the drive (press "START" on an operator station or call for a "RUN" from the control).

P. Ensure that all "STOP" circuits are functional. "STOP" from an operator station (115 VAC START/STOP circuit - 3TB49, 50 & 51) or removing a "RUN" mput from 3TB8 will not function if the drive is started from the Programmer keypad.

Q. Re-start the drive and slowly increase the speed reference until the motor starts to rotate. If the motor does not rotate, or rotates in the wrong direction, check the Diagnostic LEDs or the Programmer display. If DC tach feedback is provided, a tach polarity fault may be indicated. Refer to the fault codes in the Troubleshooting section for corrective action.

WARNING

EXERCISE CAUTION AS HIGH VOLTAGE MAY BE PRESENT.

NOTE: If experiencing difficulty in making the adjustments in Step R, it may be necessary, due to the tolerance of the tach, to verify the actual tach output voltage/RPM. Check the tachometer's actual speed and output voltage with a hand tach and voltmeter to verify the proper SW1 setting on the Process Interface card.

NOTE: If no tach is used, SFB will always be OVDC in the next step.

R. Start the drive and gradually increase speed reference to maximum. Check for 4 VDC at test points SRS and SFB on the Control and Process Interface cards. (If ASR is used, check test point ASR on the Application card.) Adjust Pl-TACH SCALE (requires optional Process Interface card), if necessary, to reach the correct top speed (only on drives with an AC or DC taach). Confirm rated motor V/Hz.

S. Slowly decrease the speed. Verify stability through the full speed range. If the drive is NOT stable, refer to the Stability Adjustment procedure in the Troubleshooting section.

T. After the drive is stable throughout the full speed range, confirm that the motor current does not exceed the nameplate rating with loading.

U. The above list of adjustments should result in a properly running drive with desired readouts at any diagnostic test on the Programmer.

Before continuing, it is recommended that the following be done:

- The list of "as shipped" parameter settings provided with the controller be checked against the actual parameters in the drive.

- All incoming signals be checked for proper value and correct connection. Refer to the "Use of Programmer Module" section, Test 20 - Electronic Wire Check.

V. Similarly, other parameters as sold and shipped with the drive can be accessed and changed as necessary. (A complete list of pertinent parameters is contained in the Custom Software Description shipped with the drive.) A partial listing of "KEY" Parameters and their descriptions can be found in the "Adjustments" section.

NOTE: Final responsibility for parameter settings resides with the customer. Document all changes to parameter settings in the space provided in the Custom Software Description,

W. Further accidental reprogramming should be prevented by subtracting a 1 from the value presently in ADR003-DGNJP (refer to Paragraph "F" this section), and entering the new value; then place the EEROM write jumper (JP13) in the "PROTECT" position (1-Z).

X. Exit the parameter mode by pressing [CLEAR/MODE] [CLEAR/MODE] [CLEAR/MODE]. OPERATE should appear. The RED keypad functions will now control the drive.

THE DRIVE IS NOW READY FOR OPERATION.

TROUBLESHOOTING

The AC-500 controller is equipped with an integral self-diagnostic system which serves as an aid in identifying potential drive problems and in servicing the equipment should repairs become necessary. Information is also provided in this book which will assist qualified personnel in performing additional tests and manual checks helpful in determining and resolving causes of possible drive problems.

CAUTION

PERSONNEL INVOLVED IN SERVICING THIS EQUIPMENT SHOULD READ AND UNDERSTAND THE INSTRUCTIONS AND INFORMATION CONTAINED IN THIS MANUAL AND OTHER PUBLICATIONS, INSTRUCTIONS, AND DRAWINGS PROVIDED WITH THE EQUIPMENT BEFORE ATTEMPTING ANY TESTS OR REPAIRS. PARTICULAR ATTENTION MUST BE PAID TO ALL WARNINGS, CAUTIONS AND NOTES PROVIDED IN THE DOCUMENTATION AND ON LABELS OR TAGS AFFIXED TO THE EQUIPMENT.

A systematic approach to the troubleshooting, localizing the problem and its cause, will reduce service time. The following should be kept in mind:

1. Is the cause internal to the AC-500 controller or the result of external controls, equipment, circuitry or conditions? What appear to be controller problems may actually be caused by the power system, external control or signal wiring, external devices or application problems.

2. Which output phases are affected?

   a) If only one phase is affected - which component, signal or card circuit is unique to that phase? Check all circuits or components which are phase-specific and any cards with seperate channels or circuitry for individual phases.

   b) If all phases are affected - a common card, programmable parameter or other shared circuitry should be investigated first as the possible cause.

The AC-500 self diagnostics and automatic tests provide information useful in evaluating and analyzing drive performance or problems. Familiarity with the automated test procedures will prove invaluable in interpreting the test results to identify specific problems indicated by the information obtained. The Diagnostic Codes of the AC-500 controller are referenced to descriptions and troubleshooting information provided in this manual. The majority of these codes may be read from the LED display provided on the Control card and visible through the rectangular window in the card enclosure door. Additional, detailed information and automatic tests are available through the use of a separate Hand-held Programmer, optional installed Programmer card, or data communications link.

Several levels of diagnostics are provided in the AC-500 controller and outlined in these instructions:

IN-PROCESS MONITORING:

The AC-500 fault detection system continually monitors drive performance and operation. Should a fault occur, the system will react as programmed and provide the appropriate Fault Code on the diagnostic LEDs, through the Programmer and over serial link, if used. Error Codes, relating only to data communications with the Programmer or serial link, are not displayed on the Diagnostic LEDs but are available through the Programmer or serial link. Refer to Table 4 "Specific Fault Troubleshooting", for interpretation of the Fault and Error Codes and their probable causes.

POWER-UP AUTOMATIC TESTS:

Upon initial application of power, the AC-500 performs a series of tests to assure the integrity of the control logic and also tests for shorted converter SCRs prior to firing them in the normal "standby" mode. Should the control not pass or an SCR test shorted, the AC-500 will halt the power-up sequence and annunciate the Fault Code on the diagnostic LED display, Programmer or over the serial link. Refer to Table 4, "Specific Fault Troubleshooting", for interpretation of the codes and their probable causes.

START-UP AUTOMATIC TESTS

There are two automatic self-checks which may be performed by the AC-500 upon receipt of the START command. To verify that the commutation circuit is precharged to an acceptable level prior to firing the inverter SCRs and drawing output current, a commutation circuit test is always performed. The second test is an optional power circuit check to verify that the SCRs in the power section are not shorted and that the converter, commutation, auxiliary and inverter negative main SCRs are not open or otherwise unable to fire. Fault codes which may result if there is a failure to pass these tests take the form of SCR TEST Fault Codes and may be found in Table 4, "Specific Fault Troubleshooting", with additional information provided in Table 5, "SCR Test Troubleshooting", as described next.

SCR TEST:

The AC-500 controller provides an automated SCR TEST sequence which may be initiated manually through the Programmer, via the serial link or through a jumper selection on the Control card. The test verifies the integrity of the controller signals and the power circuit components. A fault located by the test will be displayed through the Fault Code and may be interpretted by refering to Table 4 and Table 5 in this section. If the test indicates a possible power component failure, such as a faulted SCR cell, Fault Code FL14 SCRTF will be displayed. The specific test and related component may then be identified through use of the programmer or serial link (the programmer displays FL14 SC###, where ### is the SCR TEST code) and interpreted by refering to Table 5, "SCR Test Fault Troubleshooting".

AUXILLIARY DIAGNOSTICS:

In addition to the microprocessor based diagnostics described in the preceding paragraphs, the AC-500 controller is also provided with several auxiliary analog diagnostic circuits which monitor specific functions. These analog trips are equipped with individual LED indicators specific to that circuit. The auxiliary diagnostic indicators, as well as problems such as breaker tripping or SCR failure which may be the result of any number of internal or external causes, are reviewed in the OPERATIONAL TROUBLESHOOTING section of this manual.

POWER CIRCUIT TROUBLESHOOTING:

Inspection, test or replacement of specific components of the power circuit may be indicated by the preceding diagnostic checks or other troubleshooting instructions. Troubleshooting may involve use of a Volt-Ohm-Meter (VOM), oscilloscope (SCOPE) or chart recorder to verify component inegrity or observe waveforms or signal levels within the AC-500 controller and in external circuitry. These procedures are outlined in the "Power Circuit Troubleshooting" part of this section.

PROGRAMMER DIAGNOSTICS

The Programmer module allows several manually initiated tests and diagnsotic procedures to be performed. There are various diagnostic modes available that may be used to isolate problems, monitor signals within the drive, or to examine drive variables and parameters in RAM and EEROM memory without interference from external devices or control loops. The diagnostic modes and their use are described in Section 6, "USE OF THE OPTIONAL PROGRAMMER MODULE".

PERIODIC INSPECTION:

Maximum reliability of the AC-500 controller may be assured by routine maintenance and periodic inspection. Suggestions for periodic maintenance procedures are provided in Section 7, "MAINTENANCE AND REPAIR".

PARTS REPLACEMENT:

Parts replacement information including part identification, replacement procedures and part numbers is provided in the "PARTS REPAIR AND REPLACEMENT" and "SPARE AND RENEWAL PARTS" portions of the "MAINTENANCE AND REPAIR SECTION.

SPECIFIC FAULT CODE TROUBLESHOOTING

The main diagnostic indicators of the AC-500 controller are the Fault Codes which are displayed through six LEDs on the Control card or through the programmer module or serial link. In addition, communications diagnostics are provided by a series of Error Codes which are read either through the serial link or the programmer module. These Fault and Error Codes, when properly interpretted, provide specific information about drive performance and serve as a valuable aid to troubleshooting the AC-500 controller.

CODE INTERPRETATION

The diagnostic LEDs on the Control card provide indication of both normal operation and drive faults. In normal operation the LEDs will flash sequentially from right to left in a pattern indicating the mode of operation:

ONE at a time — "Ready-to-run" mode (no faults)

TWO at a time — Running mode (activated by RUN, JOG, START/STOP, etc.)

Should a fault be detected, the Diagnostic LEDs will cease the sequential pattern and display a series of ON and OFF LEDs which can be read as a binary number representing the Fault Code. The decimal value of the Fault Code is calculated by adding the binary bit values corresponding to each lit LED as printed on the Control card. If the display is blinking the value of 64 is added to this total.

The Programmer displays Fault Codes as a Fault number and name (e.g.; FL01 IOC, where FL01 is fault number 01 and IOC is the name of the fault, Instantaneous Overcurrent).

The Fault Codes are listed in TABLE 4 along with their possible causes and recommended troubleshooting procedures. Both the Diagnostic LED readout and the Programmer Fault Code are listed, as is a fault name or mnemonic. The LED readout is represented as a series of ones and zeros with ones indicating lit LEDs and zeros, LEDs that are unlit. An underline is used to indicate that the LED display is blinking on and off.

The following chart provides two examples of Fault Code interpretation.

FAULT CODE INTERPRETATION

LED DISPLAY BIT PATTERN 32	16	8	4	2	1	DECIMAL VALUE CALCULATION	PROGRAMMER FAULT CODE
0	1	1	1	0	0	16 + 8 + 4 = 28	FL28
0	1	0	1	0	0	16 + 4 + 64 = 84	FL84

0 = LED is OFF 1 = LED is ON 1 = LED is BLINKING

FAULT RESPONSE

The AC-500 controller's response to specific faults varies with the type of fault. Fault response is represented in Table 4 by the letters L, N, T and A which indicate the following:

LATCHED — Fault indication is maintained until the controller is RESET, even though the cause of the vault may no longer exist.

NON-LATCHED — Fault indication clears once the cause of the fault no longer exists. This type fault does not require that the drive be reset.

TRIP — The AC-500 controller removes power from the motor, requiring the controller to be RESET before operation may be resumed. All TRIP faults also cause indication to be LATCHED.

ANNUNCIATE — The drive will continue operating with the fault code displayed for reference. ANNUNCIATED faults may be either LATCHED or NON-LATCHED.

CONTROLLER RESET

As indicated, certain faults will require that the AC-500 controller be reset before drive operation may resume. There are two methods available to reset the controller.

A SOFTWARE (or SOFT) RESET may be executed by depressing the Programmer's RESET key or via a command over the serial link.

A HARDWARE (or HARD) RESET may be executed by depressing the RESET button on the Control card, momentarily connecting 6PL-1 to 6PL-4 (3TB69 to 3TB66) or interrupting incoming power to the drive.

Most LATCHED faults may be cleared by executing a SOFT reset, however certain faults, indicated by the letter H in Table 4, should be cleared with a HARD reset. In general, if a SOFT reset fails to clear the fault and restore the drive to proper operation, a HARD reset may be required.

CAUTION

DO NOT PERFORM A HARD RESET OR TURN INPUT POWER TO THE DRIVE ON OR OFF WITH THE EEROM JUMPER ON THE CONTROL CARD IN THE ENABLED (UNPROTECTED) POSITION. SUBSEQUENT MISOPERATION MAY RESULT.

TABLE 4
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000001	FL01	IOC	Instanteous AC overcurrent (L,T) • Tighten motor connections. • Check for motor ground or short. • Mechanical Problems-locked rotor, binding shaft, unreleased brake, vibration, gear box, or load problems. • ADR043-IOCTR parameter set too low. • Current feedback failure - check 1PL, NFB, CFA, CFB and CFC connectors. • Check current transformers CTA, CTB and CTC for open. • CT loading resistors incorrect. Check SW1 and SW2 settings on Power Supply/Interface card. • SCR misfiring - Check CVF or LVF waveforms. • Check for input at SJ2(4TB16 or 13PL-16). • ADR011-SJAJP set incorrectly; check per Custom Software Description. • Check all SCR RC snubbers for short - refer to "Power Section Troubleshooting". • SCR failure; try SCR test or refer to "Power Circuit Trouble-shooting" section.
000010	FL02	OVRSP	Overspeed (sensed by analog tach circuitry) (L,T) • ADR073-SPDLM set too close to ADR068-REFSF and ADR071-REFLM - set at a minimum of 5% greater than REFSF and REFLM. • Speed overshoot occuring - speed regulator response too high, reduce gain. • Improper input at SJ1 or SJ2 (4TB15 or 4TB16). • Tach feedback circuit failure - check tach connections at Process Interface card 1TB and connector 7PL.
000011	FL03	AUXFT	Auxiliary Fault Trip (check for card LED indication) (L,T) Inverter Firing Generator card - - Commutation capacitor over-voltage (XCOV-where X is phase A, B or C) Power Supply/Interface card - - analog instantaneous overcurrent (IOC) - - ground fault trip (GFT) Dynamic Braking Module - - dynamic braking trip (DBT) XCOV (X = A, B or C) • Excessive or imbalanced voltage - check LEDs to determine phase(s) • Check for failed commutation circuit components and loose or poor connections. • Check for pump up of DC link - motor impedance may be too low. • Note when COV occurs (during acceleration, decelertion or running). IOC • Tighten motor connections • 1PL damaged or shorted • Mechanical problems-locked rotor, binding shaft, unreleased brake, vibration, gear box or load problems. • Verify that Pl-IOC SCALE on the Power/Supply Interface card is set properly.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
			IOC (continued) • Check setting of SW1, SW2 and P5 - CFE GAIN on the Power Supply/Interface card. • Check for loose or damaged connection between the DC link link capacitor CT(FCC) and the Power Supply/Interface card or an open CT(FCC). • Check all SCR RC snubbers for short-refer to "Power Circuit Troubleshooting". • SCR failure try SCR test or refer to "Power Circuit Troubleshooting". **GFT** • Check motor and inverter output wiring for ground. **DBT** • Dynamic braking module trip - refer to Dynamic braking instructions for troubleshooting.
000100	FL04	DCL2L	Decel time set too low (N.A.) • ADR058-DCELT set too low - increase value. • Dynamic braking circuit failure - check DBTPL and DB contactor operation. • Load/inertia problem - check application, may require dynamic braking if not installed.
000101	FL05	TPOL	Tach polarity incorrect (L,T) • Reversed tach connections. • Switch SW1 on Process Interface card set improperly - position 7 must be open for DC tach, closed for AC tach. • Incorrect phase connection to motor - check connections.
000110	FL06	JMPER	Software jumper illegally changed while running (L,T) • Drive must be stopped to change parameters in ADROOO to ADR014.
000111	FL07	RESET	Reset while running (L,T) • Drive powered up with RUN or JOG closed - check 6PL-12 and 6PL-13 (3TB58 and 3TB57). • Momentary loss of incoming power while running. • Incorrect logic polarity, check ADR012-OPTJP.
001000	FL08	TOC	Timed overcurrent (L,T) • Motor mechanically overloaded - check for change in process load, worn bearings, gears or brakes. • ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. • Current feedback failure or 1PL disconnected or damaged. • Dynamic braking failure (if used) - i.e., failure to pick up contactor.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSED/REMEDIES
001001	FL09	LNKTR	Serial link trip (drive told to trip via serial link (L,T). • Drive received external trip signal - check for external fault. • 12PL or 8PL incorrectly connected or damaged.
001010	FL10	LNKFL	Serial link fault (communications failure (L,T) • Programmer card jumpers incorrect - refer to Table 1. • Baud rate set incorrectly - check BAUDR (ADR126). • 12PL or 8PL incorrectly connected or damaged. • Intermittent connection between 3TB32 (CTSA) and 12PL-7 or 3TB41(CTSB) and 12PL-14. • FL36 appears momentarily previous to FL10 latching - refer FL36. • Serial operator station disconnected from 3TB while power on.
001011	FL11	CMFLT	Catch module fault (L,T) • Motor current exceeded level set by ADR054-CMILV. • Check for proper setting of ADR049-CMSRT.
001100	FL12	DFPOF	Delayed firing power off when it should be on (L,T) • Q2, Q3, R33 or U14 failed on Control card - replace Control card.
001101	FL13	EEXXX	Checksum error or parameter out or range (L,T,H) where XXX equals a 3-digit number, indicating an out-of-range parameter address or altered checksum. • If the fault reads EE255 the EEROM should be checked for the application - contact Product Service for information at Area Code (814) 455-3219. • Verify JP13 is not in write enable position during power up or power down. • , Spontaneous parameter alteration - check all parameters and verify that the EEROM is not write "ENABLED" (JP13 in 2-3 position) during power up, power down or "hard" reset. • Value other than zero in ADR255 - replace EEROM.
001110	FL14	SCXXX	SCR trip fault (L,T) (where XXX equals the cell Fault code) • A device failure has been detected in the inverter power section. Cross reference the cell fault code displayed by the programmer with Table 5, "SCR Test Fault Trouble-shooting", to determine which device(s) has failed.
001111	FL15	CVFBL	Loss of converter voltage feedback (L,T) • Voltage feedback wires disconnected, switched or damaged - check CVP and CVN connections between Converter Pulse Transformer card and the converter bridge. • Check JP1 and JP2 on Converter Pulse Transformer card for proper voltage setting. • NFB disconnected or damaged. • DC Link shorted - check for short between P1 and N1. • Power Supply/Interface card failure - replace.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010000	FL16	NOTRP	Trip from N.O. interlock on Application card (L,T) • External input at 4TB8 (13PL-8). • Application card absent but selected by soft -jumper - AFR010-APPJP should be zero. • If preceded by FL07 and soft reset, check logic polarity in ADR012-OPTJP. • Microapplication card jumper JP33 set wrong or U33 installed wrong or failed. • Replace Application card.
010001	FL17	XSTOP	Coast stop trip (L,T) • Check for input at XSTOP on 3TB1 (6PL-16). • Check ADR004-STPJP - a 4 causes a cost stop when XSTOP selected.
010010	FL18	STALL	Stalled overcurrent (L,T). • ' Mechanical problems - failed brake, frozen bearings, application problem, etc. • ADR046-STLTH and ADR047-STLGN set wrong - check against "as shipped" parameter list. • * Drive running less than 1% of top speed and motor current greater than ADR046-STLTH. • Current feedback failure or 1PL, NFB, CFA, CFB or CFC disconnected or damaged. • - Dynamic braking failure (if used) - i.e. failure to open DB circuit.
010011	FL19	OVERV	Excessive link capacitor voltage (L,T) • .AC line out of tolerance - check for +10%, 5%. • ADR080-VLIM set too high - compare to parameter list. • Check JP1 and JP2 on Converter Pulse Transformer card for proper voltage setting. • Deceleration rate too fast - refer to FL04-DCL2L. • Check for overhauling load.
010100	FL20	TLOSS	Loss of tach feedback (L,T) • Tach leads disconnected. • Tach feedback circuit failure - 7PL disconnected or damaged. • Process Interface card jumpers or switches incorrect. Refer to Instruction Book, Table 1. • P7-SFB ZERO, misadjusted on Process Interface card - set test point SFB for zero volts at zero speed. • AC tach connected but open loop frequency and speed regulator beingused - remove JP7 from Process Interface card. • JP17 on Control card in 1-2 position. Move 2-3 postion. • Encoder Process Interface card installed but not enabled in ADR002-REGJP. • Process Interface card installed but Encoder option enabled. • Process Interface card failed - replace.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSED/REMEDIES
010101	FL21	NCTRP	Trip from N.C interlock Application card (L,T) • External input on 4TB7 (13PL-7). • Application card absent but selected by soft jumper ADR010-APPJP should be zero when no Application card is present. • . If preceded by FL07 and soft reset, check logic polarity on ADR012-OPTJP.
010110	FL22	PHLOS	AC phase loss (L,T) • Fuse(s) blown in AC line - FU1, 2 or 3. • JP14 on Control card in wrong position - see Table 1. • Converter Fuse open-check FUL1, 2 or 3 on the Converter Pulse Transformer card. If one or more is open, check AD1, 2 and 3 (also on the Pulse Transformer card), the filter capacitor and auxiliary commutation SCRs (500KVA frame only - ASA, ASB and ASC). • Leg fuse detection too sensitive, check ADR187-FUSLS (set to 0 to inhibit detection). • 1PL disconnected or damaged. • Low or unbalanced AC line voltage - line to line or line to neutral more than 5% low. • Excessive AC line notching. • Loose AC power connections. • 15 volt supplies outside limits (±5%) - replace Power Supply/ Interface card.
010111	FL23	OVERF	Over Frequency (L,T) *-Check setting in ADR076-FRQTL per Custom Sofware Description (should not be more then 1.2 x MAXHZ). • Check DC link voltage for pump-up on deceleration. • Check for overhauling load.
011000	FL24	u3 FL	Bad ram on U3 8155 (L,T) • Replace Control card.
011001	FL25	SPARE	
011010	FL26	SPARE	
011011	FL27	LSYNC	Loss of AC line signal (L,T • Fuse blown in AC line - FU1, 2 or 3. • 1PL disconnected or damaged. • Low or unbalanced AC line voltage - line to line or line to neutral more the 5% low. • Loose AC power connections. • 15 volt supplies outside limits (±5%) - replace Power Supply/Interface card. • AC line frequency below 45 Hz or above 63 Hz (70 Hz if 32 in ADR001-FDBJP). • Check for 50/60 Hz square wave at test point SY on Control card - replace Control card if no signal.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011100	FL28	RMODE	Illegal selection of RUN mode (L,T) • Do not command DIAGNOSTIC RUN or STATIC while running or vice versa.
011101	FL29	CNLON	Control on input opened (L,T) • Open circuit between 3TB3 (6PL-18) and 3TB4 (6PL-19). • 6PL incorrectly connected or damaged. • - Noise between 3TB3 and 3TB4. • Control card failed - replace.
011110	FL30	NOAPP	Application card failure (L,T) • Application card absent but selected by soft jumper - ADR010-APPJP should be zero when no Application card present. • Microapplication card jumper JP33 set wrong or U33 installed wrong or failed. • 3PL incorrectly connected or damaged. • Replace Application card.
011111	FL31	NOSEQ	Phase sequence inderterminate (L,T) • Excessively noisy or notch AC line. • Line unbalance, phase to ground, by more than 15%. • 15 volt supplies outside limits (±5%) - replace Power Supply/ Interface card. • 1PL incorrectly installed or damaged. • Control card software incompatable - check software revision on. PROM labels or with Diagnostic Test 10.
100000	FL32	TOCAN	Timed overcurrent annunciation (N,A) • Motor mechanically overloaded - check for change in process load, worn bearings, gears or brakes. • ADR044-TOCTH and ADR045-TOCGN set too low - check against the parameter list shipped with drive. • Current feedback failure or 1PL disconnected or damaged. • Motor wiring ground or short - check wiring. • Dynamic braking failure (if used) - i.e., failure to open DB circuit.
100001	FL33	PHSEQ	Phase sequence reversed (L,T) • Reverse any two AC input lines, recheck rotation of three phase blowers. • AC line voltage feedback jumpers set incorrectly - check JP3 thru 5 on Converter Pulse Transformer card. • Excessively noisy or.notched AC line. • Line unbalance, phase to ground, by more than 15%. • 15 volt supplies outside limits (±5%), noisy or excessive ripple, replace Power Supply/Interface card. • 1PL incorrectly installed or damaged. • ADR012-OPTJP not properly set for phase sequence insensitivity option.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100010	FL34	PLLSY	AC line PLL out of sync (N,A) • CFU1 on Power Supply/Interface card blown. • Excessively noisy or notched AC Iine. • Line unbalance, phase to ground, by more than 15%. • 15 volt supplies outside limits (±5%) -replace Power Supply/ Interface card. • On variable frequency AC lines (i.e. -alternators) — PPLGN (ADR029) too low.
100011	FL35	0 SPD	Motor rotating when RUN commanded (N,A) • RUN command initiated during motor rotation. • Motor being driven externally when RUN commanded. • Loss of DC link voltage feedback - check CVP, CVN and NFB connectors. • ADR048-RSTOP set too low. • Test point SFB contains an offset at zero speed - adjust P7-SFB ZERO, on Process Interface card. • ADR004-STPJP not set properly for "catch mode".
100100	FL36	LNKER	Bad message detected on serial link by drive or other device (N,A) • Excessive noise on 12PL-7 thru 14. • 12PL incorrectly connected or damaged. • 5 VDC power supply noisy. • Baud rate incorrectly set - ADR126-BAUDR normally set at 4800. • Check Process Interface jumpers JP1, JP11, JP12 and JP13. • Serial link checksum byte incorrect. • Serial link message too long or handshaking incorrect - refer to Serial Link Users Guide. • Replace Process Interface card. • Replace Control card.
100101	FL37	SPARE	
100110	FL38	ELOCK	Attempt to change software adjustments when soft locked (N,A) • ADR003-DGNJP must be odd to program EEROM - add one to value in DGNJP.
100111	FL39	EFAIL	Software adjustment not stored in EEROM (N,A) • JP13 on Control card in "PROTECT" position. • EEROM failed - replace.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
101000	FL40	MXSPD	Speed reference greater than REFLM speed adjustment (N,A) • Manual reference jumper JP1 on Control cardimproperly set - see Table 1. • Reference scale pot P1 on Control card misadjusted - see Table 1. • ADR071-REFLM set below ADR070-RPMSF. • Pl-ASR SCALE, on Application card misadjusted (if used) - see Table 1. • P4-PF SCALE or P5-PF OFF, misadjusted on Process Interface card (if used) - see Table 1. • ADR075-RPM/H set improperly for frequency reference (if used). • Top speed reference for parallel input (if used) is set higher than ADR068-REFSF and ADR070-RPMSF. • Preset references (RUNRF, R1RF, R2RF, SLWRF) set too high (if used).
101001	FL41	KEYOF	Keypad function disable (N,A) • Programmer key depressed for an unenabled function.
101010	FL42	RLOGC	Run logic error (N,A) • JOG, RUN, or DIAGNOSTIC RUN called for with XSTOP open. • Keypad STOP called for during an external run cycle. • Incorrect logic polarity - check ADR012-OPTJP.
101011	FL43	U11FL	A-D converter failure (L,T) • Check for noise on ACOM, move wiring to PCOMX. • Control card failed - replace.
101100	FL44	U4LAT	U4 to U5 communication error (L,T) • Check PROM revision compatibility. • Check for noise on power supply or analog inputs. • Replace Control card.
101101	FL45	SLOOP	Speed tracking fault (L,T) • Drive entered current limit or CEMF limit when tracking enabled. • ADR036-SERLM set *too* low - should be between 1 and 7. • Speed regulator unstable. • Check software revision (Diagnostic Test 10) - if earlier than 33 replace software.
101110 .	FL46	VCUFL	Voltage or current unbalanced between paralleled drives (L,T) • Mechanical problems - locked rotor, failed brake, frozen bearings, application problem, etc., cause excessive starting current. • Check setting of SW1 and SW2 on Power Supply/Interface card for proper CT loading. • Check converter voltage feedback for proper SCR firing. • Check AC output lines for grounds or shorts.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
101111	FL47	OADJ>	A-D converter offset too low (N,A) • P4-OFFSET, on Control card misadjusted - turn CW. • Check for noise on ACOM.
110000	FL48	OADJ<	A-D converter offset too high (N,A) • P4-OFFSET, on Control card misadjusted - turn CW. • Check for noise on ACOM.
110001	FL49	NO U4	U4 to U6 communications error (L,T) • Check for PROM revision compatibility. • Check for noise on power supplies or analog inputs. • Replace Control card.
110010	FL50	U4BAD	U4 RAM, timer, or ALU is bad (L,T) • Verify EEROM parameters ADR011, 014, 056, 088, 187,209 and 210. • U12 incorrect or failed - replace. • U4 failed - replaced Control card.
110011	FL51	U1ROM	U1 ROM is bad (L,T) • Replace U1 on Control card. • Replace Control card.
110100	FL52	NCLSD	Annunciate N.C. interlock on Application card (N.A) • Check external input on 4TB7 (13PL-7). • Application card absent but selected by soft jumper - ADR010-APPJP should be 0 when no Application card is present.
110101	FL53	IFTST	Inverter Failure to commutate DC link shoot through (IFT LED on Power Supply/Interface card lit), L,T) • Check for motor or inverter output ground or short. • Verify that relay coils are properly suppressed. • Verify proper setting of SW1, SW2 and P5-CFB GAIN on the Power Supply/Interface card. • 1PL damaged or shorted. • Check commutation circuits for failed components or wiring - run SCR test (Test 12) or refer to "Power Circuit Trouble-shooting section. • Check inverter circuits for failed SCRs or wiring- run SCR test (Test 12) or refer to "Power Circuit Troubleshooting". • Check for proper wiring, failed components or signs of over-heating of the inverter SCR RC snubber circuits - refer to "'Power Circuit Troubleshooting". • Inverter SCR gate current problem - verify that the gate currents are consistent. If one or more varies, check pulse transformer circuitry and replace card if necessary. • If gate pulse transformer okay, replace SCR(s) related to unbalanced gate pulse.
110110	FL54	NO US	U5 to U4 communication error (L,T) • Check for PROM revision compatibility. • Replace Control card on U4 and U5.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
110111	FL55	U5BAD	U5 RAM, timer or ALU is bad (L,T) • U5 failed - replace. • Replace Control card.
111000	FL56	U5ROM	U5 ROM is failed (L,T) • U5 failed - replace.
111001	FL57	NOPEN	Annunciated N.O. interlock on Application card (N,A) • External input at 4TB8 (13PL-8). • Application card absent but selected by soft jumper - ADR010-APPJP should be 0 when no Application card is present.
111010	FL58	U5COM	U4 not communicating correctly with U5 (L,T) • Check for PROM revision compatibility. • Noise on tach input. • Replace Control card.
111011	FL59	NO U6	U6 does not communicate properly with U4 (L,T) • Check for PROM revision compatibility. • - Replace Control card.
111100	FL60	U6BAD	U6 RAM, timer or ALU is bad (L,T) • U6 failed - replace. • Control card failed - replace.
111101	FL61	U6ROM	U6 ROM is bad (L,T) • U6 failed - replace. • c Control card failed - replace.
111110	FL62	MIFPS	Multiple Power Supply/Interface card faults (L,T) • Replace Control card.
111111	FL63	U6COM	U4 not communicating correctly with U6, (L,T) • Check for PROM revision compatibility. • Check for noise on tach input signal. • U4 and U5 on Control card in wrong sockets. • Replace Control card.
000000	FL64	SPARE	
000001	FL65	AIFOP	Converter SCR 1SP is not operating (L,T) • Gate circuit 1SP disconnected or open. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal - replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present - replace Power Supply/ Interface card if no feedback is present. • Failed SCR - check and replace (refer to "Maintenance and Repair" section). • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000010	FL66	A2FOP	Converter SCR 2SP is not operating (L,T) • Gate circuit 2SP disconnected or open. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal - replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present - replace Power Supply/ Interface card if no feedback is present. • Failed SCR - check and replace (refer to "Maintenance and Repair" section). • Replace Control card.
000011	FL67	A3FOP	Converter SCR 3SP is not operating (L,T) • Gate circuit 3SP disconnected or open. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnsotic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal - replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify feedback pulses are present - replace Power Supply/ Interface card if no feedback is present. • Failed SCR - check and replace (refer to "Maintenance and Repair" section). • Replace Control card.
000100	FL68	A4FOP	Converter SCR 1SN is not operating (L,T) • Gate circuit 1SN disconnected or open. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal - replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present - replace Power Supply /Interface card if no feedback is present. • Failed SCR - check and replace (refer to "Maintenance and Repair" section). • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000101	FL69	A5FOP	Converter SCR 2SN is not operating (L,T) @ Gate circuit 2SN disconnected or open. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal ▪ replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present ▪ replace Power Supply/Interface card if no feedback is present. • Failed SCR ▪ check and replace (refer to "Maintenance and Repair" section). • Replace Control card.
000110	FL70	A6FOP	Converter SCR 3SN is not operating (L,T) • . Gate-circuit 3SN disconnected (L,T) • 5PL, PFPL, NPT or NFB disconnected or damaged. • , Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal ▪ replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present ▪ replace Power Supply/Interface card if no feedback is present. • Failed SCR ▪ check and replace (refer to "Maintenance and Repair" section). • Replace Control card.
000111	FL71	FWDOP	All converter SCRs are not operating (L,T) • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cells (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for gate pulse signals ▪ replace Converter Pulse Transformer card if signals are absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present ▪ replace Power Supply/Interface card if no feedback is present. • Failed SCR ▪ check and replace (refer to "Maintenance and Repair" section). • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
001000	FL72	SPARE	
001001	FL73	SPARE	
001010	FL74	SPARE	
001011	FL75	SPARE	
001100	FL76	SPARE	
001101	FL77	SPARE	
001110	FL78	SPARE	
001111	FL79	SPARE	
010000	FL80	SCTFL	Converter SCR short circuit test failed (L,T) • Check for short across P1 and N1 output of converter. • Check for multiple shorted converter SCRs. • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signals to see if they are turned on continuously - if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.
010001	FL81 .	A1FSH	Converter SCR 1SP is shorted (L,T) • Check SCR 1SP for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair" section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal, 1SP to see of it is turned on continuously - if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010010	FL82	A2FSH	Converter SCR 2SP is shorted (L,T) • Check SCR 2SP for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair " section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal, 2SP, to see if it is turned on continuously - if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.
010011	FL83	A3FSH	Converter SCR 3SP is shorted (L,T) • Check SCR 3SP for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair" section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • .If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal, 3SP, to see if it is turned on continuously -if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.
010100	FL84	A4FSH	Converter SCR 1SN is shorted (L,T) • Check SCR 1SN for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair" section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFR for damage or shorts. • If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal, 1SN, to see if it is turned on continuously - if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
010101	FL85	A5FSH	Converter SCR 2SN is shorted (L,T) • Check SCR 2SN for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair" section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • sIf fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal 2SN, to see if it is turned on continuously -if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • - Replace Control card.
010110	FL86	A6FSH	Converter SCR 3SN is shorted (L,T) • Check SCR 3SN for short with ohmmeter on X10 scale - if shorted replace (refer to "Maintenance and Repair" section). • Check all connectors to the Converter Pulse Transformer card for damage, crimped or shorted wires, etc. - replace if necessary. • Check 5PL, PFPL, NPT and NFB for damage or shorts. • If fault occurs on applying power to controller, "soft" reset the fault with a Programmer. Run Diagnostic Static (refer to "Use of the Optional Programmer Module" section) and monitor the gate signal, 3SN, to see if it is turned on continuously - if so replace Converter Pulse Transformer card. • While in Diagnostic Static check test point CVF on the Power Supply/Interface card - if continuously locked high, replace card. • Replace Control card.
010111	FL87	NOUAP	Microapplication card failure (L,T) • Microapplication card absent but selected by soft jumper - ADR014-SPCJP should be set to disable Microapplication card interface (refer to Custom Software Description). • 3PL incorrectly connected or damaged. • U30 or U33 on Microapplication card failed - replace.
011000	FL88	UAPWD	MUP watchdog misoperated (L,T) • Application card (G2 installed when Microapplication card (M3, M4 or M5) is called for - check setting in ADR014-SPCJP per Custom Software Description. • U30 on Microapplication card failed or jumpers set incorrectly, replace card if necessary.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
011001	FL89	MUPTO	RUP to MUP communication failure (L,T) • U30 on Microapplication card failed - replace. • - Replace Microapplication card. • Check for excessive message loading on LAN. • Intermittent LAN connection or improper LAN termination - refer to LAN Users Guide.
011010	FL90	NOMSG	RUP or MUP watchdog detected no message received within time period expected (L,T) • Serial Link or Local Area Network transmission lines disconnected. • 12PL damaged or disconnected. • External device failed to send message in time period set by ADR005-SERJP.
011011	FL91	SPARE	
011100	FL92	SPARE	
011101	FL93	SPARE	
011110	FL94	SPARE	
011111	FL95	WATCH	FUP watchdog detected RUP has misoperated (L,T) • Same as FL58
100000	PASSED	PASSED	Annunciates when SCR test has passed.
100001	FL97	OCTFL	Converter SCR open circuit test failed (L,T) • 5PL, PFPL, NPT or NFB disconnected or damaged. • Converter Gate Pulse Transformer card failed - replace. • Power Supply/Interface card failed - replace. • Control card failed - replace.
100010	FL98	OCTMF	Multiple open Converter SCRs (L,T) • Check all gate plugs. • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose, or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for gate pulse signal - replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present - replace Power/ Supply/Interface card if no feedback is present. • Failed SCR(s) - check and replace (refer to Maintenance and Repair" section). • Replace Control card.

TABLE 4 (continued)
SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
100011	FL99	ALLOP	All Converter SCRs tested open (L,T) • 5PL, PFPL, NPT or NFB disconnected or damaged. • Loose or open bus bar connections to cell (200 KVA Frame only). • Run the controller in Diagnostic Static (refer to "Use of the Optional Programmer Module" section). Check for a gate pulse signal · replace Converter Pulse Transformer card if signal is absent. • While still in Diagnostic Static monitor test point CVF on the the Power Supply/Interface card with an oscilloscope and verify that feedback pulses are present · replace Power Supply/Interface card if no feedback is present. • Failed SCR(s) · check and replace (refer to "Maintenance and Repair" section). • Replace Control card.

TABLE 4 (continued)
SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
	ER01	BUART	Bad UART on the 8031 U.C. • This is normal response to Test 50. • Check JP3 & JP4 on Programmer card. • Programmer card failed, replace.
	ER02	BKPAD	The keypad has shorted leads. • , Replace ribbon cable. *-A key pressed during power up or reset. • Failed Programmer.
	ER03	BDMEM	The random access memory is faulty. • Programmer card failed, replace.
	ER04	BPRGM	The program memory checksum is incorrect. • The Programmer card failed, replace.
	ER05	BTIMR	The timers in the 8031 U.C. are faulty. • Programmer card failed, replace.
	ER06	NADDR	An address is not specified. • "ENTER" key pressed without specifying a parameter address
	ER07	2 KEYS	Two keys were pressed simultaneously. • Press keys - one at a time. • . Failed keypad. • Failed Programmer card.
	ER08	ILBRD	Illegal broadcast. • Failed Progammer card.
	ER11	NOACK	No acknowledgement received. • Device connected to the serial link not responding to the acknowledge requests from the Programmer card. • Wires to the serial link disconnected. • Driver or receiver chip failed on Programmer, Process Interface card, or external device. • Processors locked up, use hard reset. • If Hand-held Programmer, may be failed U6 port on Control card - replace. • Noise on transmission wires. • RS232 or Hand-held programmer cables damaged.
	ER12	N-CTS	No clear to send. • Serial link transmission Iines disconnected. • Check JP3 and JP4 on Programmer card. • Noise on 5 volt power supply. • Unsuppressed relays or ground loops. • Programmer card failed.

TABLE 4 (continued)
SPECIFIC ERROR TROUBLESHOOTING

DIAGNOSTIC READOUT	PROGRAMMER CARD	NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
	ER13	ORBUF	Receive buffer has overflowed. • Message sent to the drive which exceeds the maximum number of bytes (11 data bytes). • Baud rates set incorrectly. • Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. • Message protocol used incorrectly - refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
	ER14	CKSUM	Incorrect checksum was received. • The serial link message checksum has been calculated improperly by external Serial Link device - refer to Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device - refer to Serial Link User's Guide. • ' Message protocol used incorrectly - refer to Serial Link User's Guilde. • Hardware configuration incorrect for serial link operation. • External serial link device failed.
	ER15	BDMSG	Something other than a "start of transmission" was received when "S.O.T" was expected. • "Start of transmission" byte of the message sent over the serial link has been determined incorrectly - refer to Serial Link User's Guide. • Baud rates set incorrectly. • Byte protocol set incorrectly on external device (1 start bit, 8 data bits, and 1 stop bit). • Message protocol used incorrectly - refer to Serial Link User's Guide. • Hardware configuration incorrect for serial link operation. • - External serial link device failed.

SCR TEST TROUBLESHOOTING

The AC-500 controller SCR TEST consists of a series of sequential SCR firings which are monitored to locate system faults. These SCR firings are carefully controlled so as to minimize any chance of further damage to the drive in the event of an existing problem. Although the tests are automatic once initiated, a basic understanding of the nature of the tests will aid in interpreting the results.

WARNING

REFER TO THE "USE OF THE OPTIONAL PROGRAMMER MODULE" SECTION FOR COMPLETE INFORMATION ON RUNNING THE SCR TEST. POWER WILL BE APPLIED THROUGHOUT THE POWER CIRCUIT AND MOTOR DURING THIS TEST.

SCR TEST DESCRIPTION

There are five specific tests which make up the SCR TEST program:

1. CONVERTER TEST
2. DC LINK TEST
3. POWER CIRCUIT SINGLE SHORTED SCR TEST
4. MAIN SCR OPEN CIRCUIT TEST
5. COMMUTATION CIRCUIT TEST

The AC-500 controller will perform these tests in sequence, displaying Fault Code 96 ("PASSED" on the Programmer) if all tests are sucessfully completed.
Should a test detect a fault, the appropriate Fault Code will be displayed. When the test indicates a failed device in the inverter or commutation circuit, a Fault 14 (FL14) will be displayed. Use of the Programmer will provide an additional Ceil Fault Code, such as "FL14 SCXXX", where XXX is referenced to specific information in Table 5, SCR FAULT TEST TROUBLESHOOTING.

Completing and passing the SCR TEST verifies:

a. Short circuits are not present in the power circuit.
b. SCR firing can be achieved by the controller cards.
c. Feedback and firing signals are present.

Generally, once the drive has passed on the SCR TEST, it should be ready to run. Should misoperation continue, other procedures outlined in the troubleshooting section of this manual may be required to locate and correct the problem.

The integrity of the SCR snubber circuits is not verified by the SCR TEST and if trouble persists the snubber circuit wiring and components should be checked manually.

1. CONVERTER TEST

The first part of the SCR TEST program checks the integrity of the converter section power components and the converter voltage feedback. Tests are made for both open and short circuits. SCR firings are at low conduction angles early in the AC input cycle to prevent excessive currents in the event of a short circuit.

SCR-SHORT CIRCUIT

Each converter SCR is fired individually. If converter output voltage is sensed, the test is aborted and a Fault Code displayed indicating which SCR(s) caused the voltage to appear.

SCR-OPEN CIRCUIT

Converter SCRs are fired in pairs. If a firing SCR pair do not produce the expected converter output voltage the test is aborted and a Fault Code displayed indicating the Faulted SCR pair.

2. DC LINK TEST

The second part of the SCRTEST program checks for short circuits in the DC Link and verifies that the DC Link filter is functional. Three tests check for link shorts, proper filter time constant, and correct feedback polarity.

- SHORT CIRCUIT

The converter SCRs are fired at a fixed low conduction angle. If a preset link voltage is not reached within the expected period, a time-out condition occurs, the test is aborted and the appropriate Fault Code is displayed. Failure to pass this test indicates either a shorted capacitor or a shorted pair of Inverter SCRs and requires that a manual ohmmeter check of the inverter components be made.

TIME CONSTANT

Once the DC link short circuit test is successfully completed the time required for the link to discharge is measured. If the DC Link time constant is too long or too short the test is aborted and the appropriate Fault Code is displayed.

FEEDBACK POLARITY

Upon completion of the preceding checks the polarity of the voltage feedback is verified and if incorrect the test is aborted and the appropriate Fault Code displayed.

3. POWER CIRCUIT SINGLE SCR SHORT TEST

The third part of the SCR TEST program identifies if any single inverting or commutating circuit SCR is shorted.

POSITIVE MAIN • NEGATIVE COMMUTATION

All inverter phases go through a positive commutation and all negative inverter mains are gated on. Loss of link voltage followed by a recovery indicates a shorted negative commutation SCR, while loss of link voltage with a failure to recover indicates a shorted positive main SCR. In either case, the test is aborted and the Fault Code displayed. A manual check with an ohmmeter is required to determine which phase and specific SCR is shorted.

NEGATIVE MAIN • POSITIVE COMMUTATION

Upon successful completion of the previous procedure all inverter phases go through a negative commutation and all positive inverter mains are gated on. Loss of link voltage followed by a recovery indicates a shorted positive commutation SCR, while loss of link voltage with no recovery indicates a shorted negative main SCR. In either case, the test is aborted and the Fault Code displayed. A manual check with an ohmmeter is required to determine which phase and specific SCR is shorted.

4. MAIN SCR OPEN TEST

The fourth part of the SCR TEST program verifies that each of the main inverter SCRs can be gated on and will conduct.

MOTE

THE FOLLOWING TESTS REQUIRE THAT A MOTOR OR OTHER THREE PHASE LOAD BE PRESENT ON THE OUTPUT OF THE AC-500 CONTROLLER. WITHOUT A LOAD A FL14-SC039 FAULT WILL RESULT AND THE TEST WILL BE ABORTED. IF THE MOTOR OR OTHER APPROPRIATE THREE PHASE LOAD IS NOT AVAILABLE, THE THREE OUTPUT PHASES (T1, T2 and T3) MAY BE SHORTED TOGETHER.

The output phases are enabled with the polarity of one phase different from the other two and the DC link voltage is checked. The test continues in this fashion until all combinations have been checked. If the link fails to discharge, an open main SCR or lost gate signal is indicated. The test is aborted and the appropriate Fault Code is displayed. A cell fault code of 39 (multiple open SCRs) may be displayed if there is no motor connected or if the starter has failed to close.

5. COMMUTATION CIRCUIT CHECK

The final part of the SCR TEST program checks that each phase has the ability to generate the correct commutation capacitor voltage magnitude and polarity. The following are checked by this test:

- Commutation power supply
- Commutation SCRs
- Freewheeling diodes
- Commutation capacitor voltage feedback
- Commutation capacitors
- Commutation inductors

If any phase fails this test with the display of a code that shows the Commutation Power Supply (CPS) is at fault, indicates that the above items should be checked.

The following table has three individual parts; Short Circuit Test, Inverter Circuit Test (open circuit test) and Commutation Circuit Test (also an open circuit test). The table lists all of the fault codes in numerical sequence from 1 to 103. The Short Circuit Test defines a group of cells which may be suspect of having shorts in a particular circuit. The Inverter Circuit Test indicates if one or more cells (up to four) in the inverter section has failed open. The Commutation Circuit Test indicates if one or more cells in groups of cells in the commutation section have failed open. The inverter bypass diodes are also checked in this section. Also note that there is a distinction between basic KVA ratings with regard to the Commutation Power Supply (CPS) columns. As there is no commutation power supply on the 200 KVA frame size, an F in this column refers to a possible open commutation SCR. An F in this column for a 500 KVA frame size indicates that the commutation power supply is failed or the auxiliary commutation cell for the phase is open.

NOTE: A converter cell failure is indicated by a normal fault display (i.e.: FL-65-A1FOP). Converter cell faults include FL65 to FL71, FL80 to FL36 and FL97.

When a cell failure is indicated the following should be checked prior to replacing the cell:

1. Motor must be connected and output starter (if used) picked up.
2. Check all connections to the device including gate leads on SCRs.
3. Connections to other devices in the same part of the circuit as the indicated device.
4. Gate pulse signals to the device (SCRs only).

Refer to Figures 1.4 thru 1.9 to determine the location of the cell in the circuit. Refer to "Power Section Troubleshooting" for procedures to verify that a device has failed and the "Maintenance and Repair" section in this book for instructions on replacing a failed cell.

TABLE 5 (continued)
SCR TEST FAULT TROUBLESHOOTING
INVERTER SHORT CIRCUIT TEST

CELL FAULT CODE	DESCRIPTION
001	Short circuit test can not make 50 VDC on the link within one minute. - Check for shorts in the filter capacitor bank (Cl) per "Power Circuit Troubleshooting". - Shorted positive or negative inverting SCRs or diodes in the save phase (or both diodes reversed in the same phase) - check per "Power Circuit Troubleshooting". - Shorted positive and negative commutation SCRs in the same phase (or both diodes reversed in the same phase) - check per "Power Circuit Troubleshooting". - Shorted DC Link RC snubber. - 5PL, PFPL or NFB disconnected or damaged. - Improper link voltage feedback - check test point CVF on the Power Supply/Interface card. - Replace Power Supply/Interface card.
002	Link filter time constant too long. - Filter capacitor (Cl) load sharing resistor(s) (R1) failed or connection loose or open. - Filter capacitor(s) (Cl) failed - refer to "Power Circuit Troubleshooting", DC Lmk Filter Troubleshooting. - Check motor output leads and motor for ground fault.
003	Link filter time constant too short. - Filter capacitor (Cl) load sharing resistor(s) (R1) failed or connection loose or open. - Filter capacitor(s) (Cl) failed - refer to "Power Circuit Troubleshooting", DC Link Filter Troubleshooting. - Check motor output leads and motor for ground fault.
004	500KVA Frame - Positive inverter SCR shorted (any phase). - 5PL, PFPL or NFB damaged or shorted. - Positive inverter SCR snubber shorted - check per "Power Circuit Troubleshooting". - Inverter pulse transformer failed - check per "Power Circuit Troubleshooting". - Positive bypass diode shorted or installed backward - check per "Power Circuit Troubleshooting". - Positive inverter SCR shorted - check per "Power Circuit Troubleshooting". - Feedback failure - run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. - Replace the Power Supply/Interface card.

TABLE 5 (continued)
SCR TEST FAULT TROUBLESHOOTING
INVERTER SHORT CIRCUIT TEST

CELL FAULT CODE	DESCRIPTION
004	200KVA Frame – Positive inverter SCR shorted, negative commutation SCR (any phase) or any commutation capacitor shorted. – 5PL, PFPL or NFB damaged or shorted. – Positive inverter OR negative commutation SCR snubber shorted – check per "Power Circuit Troubleshooting". – Inverter or commutation pulse transformer failed – check per "Power Circuit Troubleshooting". – Positive bypass diode shorted or installed backward – check per "Power Circuit Troubleshooting". – Positive inverter or negative commutation SCR shorted – check per "Power Circuit Troubleshooting". – Feedback failure – run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. – Replace the Power Supply/Interface card.
005	500KVA Frame – Negative commutation SCR shorted (any phase). – 5PL, PFPL or NFB damaged OR shorted. – Negative commutation SCR snubber shorted – check per "Power Circuit Troubleshooting". – Commutation pulse transformer failed – check per "Power Circuit Troubleshooting". – Negative commutation SCR shorted – check per "Power Circuit Troubleshooting". – Feedback failure – run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. – Replace the Power Supply/Interface card.
006	500KVA Frame – Negative inverter SCR shorted (any phase). – 5PL, PFPL or NFB damaged or shorted. – Negative inverter SCR snubber shorted – check per "Power Circuit Troubleshooting". – Inverter pulse transformer failed – check per "Power Circuit Troubleshooting". – Negative bypass diode shorted or installed backward – check per "Power Circuit Troubleshooting". – Negative inverter SCR shorted – check per "Power Circuit Troubleshooting". – Feedback failure – run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. – Replace the Power Supply/Interface card.

TABLE 5 (continued)
SCR TEST FAULT TROUBLESHOOTING
INVERTER SHORT CIRCUIT TEST

CELL FAULT CODE	DESCRIPTION
006	200KVA Frame – Negative inverter SCR shorted, Positive commutation SCR (any phase) or any commutation capacitor shorted. – 5PL, PFPL or NFB damaged or shorted. – Negative inverter or positive commutation SCR snubber shorted – check per "Power Circuit Troubleshooting". – Inverter or commutation pulse transformer failed – check per "Power Circuit Troubleshooting". – Negative bypass diode shorted or installed backward – check per "Power Circuit Troubleshooting". – Negative inverter or positive commutation SCR shorted – check per "Power Circuit Troubleshooting". – Feedback failure – run Diagnostic static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. – Replace the Power Supply/Interface card.
007	500KVA Frame – Positive commutation SCR shorted (any phase). – 5PL, PFPL or NFB damaged or shorted. – Positive commutation SCR snubber shorted – check per "Power Circuit Troubleshooting". – Commutation pulse transformer failed – check per "Power Circuit Troubleshooting". – Positive commutation SCR shorted – check per "Power Circuit Troubleshooting". – Feedback failure – run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, check the NFB cable for damage. – Replace the Power Supply/Interface card.
008	Link voltage feedback reversed. – NFB cable damaged. – Feedback failure – run Diagnostic Static (refer to "Use of the Programmer Module" section) and monitor test points CVF, LVF, CVCO and LVCO on the Power Supply/Interface card. The voltage on CVF and LVF and the pulse frequency on CVCO and LVCO should change as the reference is changed. If any of these points do not vary with the reference, replace the Power Supply/Interface card.
009	SPARE
010	SPARE
011	SPARE

TABLE 5 (continued)
SCR TEST FAULT TROUBLESHOOTING
INVERTER SCR OPEN CIRCUIT TEST

CELL FAULT CODE	PHASE A		PHASE B		PHASE C	
	ISPA	ISNA	ISPB	ISNB	ISPC	ISNC
12	F	-	-	-	-	-
13	-	-	F	-	-	-
14	-	-	-	-	F	-
15		F	-	-	-	-
16	F	F	-	-	-	-
17		F	F	-	-	-
18		F	-	-	F	-
19		F	F	-	F	-
20	-	-	-	F	-	-
21	F	-	-	F	-	-
22		-	F	F	-	-
23	-	-	-	F	F	-
24	F	-	-	F	F	-
25	-	F	-	F	F	-
26	F	F	-	F	F	-
27	-	F	F	F	F	-
28	-	-	-	-	-	F
29	F	-	-	-	-	F
30	-	-	F	-	-	F
31	F	-	F	-	-	F
32	-	-	-	-	F	F
33	-	F	F	-	-	F
34	F	F	F	-	-	F
35		F	F	-	F	F
36	F	-	-	F	-	F
37	F	-	F	F	-	F
38	F	-	-	F	F	F
39	MULTIPLE OPEN SCRs - REFER TO "POWER CIRCUIT TROUBLESHOOTING"					

ISP_ = Positive Inverter SCR, Phase_ **F = Cell(s) Failed Open**
ISN_ = Negative Inverter SCR, Phase- **- = Passed**

TABLE 5 (continued)

SCR TEST FAULT TROUBLESHOOTING

COMMUTATION CIRCUIT FUNCTIONAL TEST

!ELL	PEASE A			PHASE B			PHASE C		
'AULT :ODE	DPA, ICSPA	DPA, DNA, CSPA,CSNA	CPS*	DPB, CSPB	DPB, DNB CSPB,CSNB	CPS*	DPC, CSPC	DPC, DNC CSPC,CSNC	CPS*
40	-	-	F	-	-	-	-	-	-
41	-	F	-		-	-	-	-	-
42	F	-	-		-	-	-	-	-
43	-		-		-	F	-	-	-
44	-	-	F		-	F	-	-	-
45	-	F	-	-	-	F	-	-	-
46	F		-	-	-	F	-	-	-
47	-	-	-	-	F	-	-	-	-
48	-	-	F	-	F	-	-	-	-
49	-	F	-	-	F	-	-	-	-
50	F	-	-	-	F	-	-	-	-
51	-	-	-	F	-	-	-	-	-
52	-	-	F	F	-	-	-	-	-
53	-	F	-	F	-	-	-	-	-
54	F	-	-	F	-	-	-	-	-
55	-	-	-	-	-	-	-	-	F
56	-	-	F	-	-	-	-	-	F
57	-	F	-	-	-	-	-	-	F
58	F	-	-	-	-	-	-	-	F
59	-	-	-	-	-	F	-	-	F
60	-	-	F	-	-	F	-	-	F
61		F	-	-	-	F	-	-	F
62	F	-	-	-	-	F	-	-	F
63	-		-	-	F	-	-	-	F
64	-		F	-	F	-	-	-	F
65	-	F		-	F	-	-	-	F
66	F			-	F	-	-	-	F
67	-	-	-	F		-	-	-	F
68	-	-	F	F	-	-	-	-	F
69	-	F	-	F		-	-	-	F

DP_ = Positive Diode, Phase-
DN_ = Negative Diode, Phase_
CSP_= Positive Commutation SCR, Phase-
CSN_= Negative Commutation SCR, Phase_
CPS_= Commutation Power Supply

- = Passed
F = Failed
* - For 200 KVA units, CSP_ or CSN_ open for particular phase.
* - For 500 KVA units, Commutation Power Supply failure or AS_ for particular phase open.

TABLE 5 (continued)
SCR TEST FAULT TROUBLESHOOTING
COMMUTATION CIRCUIT FUNCTIONAL TEST

CELL	PHASE A			PHASE B			PHASE C		
F'AULT CODE	DPA, CSPA	DPA, DNA, CSPA, CSNA	CPS*	DPB, CSPB	DPB, DNB CSPB, CSNB	CPS*	DPC, CSPC	DPC, DNC CSPC, CSNC	CPS*
70	F	–	–	F	–	–	–	–	F
71	–	–	–	–	–	–	–	F	–
72	–	–	F	–	–	–	–	F	–
73	–	F	–	–	–	–	–	F	–
74	F	–	–	–	–	–	–	F	–
75	–	–	–	–	–	F	–	F	–
76	–	–	F	–	–	F	–	F	–
77	–	F	–	–	–	F	–	F	–
78	F	–	–	–	–	F	–	F	–
79	–	–	–	–	F	–	–	F	–
80	–	–	F	–	F	–	–	F	–
81	–	F	–	–	F	–	–	F	–
82	F	–	–	–	F	–	–	F	–
83	–	–	–	F	–	–	–	F	–
84	–	–	F	F	–	–	–	F	–
85	–	F	–	F	–	–	–	F	–
86	F	–	–	F	–	–	–	F	–
87	–	–	–	–	–	–	F	–	–
88	–	–	F	–	–	–	F	–	–
89	–	F	–	–	–	–	F	–	–
90	F	–	–	–	–	–	F	–	–
91	–	–	–	–	–	F	F	–	–
92	–	–	F	–	–	F	F	–	–
93	–	F	–	–	–	F	F	–	–
94	F	–	–	–	–	F	F	–	–
95	–	–	–	–	F	–	F	–	–
96	–	–	F	–	F	–	F	–	–
97	–	F	–	–	F	–	F	–	–
98	F	–	–	–	F	–	F	–	–
99	–	–	–	F	–	–	F	–	–
100	–	–	F	F	–	–	F	–	–
101	–	F	–	F	–	–	F	–	–
102	F	–	–	F	–	–	F	–	–
103	MULTIPLE OPEN SCRs ANDDIODES – REFER TO "POWERCIRCUITTROUBLESHOOTING"								

POWER CIRCUIT TROUBLESHOOTING

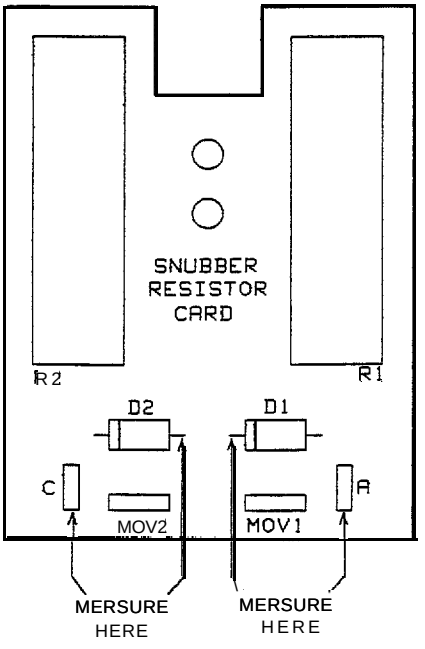
The following describes procedures for testing the power section components in the AC-500 controller. These procedures should be used in conjunction with the "Specific Fault Troubleshooting" and "SCR Test Fault Troubleshooting" tables to properly determine the exact device(s) that have failed. Refer to the Power Circuit Drawings (Figures 1.5 and 1.9 and Power Section Layout (Figures 5.1 thru 5.3).

Equipment required:

Volt Ohmmeter	Analog or digital with diode test position. 20 kohm/Volt minimum input impedance. (refer to meter manual for expected readings in diode test mode).
Oscilloscope	Dual trace perferred with clamp on amp probe.

CHECKING SNUBBERS

Whenever an SCR(s) is suspected of failure or intermittent power faults (i.e.: IOC, IFT, etc.), the RC Snubber(s) for the SCR or stack of SCRs should be checked for shorts. To check fur a shorted snubber, turn power OFF to the controller, check from center point of the snubber diodes to each connector (see sketch) with an ohmmeter set on the X10 scale. The resistance should be balanced for the two readings and not zero.



CHECKING PULSE TRANSFORMER CARDS

When a pulse transformer card is suspected of failed components the following procedure can be used to determine if there is a card failure.

Turn power to the drive OFF. Open the power section door and disconnect the NPT cable from all but the suspect pulse transformer card (i.e.: when checking the Converter Pulse Transformer card, disconnect NPT from the Inverter and on 500KVA Frames, Commutation Pulse Transformer card). Also disconnect the motor leads or remove control to the motor starter, if used.

WARNING

HIGH VOLTAGE WILL BE PRESENT THROUGHOUT THE POWER CIRCUIT WHEN MAKING THE FOLLOWING CHECKS:

Reapply power to the controller and check for +24 VDC (±5%) between the +24 and PCOMR test points on the suspect card. If no voltage is present, check cable NFB for loose connection or damage to the cable. Replace if necessary.

Next, run the drive in DIAGNOSTIC mode (Diagnostic Static to check the converter, Diagnostic Run to check the inverter or commutation section) at approximately half reference. Check the gate pulse signals from the card with an oscilloscope using an amp probe. Pulses should be present if the gate circuit is good. Check the gate connector and wiring if no pulses are present. If the connector and wiring are good, check for a gate pulse signal from the driver chip by connecting the common of the scope to PCOMR and the scope probe to the 15K ohm resistor to the pulse transformer. Replace the card if no signal is present. Replace the SCR if the signal is present. Stop the controller, remove power and replace the necessary component. Remember to reconnect any cable disconnected at the beginning of this procedure.

CHECKING CONVERTER SCRS

The converter SCRs can be checked with the power off, without disconnecting components. Wait one minute after disconnecting power and until the DC Link (P2 to N2) reads below 10 VDC before making any checks.

The measurement points are as follows:

L1 to P2	Checks positive phase 1 SCR (1SP)
L1 to N2	Checks negative phase 1SCR (1SN)
L2 to P2	Checks positive phase 2 SCR (2SP)
L2 to N2	Checks negative phase 2 SCR (2SN)
L3 to P2	Checks positive phase 3 SCR (3SP)
L3 to N2	Checks negative phase 3 SCR (3SN)

The actual physical location of the measurement points for either **frame size** (200 KVA or 500 KVA) can be found in Figures 5.1 and 5.2 (the P2 and N2 points in the 500 KVA power section are the inverter heatsinks).

Using a volt-ohmeter selected to read ohms on the 1K scale or diode check (refer to meter manual for expected readings in diode check mode), check across all six SCRs in both directions. Good SCRs should read high (over 100k ohms) in both the forward and reverse directions, while failed SCRs (shorted) will read low or zero ohms in one or both directions.

NOTE: Capacitor charging may be noted as an increasing resistance while the meter is connected to the circuit.

If any SCRs appear to be failed, the converter section should be disconnected from the power circuitry at terminal points L1C, L2C, L3C, P1 and N1 (on 500 KVA units, the stack may be removed per the replacement procedure **in the** "Maintenance and Repair" section and tested on a bench). Recheck the above readings at the disconnected terminals. If the readings still indicate a failed device, refer to the "Maintenance and Repair" section for replacement information. The individual SCRs should be rechecked when they are disconnected from the converter circuit to ensure that a short in one device does not produce a faulty reading across another device. See the checking SCR portion of this section.

Whenever a converter SCR has been removed for replacement, the RC Snubber cards and wiring should be inspected for damage.

CHECKING COMMUTATION POWER SUPPLY AND COMMUTATION SCRS

The commutation power supply (CPS) circuit and components vary between the 200 KVA and 500 KVA frame sizes. The 200 KVA frame CPS consists of the commutation power diode bridge, fuses, filter capacitor resistors (R2P and R2N), SCR pulse transformers (all located on the Converter Pulse Transformer card) and the commutation SCR modules and diodes (DCD) (mounted on the heatsink assembly). The **500 KVA** Frame CPS also has the diode bridge and fuses located on the Converter Pulse Tranformer card. In addition, the CPS utilizes a Commutation Power Supply card and auxiliary commutation SCRs (ASA, ASB and ASC) mounted on the inside of the power section door and an individual Commutation Pulse Transformer card located on the front of the commutation SCR stack. The following procedure indicates any difference between frame sizes when checking the CPS.

The CPS can be checked with power off. Wait one minute after disconnecting power and until the DC Link (P2 to N2) is below 10 VDC before making any checks.

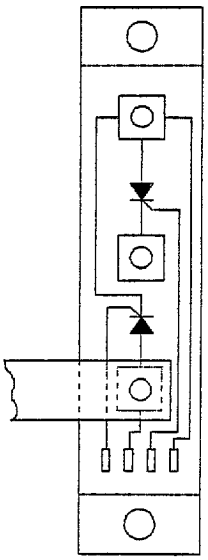
A. Diodes AD1, AD2 and AD3 Converter Pulse Transformer Card

The CPS diodes may be checked with a volt-ohmmeter selected to read ohms on the X10 scale or diode check. First lift one end of Fuses FUL1, FUL2 and FUL3 to isolate the diodes from the AC line, and check for open fuses. Check across each of the diodes in both the forward and reverse direction. (Each phase of the rectifier contains three diodes. Check across the individual diodes **in** each phase). Good diodes will provide almost infinite resistance in the reverse direction and a low reading (less than 100 ohms) in the forward direction. Failed diodes will read almost zero resistance in both directions (shorted) or infinite resistance in both directions (open).

If any diodes appear to be failed, replace the Converter Pulse Transformer card.

B. SCRs ASA, ASB and ASC (500 KVA Frame only)

The auxiliary commutation SCRs may be checked with a volt-ohmmeter selected to read ohms on the X1K scale or diode check. Each module contains two SCRs as shown in the sketch below:



B. SCRs ASA, ASB and ASC (500 KVA Frame only) (continued)

Check across all six SCRs in both directions. Good SCRs should read over 100k in both directions, while failed SCRs (shorted) will read zero ohms in one or both directions.

If any SCRs appear to be failed, the module should be disconnected and removed and the above readings rechecked. If the readings still indicate a failed device, replace the module. Whenever an auxiliary commutation SCR is replaced, the CPS card and related SCR wiring should be inspected.

C. Commutation SCRs

Using a volt-ohmmeter selected to read ohms on the X1K scale or diode check, read across the following measurement points:

P2 to MCA	Check positive phase A SCR (CSPA)
P2 to MCB	Check positive phase B SCR (CSPB)
P2 to MCC	Check positive phase C SCR (CSPC)
N2 to MCA	Check negative phase A SCR (CSNA)
N2 to MCB	Check negative phase B SCR (CSNB)
N2 to MCC	Check negative phase C SCR (CSNC)

Check across all six SCRs in both directions. Good SCRs should read high (over 100K ohms) in both directions, while failed SCRs (shorted) will read low or zero ohms in one or both directions.

NOTE: Capacitor charging may be noticed as an increasing resistance while the meter is connected to the circuit.

If any SCRs appear to be failed, the commutation section should be disconnected from the power circuitry at terminal points CP and CN (on 500 KVA units, the stack may be removed per the replacement procedure in the "Maintenance and Repair" section and tested on a bench). Recheck the above readings at the disconnected terminals. If the readings still indicate a failed device, refer to the "Maintenance and Repair" section for replacement information. The individual SCRs should be rechecked when they are disconnected from the commutation circuit to ensure that a short in one device does not produce a faulty reading across another device. See the checking SCRs portion of this section.

Whenever a commutation SCR has been removed for replacement, the RC snubber cards and wiring should be inspected for damage.

D. Filter capacitors C2P and C2N, resistor(s) (CPSR on 500 KVA or R3 and R4 on 200 KVA) and commutation capacitors CCA, CCB and CCC.

The filter capacitor may be checked with a volt-ohmmeter to determine if it charges up or is shorted. Also refer to DC Link Filter Troubleshooting in this section for further information on electrolytic capacitor inspection.

The filter resistor(s) and commutation capacitors may be checked with a volt-ohmmeter to determine if they are open or shorted. The filter resistor(s) should read as follows:

R3 (200 KVA)	500 ohms
R4 (200 KVA)	5 ohms
CPSR (500 KVA)	10 ohms

The commutation capacitors can be checked on the X1K scale and should read over 100K both forward and reverse when measuring across the following points:

T1 to MCA	Checks CCA
T2 to MCB	Checks CCB
T3 to MCC	Checks CCC

CHECKING INVERTER SCRs AND DIODES

The inverter SCRs and diodes can be checked with the power off, without disconnecting any circuitry. Wait one minute after disconnecting power and until the DC Link (P2 to N2) reads below 10 VDC before making any checks.

The measurement points are as follows:

T1 to P2	Check positive phase A SCR and diode (ISPA and DPA)
T1 to N2	Check negative phase A SCR and diode (ISNA and DNA)
T2 to P2	Check positive phase B SCR and diode (ISPB and DPB)
T2 to M2	Check negative phase B SCR and diode (ISNB and DNB)
T3 to P2	Check positive phase C SCR and diode (ISPC and DPC)
T3 to N2	Check negative phase C SCR and diode (ISNC and DNC)

The actual physical location of the measurement points for either frame size (200 KVA and 500 KVA) can be found in Figues 5.1 and 5.2 (the P2 and N2 points in the 500 KVA power section are the inverter heatsinks).

Using a volt-ohmmeter selected to read on X10 scale or diode check (refer to meter manual for expected readings in diode check mode), check across all six measurement points in both directions. The normal readings indicating good devices are as follows, with 'a positive meter lead connected to the first point.

P2 to T1, 2 or 3	High resistance
T1, 2 or 3 to P2	Low resistance
N2 to T1, 2 or 3	Low resistance
T1, 2 or 3 to N2	High resistance

NOTE: Capacitor charging may be noticed as an increasing resistance while the meter is connected to the circuit.

If any of the above readings are zero or inconsistent with the list, the suspect inverter section (positive or negative) should be disconnected from the rest of the circuitry and rechecked. Positive modules can be isolated by disconnecting T1, T2 or T3 (or all) and P2, negative modules by disconnecting T1, T2 or T3 (or all) and N2. (On 500 KVA units, the stack may be removed per the replacement procedure in the "Maintenance and Repair" section and tested on a bench). Recheck the above readings at the disconnected terminals. If the readings still indicate a failed device, refer to the "Maintenance and Repair" section for replacement information. The individual SCRs and diodes should be rechecked when they are disconnected from the inverter circuit to ensure that a short in a device does not produce a faulty reading across another device. See the checking SCR portion of this

Whenever an inverter SCR or diode has been removed for replacement, the RC snubber cards and wiring should be inspected and checked for damage.

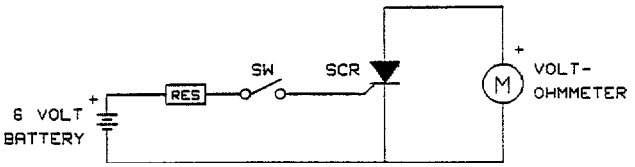
CHECKING SCRs AND DIODES

Two types of SCR packages are used in the AC-500 power circuit, Press-Pak SCR and dual cell module (two devices in one module, used in the 200 KVA frame and for the auxiliary commutation SCRs in the 500 KVA frame). Pressure must be applied to the Press-Pak cell to ensure good conduction through the internal cell structure while the dual cell module SCR requires no pressure on the terminal points.

Disconnect the suspected SCR as much as possible from the remainder of the power circuitry. Using a volt-ohmmeter selected to read ohms on the times 1K scale, check the forward and reverse resistance of each individual SCR cell. Typical readings for good to faulty SCR's foilowing:

SCR DESCRIPTION	FORWARD READING	REVERSE READING
Good SCR	100K to Infinity	100k to Infinity
Shorted SCR	Zero	Zero
Inoperative SCR	1 to 2K	100K to Infinity
Open SCR	100K to Infinity	100K to Infinity

Since an open SCR will give about the same resistance reading as a good SCR, another method must be used to find this type of fault. It should be point out, however, that most cells fail by shorting and very few by opening. If an open SCR is suspected, or it it is desired to check the switching operation of an SCR, the following circuit should be used:



The volt-ohmmeter is selected to read ohms on the 1K scale, and is connected to read the forward resistance of the SCR. When switch SW is closed, the forward resistance of a good SCR will change from a high value (100K to infinity) to a low value (1 to 10K). When the switch is opened, a good SCR will revert to its high forward resistance or blocking state if the holding current source (volt-ohmmeter battery) is momentarily removed or latching current is not exceeded. A faulty SCR will not switch, remaining in either an open or conducting state.

If any SCRs are suspected of being faulty from the above resistance checks, the SCR module or stack assembly should be removed from the case. After the SCR gate leads have been disconnected, recheck the forward and reverse resistances before replacing the SCR.

This should be done before the SCR is definitely classified as damaged or faulty, since a fault in another SCR or another part of the circuitry can produce a faulty reading from a good SCR before it is disconnected from the circuit. After a Press-Pak SCR is removed from the heatsink it may read open due to lack of pressure against the internal cell structure. Apply pressure to obtain a true reading.

DC LINK FILTER TROUBLESHOOTING

The DC link filter consists of the L1 and L2 filter chokes and the Cl capacitor assembly.

A. Filter Capacitor Assembly(s)

This consists of one or more assemblies of paralleled, (230 volt AC drives), or series and paralleled (460 volt AC drives) electrolytic capacitors. When the drive has not been operated for 6 months or more, these capacitors start to degrade and their leakage current increases. A procedure called forming is required to return the electrolytic capacitors to their rated operating capability. Refer to Apply AC Power of the "Start Up Procedure" section, step K, for proper forming procedure.

Electrolytic capacitors can fail by shorting, can exhibit excessive leakage current, or can dry up and lose their capacitance. The latter usually results from a ruptured vent plug due to "gassing" from excessive current and/or temperature.

The filter capacitor assembly can be checked for shorted capacitors using a volt-ohmmeter after the power has been off for more than 1 minute and the P2 to N2 voltage is less than 10 volts. On 460 volt AC drives with seriesed capacitors, the assembly can be checked for excessive leakage capacitors by checking the midpoint, or Q point, voltage when the drive is operating. Using a volt-ohmmeter, check the difference between the P2 to Q and Q to N2 voltages at maximum DC link voltage. This difference should not exceed 5% of the P2 to N2 voltage. If the above tests indicate either a shorted or leaking capacitor, the filter assembly should be removed and disassembled to the point where the resistance of each capacitor can be individually checked. Any shorted or leaking capacitors should be replaced. The remaining capacitors should be inspected for ruptured vent plugs according the the following instructions.

The best way of evaluating the condition of the electrolytic capacitors is to visually inspect their vent plugs. These are 3/16" diameter plugs in the top cover of the capacitor case. Internal gas pressure can cause a bubble to form in this plug and the red color will lighten until it is almost white. Eventually the plug will rupture.

However, this does not cause an immediate capacitor failure, but will result in a gradual loss of capacitance. Any electrolytic capacitors which are found to have ruptured plugs should be replaced as soon as conveniently possible. If any capacitor vent plug contains a bubble larger than 1/16" in diameter, the capacitor assembly should be inspected at the next scheduled shutdown or planned maintenance for ruptured vent plugs.

If more than 25% of the capacitors have broken vent plugs, and the drive has been operated over 20,000 hours, consideration should be given to replacing all of the capacitors in the filter assembly.

B. L1 and L2 Filter Chokes

These chokes should be visually checked for signs of overheating, damaged insulation or loose connections.

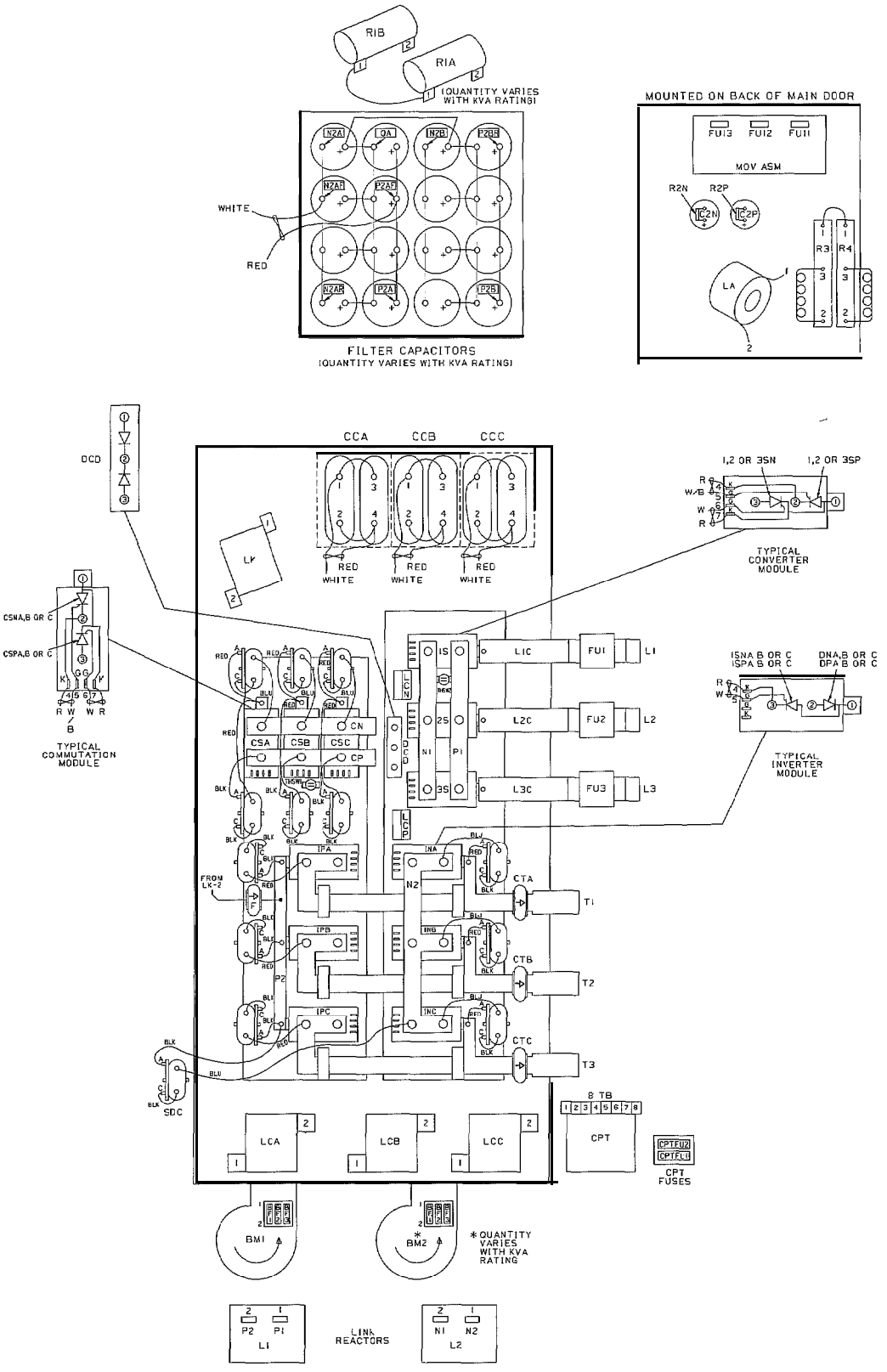


FIGURE 5.1
200 KVA POWER SECTION
(REFER TO DRAWING 36D870045AA INCLUDED WITH CONTROLLER)

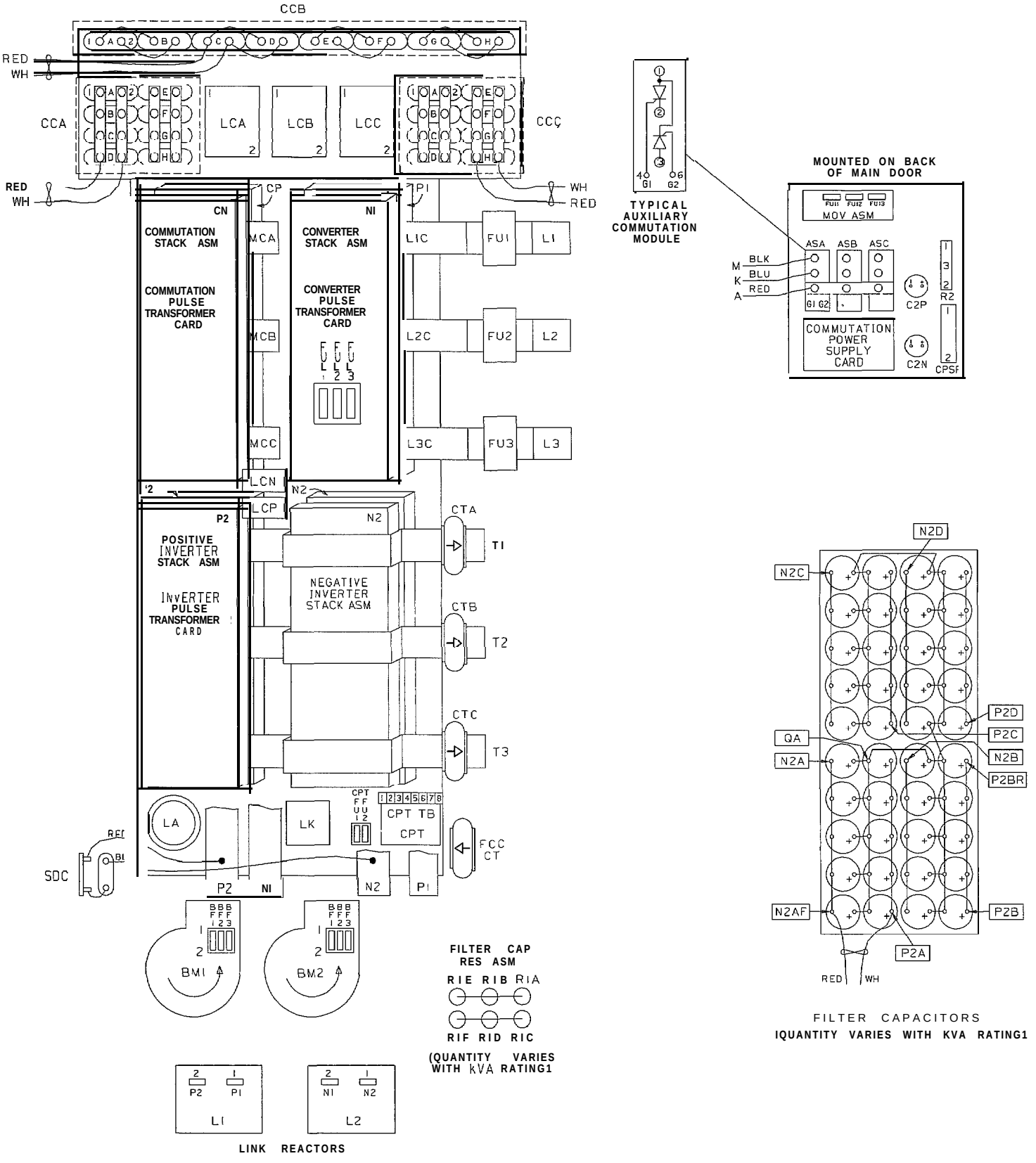


FIGURE 5.2

500 KVA POWER SECTION

(REFER TO DRAWING 36D871058AA INCLUDED WITH CONTROLLER)

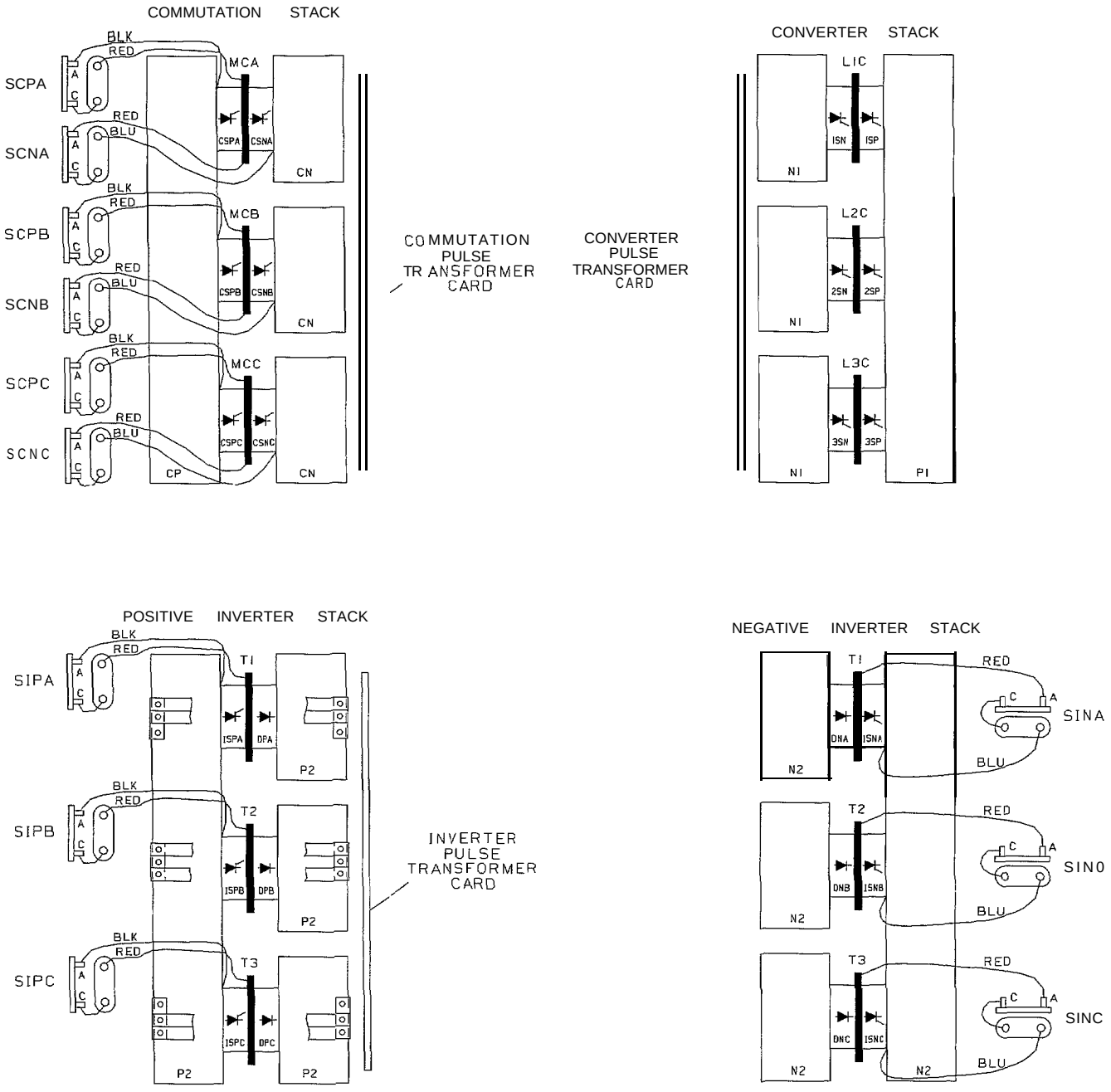


FIGURE 5.3

500 KVA CELL LOCATION

(REFER TO DRAWING 36D871058AA INCLUDED WITH CONTROLLER)

TYPICAL WAVEFORMS

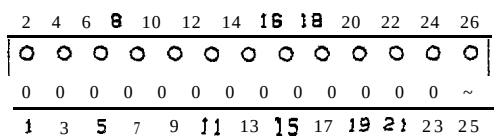
The following are examples of typical waveforms that can be seen on an oscilloscope. Each picture is accompanied by the names of the signals in order from top to bottom, the test points that they are derived from (in parenthesis), the channel amplitude settings for the individual waveforms, and the mode of operation. Also included is the proper time setting for the waveform.

Multiple waveforms are shown in one figure to allow comparison for phase relationship or to show how multiple events interact (inverter, commutation, and auxiliary commutation SCR firing signals and commutation capacitor voltage feedback with respect to each other). Some waveforms cross over others. Therefore, points are labeled on the left and right of the waveforms to indicate a particular waveform's zero point and resultant position. Zero points are shown on the left side of the waveform and the resultant position is indicated on the right.

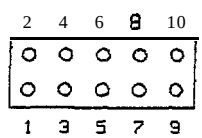
The waveforms are found on various test points on different cards. The test points for the SCR firing signals are found in the test plugs on the Inverter Firing Generator (IFG) card labeled TCN1, TCN2, TCA1 and TCA2. These plugs can be used to electronically rotate the output phases by rotating the pins on which the firing signals are sent out to the Inverter and Commutation Pulse Transformer cards. These pins can also be used to monitor the converter, inverter and commutation firing signals. In this case, the plug and pin number that the firing signal can be found on will be shown instead of a test point name (example: TCN1-3 is the positive A-phase inverter firing signal). The following lists the pinouts for the test plugs and the relative location of pins when looking at the IFG card installed with the card holder folded out.

PIN NO.	TCN1	TCN2
1	ISPC	I S P B
2	ISNC	ISNB
3	ISPA	ISPC
4	ISNA	ISNC
5	ISPB	ISPA
6	ISNB	ISNA
7	CSPC	CSPB
8	CSNC	CSNB
9	CSPA	CSPC
10	CSNA	CSNC
11	CSPB	CSPA
12	CSNB	CSNA
13	1SP	1SP
14	1SN	1SN
15	2SP	2SP
I6	2SN	2SN
1 7	3SP	3SP
18	3SN	3SN
19 .	CVIC	CVIB
20	cvcc	CVCB
21	CVIA	CVIC
22	CVCA	cvcc
23	CVIB	CVIA
24	CVCB	CVCA
25	No connection	No connection
26	No connection	No connection

PIN NO.	TCA1	TCA2
1	ASC	ASB
2	ASC	ASB
3	ASA	ASC
4	ASA	ASC
5	ASB	ASA
6	ASB	ASA
7	No connection	No connection
8	No connection	No connection
9	No connection	No connection
10	No connection	No connection



NPT, TCN1 OR TCN2 PINOUTS

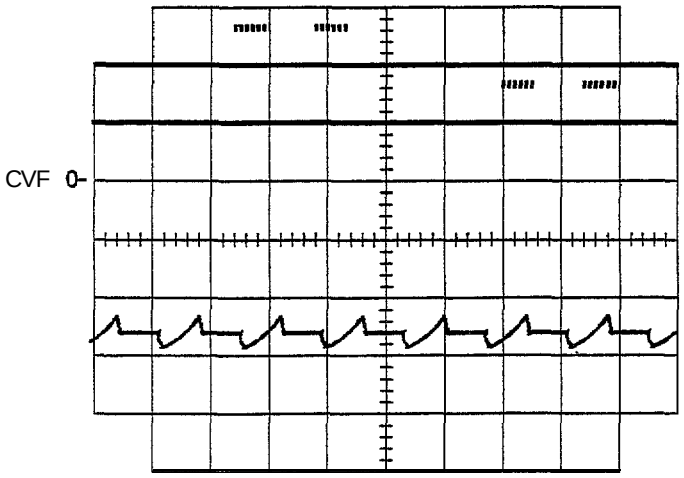


APT, TCA1 OR TCA2 PINOUTS

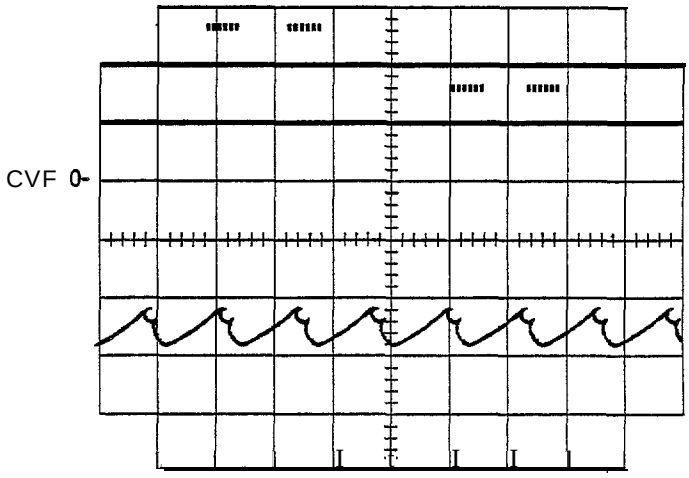
Refer to NPT and APT connector descriptions for full description of printouts.

FIGURE 5.4
CONVERTER FIRING SIGNALS AND FEEDBACK WAVEFORMS

Top trace	·	Positive converter firing signal, 1SP (TCN1-13) · 20 volts/div
Middle trace	·	Negative converter firing signal, 1SN (TCN1-14) · 20 volts/div
Bottom trace	·	Converter Voltage Feedback (CVF) · 2 volts/div



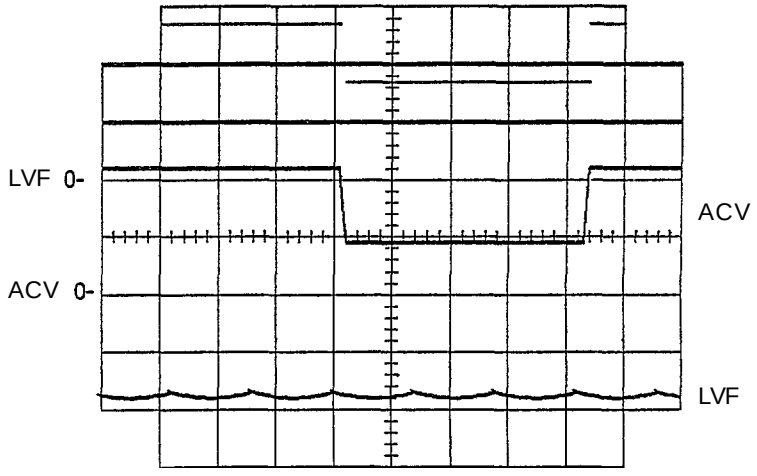
2 msec/div
&Load, 60 Hz



2 msec/div
Full Load, 60 Hz

FIGURE 5.5
LINK VOLTAGE FEEDBACK

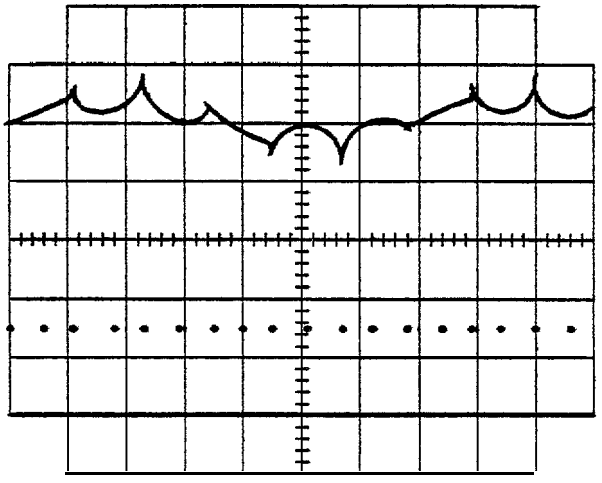
Top trace	·	Positive inverter firing signal, ISPA (TCN1-3) · 20 volts/div
Second trace	·	Negative inverter firing signal, ISNA (TCN1-4) · 20 volts/div
Third trace	·	Commutation capacitor voltage feedback (ACV) · 5 volts/div
Bottom trace	·	Link Voltage Feedback (LVF) · 2 volts/div



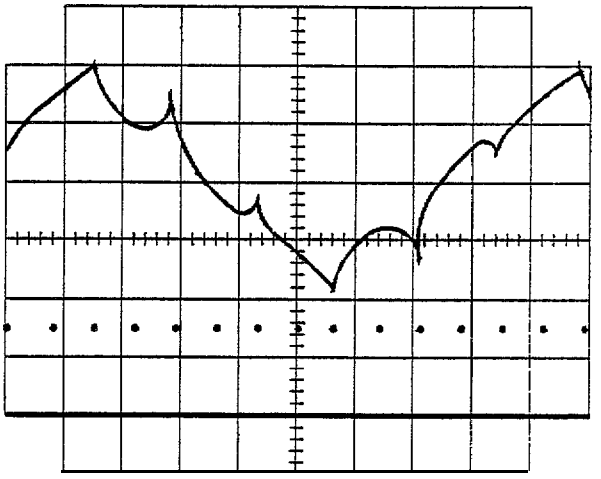
2 msec/div
Full Load, 60Hz

FIGURE 5.6
MOTOR CURRENT WAVEFORMS

Top trace - Motor CT feedback signal (CTA) - 1 volt/div
Bottom trace - Twelve times inverter frequency (12XIF) - 10 volts/div

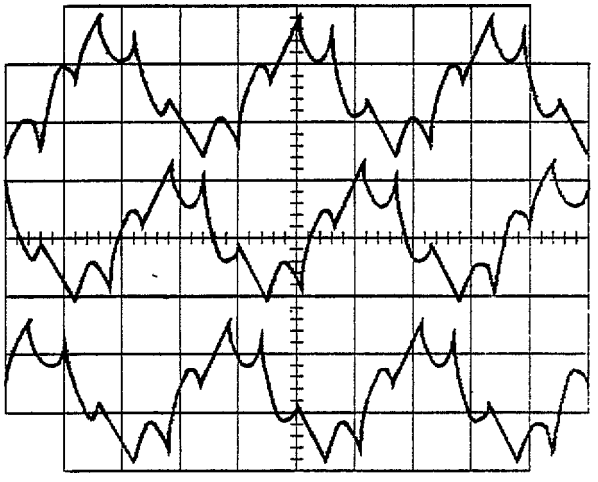


5 msec/div
20% Load, 30Hz



2 msec/div
Full Load, 60Hz

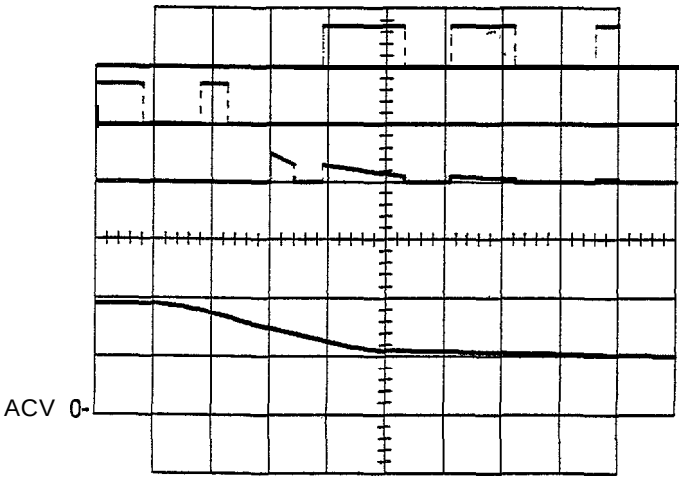
Top trace - Phase A motor CT feedback signal (CTA)
Middle trace - Phase B motor CT feedback signal (CTB)
Bottom trace - Phase C motor CT feedback signal (CTC)



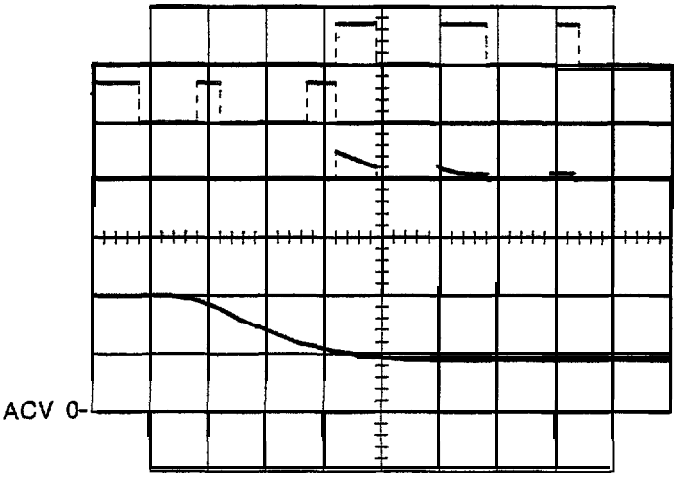
2 msec/div
1 volt/div
50% Load, 30Hz

FIGURE 5.7
INVERTER AND COMMUTATION FIRING SIGNALS

Top trace - Negative inverter firing signal, ISNA (TCN1-4) - 20 volts/div
Second trace - Positive commutation firing signal, CSPA (TCN1-9) - 20 volts/div
Third trace - Auxiliary commutation firing signal, ASA (TCA1-3) - 20 volts/div
Bottom trace - Commutation capacitor voltage feedback (ACV) - 5 volts/div

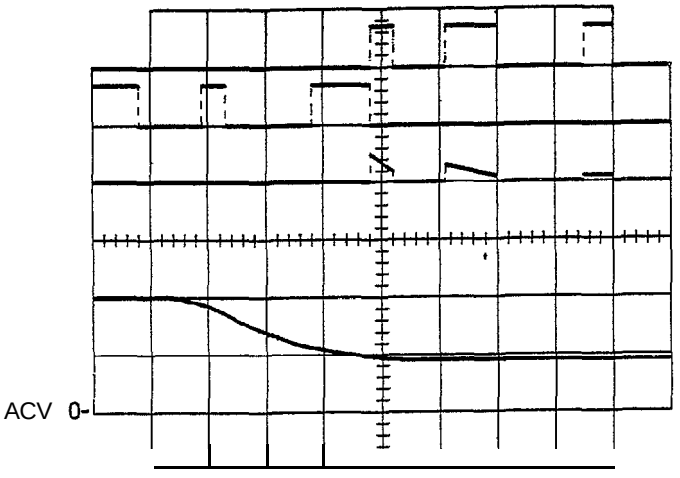


2 μsec/div
No Load, Zero Speed

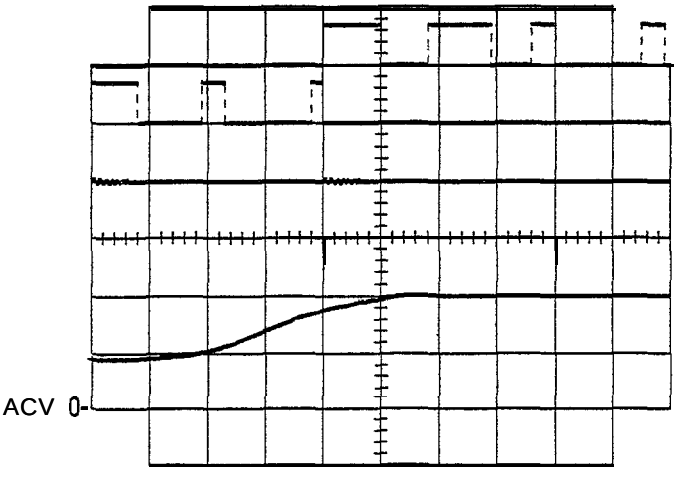


2 μsec/div
No Load, 60 Hz

Top trace - Positive inverter firing (TCN1-3)
Second trace - Negative commutation firing (TCN1-10)
Third and bottom trace - same as above.



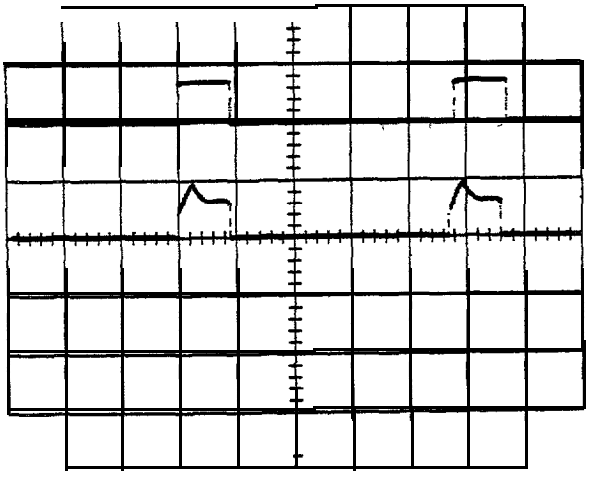
2 μsec/div
Full Load, 60Hz



2 μsec/div
No Load, 60Hz

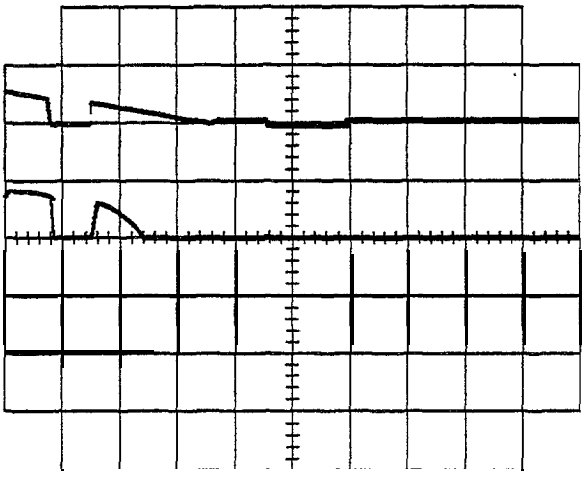
FIGURE 5.8
FIRING SIGNALS VS GATE PULSE TRANSFORMER OUTPUT

Top trace - Positive inverter firing signal, ISPA (TCN1-3) - 20 volts/div
Bottom trace - Positive inverter pulse transformer output - I amp/div



10 usec/div

Top trace - Auxiliary commutation firing signal, ASA (TCA1-3) - 20 volts/div
Bottom trace - Auxiliary commutation pulse transformer output - 1 amp/div



10 usec/div

NOTE: The gate pulse transformer output waveforms were generated using a clamp-on type current probe at the pulse transformer output plug. All amplitude settings reflect probe multiplier (i.e., 20 volts/div is the same as 2 volts/div with a times 10 probe.)

USE OF OPTIONAL PROGRAMMER MODULE

The Programmer assembly is available in two configurations, as pictured in Figures 6.1 and 6.2. The Local configuration is mounted on the Control card and plugged into 18PL. The Hand-held configuration is contained in a box and plugged directly into the Control card 18PL.

CAUTION

THE HAND-HELD PROGRAMMER CABLE MUST BE ORIENTED PROPERLY OR THE CONTROL AND PROGRAMMER CARDS WILL BE DAMAGED.

The Hand-held Programmer uses a different cable than the Local Programmer and care must be exercised when the Hand-held Programmer is plugged in.

To properly orient the cable, hold the unit in front of the drive and extend the ribbon cable back to 18PL, so that the cable is not twisted, and plug into 18PL. The red strip in the ribbon cable should be on the left when looking at the front of the Hand-held Programmer and should be on the left when plugged into 18PL.

The local Programmer cable is oriented with the red wire on the left side of the cable when the operator%-facing the drive.

The Programmer provides an alphanumeric readout of fault codes (instead of the diagnostic LEDs) and operation/adjustment of the drive through three modes - - OPERATE, PARAMETER, and DIAGNOSTIC modes.

THREE MODES OF OPERATION

A. **OPERATE** Mode - The "normal" mode, allows the red and black keypad functions to be active (the **RED** keys are dominant). This is the mode the drive will enter when power is applied.

B. **PARAMETER** Mode - Allows the user to examine and change as necessary any of 254 EEROM (E^2) Parameter addresses to "customize" the drive for a given application. In this mode all **BLACK** keys are active plus [INC], [DEC] and [STOP].

C. **DIAGNOSTIC** Mode - Allows the user to utilize any of the internal digital displays, plus run a series of diagnsotic checks and self-tuning procedures. In this mode, both the **BLUE** and **BLACK** key functions are active (with **BLUE** keys dominant) plus [INC], [DEC] and [STOP].

CHANGING BETWEEN MODES

NOTE: Throughout this section a number or function encased in brackets [ ] indicates a Programmer key pressed.

A. To enter the PARAMETER mode, first confirm that the display reads "OPERATE", then press [SET], [DRV#], [7], [7], and [ENTER]. The display should now show "PARAMETER".

B. To enter the DIAGNOSTIC mode, the drive must first be in the PARAMETER mode. If the display shows "PARAMETER", simply press [CLEAR/MODE] and the display will immediately show "DIAGNOSTIC". If a particular parameter is being displayed, press [CLEAR/MODE] causing "PARAMETER" to be displayed then press [CLEAR/MODE] again; "DIAGNOSTIC" will appear.

C. To return to the OPERATE mode, simply press [CLEAR/MODE] once in the DIAGNOSTIC MODE, or if the drive is in the PARAMETER mode, press [CLEAR/MODE] either two or three times, as required until the display returns to OPERATE.

NOTE: The Programmer may appear to "hang up" due to programming errors. In such cases it may be necessary to initiate a "hardware reset" using the RESET switch (SW2) on the Control card.

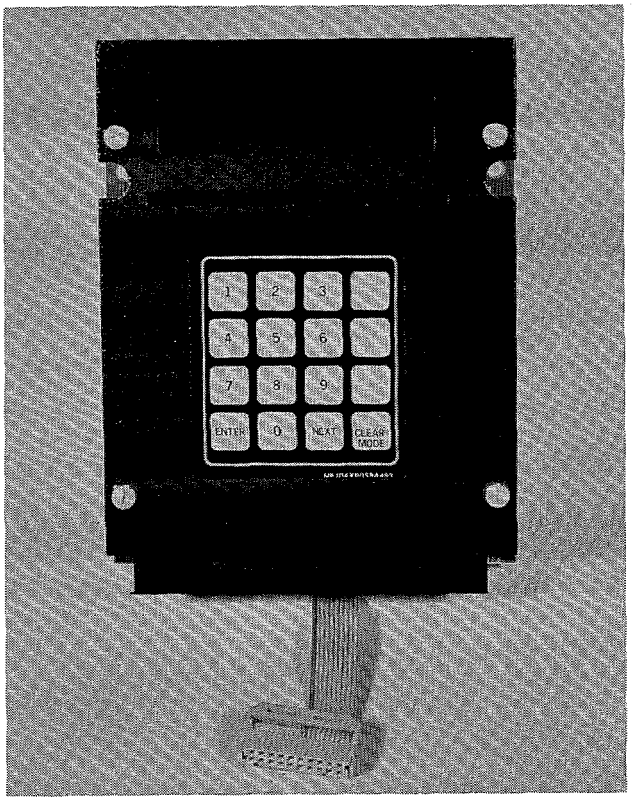


FIGURE 6.1
"LOCAL" PROGRAMMER MODULE

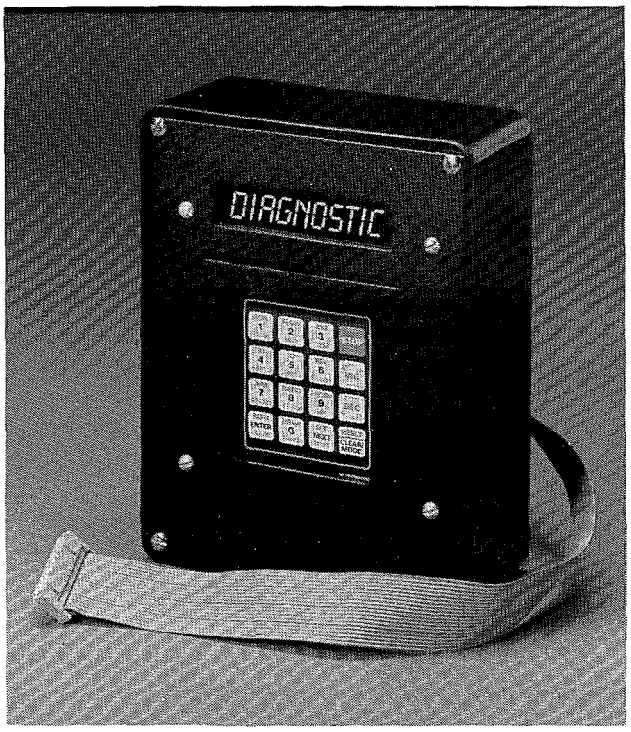


FIGURE 6.2
"HAND-HELD" PROGRAMMER MODULE

OPERATE MODE - (RED Keys dominant)

In the OPERATE mode, any of the RED key functions, if activated (if key is not activated, the Programmer will display "FL41 KEY OF" momentarily), can be used as follows:

[RUN] - Run at speed set by RUNRF.

[SLOW] - Run at speed set by SLWRF.

[JOG] - Jog at speed set by JOGRF.

[STOP] - Controlled stop.

[R1] - Run at speed set by R1RF.

[R2] - Run at speed set by R2RF.

[REV] - Reverse RUN or JOG as required by first pressing [REV], then either [RUN], [SLOW], [JOG], [R1], [R2] , or [REFR].

[A/M] - Changes drive from AUTO to MANUAL, or vice-versa, enabling use of either ASR or MSR external speed reference inputs.

[RATIO] - Displays the drive run ratio, or the percentage of speed reference at which the drive will run. Ratio is used to set a draw between multiple drives monitoring the the same external speed reference. Normally set at 10,000 indicating ratio of 1:1.

[PRGSV] - For future use.

[REFR] - Run at speed reference at which drive was last run, (includes preprogrammed keypad references or changes caused by using the [INC] and [DECJ keys, MSR or ASR).

[DRV#J - Used to examine the drive number (in most cases "00"). Also part of access code to PARAMETER mode.

[SET] - Used to adjust the settings of various keypad functions; activated by pressing [SET], then [RUN], [SLOW], [JOG], [Rl], [R2], or [RATIO], then the desired setting in decimal numbers, then [ENTER]. Also, used for entering Parameter mode. The EEROM must be write "ENABLED" to make these adjustments (see Part IV).

[INC] & [DEC] - Increases or decreases the speed of any of the keypad Run or Jog functions, or the Ratio function; simply enter the desired function and then press [INC] or [DEC] as desired.

[RESET] - Resets the drive after fault or error messages appear. The reset key is referred to as a "Software Reset"

The normal display in the OPERATE Mode will be "M XX% YY%" where "M" indicates Manual mode using the Manual System Reference, MSR (alternatively an "A" will indicate Auto mode using the Auto System Reference, ASR on the optional Application card), XX% indicates percent of rated speed, and YY% indicates percent of rated load.

PARAMETER MODE (BLACK keys, plus INC, DEC & STOP active)

A. To enter the PARAMETER mode, press [SET], [DRV#], [7], [7], and [ENTER].

B. To **examine** the present value of any parameter simply key in any three-digit parameter address (do not press [ENTER]). The name for that address will immediately be displayed along with the present contents of that address. Press [CLEAR/MODE] to clear present parameter (displays "PARAMETER") then the address number for the next parameter desired.

Occasionally, it may become necessary to check a series of parameters. Simply call up the lowest parameter address and press [NEXT] as often as required until the highest parameter number is reached.

C. To change parameters, two conditions must be present:

1. The EEROM (E^2) enable device must be in the "ENABLE" position.

   JP13: 1-2 position; E^2 write "PROTECT"
   2-3; E^2 write "ENABLE"

CAUTION

DO NOT APPLY OR REMOVE POWER FROM THE DRIVE OR PERFORM A "HARDWARE RESET" WITH THE E^2 ENABLE DEVICE IN THE "ENABLE" POSITION.

2. Parameter address #3 (designated as ADR003-DGNJP) must contain an odd number (1, 3, 5, etc). If an even number exists, add "1" to the present value and enter the new number.

Example: Examine and change the drive acceleration and deceleration rates. **Assume** the drive acceleration rate is set at 2 seconds and the deceleration rate is set at 10 seconds. Change the acceleration rate (ADR057-ACELT) from 2 seconds to 7.5 seconds. Reduce the drive deceleration rate (ADR058-DCELT) from 10 to 5.2 secs.

1. Confirm that the E^2 enable device is in the "ENABLE" position and that ADR003-DGNJP contains an odd number.

2. After examining ADR003, press [CLEAR/MODE] then [O], [5], [7]. The display shows "ACELT. 20", corresponding to a 2 second acceleration rate.

3. Either press [INC] until the display value increases to "75" or press [7], [5]. In either case the display should now show "ACELT. 75"

4. Press [ENTER], placing the new value into EEROM memory.

5. Press [NEXT], commanding the drive to go to the next parameter, ADR058-DCELT; the display shows "DCELT. 100", corresponding to a 10 second deceleration rate.

6. Press [DEC], until the display shows "DCELT. 52" or press [5], [2].

7. Press [ENTER] placing the new value into memory.

8. Press [CLEAR/MODE]: the display will now show "PARAMETER".

9. To examine other addresses simply key in the appropriate three digit parameter address number.

NOTE: This next step is **extremely important** to prevent accidental reprogramming of the drive.

10. To **seal** the new parameter values, **remove** the "1" from ADR003-DGNJP returning it to the "as shipped" **even** number, and return the E^2 enable device to the "PROTECT" position.

D. As in the OPERATE mode, the display may return to "'M. XX% YY%" after a period of inactivity, but the drive is still in the PARAMETER mode (indicated by the apostrophe before the M) awaiting further keypad entries. Simply enter the next parameter address to be examined.

E. To **exit** the PARAMETER mode, press [CLEAR/MODE] either once or twice to go to the DIAGNOSTIC mode. To re-enter the OPERATE mode, press [CLEAR/MODE] again.

DIAGNOSTIC MODE

The Diagnostic Tests listed below will aid in the start-up, tune-up and troubleshooting of the DC-300 drive. To use these diagnostic features either a Local or Hand-held Programmer is required. Availability of certain tests may be dependant on hardware or software configurations provided.

STANDARD DIAGNOSTICS TESTS:

Test 1	[DGS] - Diagnostic Static.
Test 2	[DGR] - Diagnostic Run.
Test 3	[DVM] - Displays Digital Voltmeter.
Test 4	[ARM V] - Displays AC output voltage.
Test 5	[ARM I] - Displays AC output current.
Test 6	[FIELD I] - Displays output frequency in hertz.
Test 7	[RPM] - Displays motor speed.
Test 8	[TORQ] - Displays motor torque.
Test 9	[HP] - Displays motor horsepower.
Test 10	Displays software revision of Control card microprocessors.
Test 11	Displays diagnostic variables (selectable).
Test 12	SCR test.
Test 51	Revision date of Programmer software.
Test 52	Revision level of Programmer software.
RAM TEST	Displays single diagnostic variable (selectable).
DACO, DAC8 & DAC12	Digital to Analog converter outputs.
BITS 1, 2 & 3	Status bit outputs.

OPTIONAL ADVANCED DIAGNOSTIC TESTS:

Test 20	Electronic Wire-Check.
Test 21	Diagnostic Variable pairs.
Test 22	Diagnostic Variables through DAC Outputs.
Test 25	Square-Wave Reference

A. STANDARD DIAGNOSTIC TESTS

NOTE: For Tests 1 through 9 simply enter the DIAGNOSTIC mode and press the appropriate key. To enter all other diagnostic tests, press [TEST], [X], where "X" is the number of the particular test to be performed, and then press [ENTER].

AI. TEST 1 - DIAGNOSTIC STATIC (DGS)

Diagnostic Static (DGS) allows observation and adjustment of the Diagnostic Run reference (Test 2) without actually running the drive. The Diagnostic Run reference is selected by ADR003-DGNJP and may set either MSR, the Programmer INCREASE/ DECREASE keys or the internal programmed references, ADR092 and ADR093 (TSTR1 and TSTR2)as the reference. Refer to the Custom Software Description to determine which reference is used.

Diagnostic static is initiated from a stopped condition. When initiated, the drive runs the converter section open loop and does not run the inverter section. The RUN AUXILLIARY relay (IS2 on the Power Supply/Interface card) is not picked-up, so any AC output contactor (if used) will not be energized.

1. Only the converter section will be enabled and it's SCRs synchronously fired. The inverter section will be disabled and not be fired.

2. The drive will use the diagnostic reference for phase angle control as selected by ADR003-DGNJP. Refer to the Custom Software Description.

3. MAX output starter control contact will not close; if used the output starter will not pickup. If no starter is used then the output may be disabled by removing power cable T1, T2, and T3 from the inverter output if desired.

4. The Programmer display will display the reference input.

Diagnostic Static will run only the converter section and allows adjustment of DC Link voltage with reference. The Converter and Link Voltage Feedbacks (analog and digital VCOs) may be measured without being in the converter and inverter control loops.

Pressing [STOP] discontinues Diagnostic Static.

A2. TEST 2 - DIAGNOSTIC RUN (DGR)

Diagnostic Run (DCR) causes the drive to run as a converter voltage regulator with the frequency and speed (if used) regulators open loop. As in the Diagnostic Static, the RUN AUXILIARY is not picked up. The Diagnostic Run reference is determined by parameter ADR003-DGNJP (see Diagnostic Static).

1. Any exteral speed or process loop will be disabled.

2. The drive will use the diagnostic reference for frequency as selected by ADDR003-DGNJP such as MSR or the INC/DEC buttons on the Programmer Module. Refer to the Custom Software Description.

3. MAX output starter control contact will not close. Therefore, if used the output starter will not pickup. If no starter is used then the output can be disabled by removing power cables T1, T2 and T3 from the inverter.

Diagnostic Run will allow operation of the drive with or without the motor to measure or view firing, control and feedback signals.

Pressing [STOP] will initiate a coast stop. These features, open frequency/speed loop regulation and coast stopping, will aid significantly when trying to separate mechanical problems from drive stability problems.

A3. TEST 3 - DVM MODE

(See also Section VI, DVM Mode for Drive Setup)

Test 3 provides the capability of utilizing the Local or Hand-held Programmer and Process Interface card as a DC digital voltmeter (DVM) for checkingvarious analog test points (TP) on the drive cards. DC voltage in the +/-40 volt range can be measured. This requires a test probe wire (supplied in the miscellaneous parts packet shipped with the drive) from test point DVM on the Process Interface card to the particular test point to be monitored. The DVM may be calibrated by monitoring the +5 volt test point and adjusting P3 -DVM TRIM on the Process Interface card.

Pressing [TEST] discontinues DVM mode and displays "CLR TEST".

A4. TESTS 4 thru 9 VARIOUS DISPLAYS

(ARM V, ARM I, FIELD I, SPEED, TORQ, HP)

Tests 4 thru 9 are accomplished by setting the drive in DIAGNOSTIC mode and then entering the desired test number. To change to a different test simply push another number from 4 thru 9. To cancel the last test push [TEST] once, the Programmer will display "CLR TEST".

A5. TEST 10 - SOFTWARE REVISION

Test 10 will display the revision level of the microprocessors on the Control card. For example, "R41F41T41" indicates that the regulator microprocessor (RUP) is version41 software, the firing microprocessor (FUP) is version 41 and the tachometer microprocessor (TUP) is version 41. This can be verified by comparing the information with white labels on the individual microprocessor chips on the Control card.

A6. TEST 11 - DIAGNOSTIC VARIABLE DISPLAY

(also see Tests 21 and 22)

Test 11 can be used to display, in hexadecimal numbers, certain drive variables for the purpose of comparison. For example, speed reference and speed feedback may be compared. To accomplish this, several things must be done. First, place the drive in PARAMETER mode as described in "Changing Between Modes". Next determine, from the list which follows, the RAM address of the regulator variables to be monitored. Then put the first variable RAM address number into parameter ADR122-DIAG1 and the second variable RAM address into parameter ADR123-DIAG2. The last step is to put the drive in DIAGNOSTIC mode and select Test 11 as described above. The Programmer will then display "DGXXXX.YYYY", where "XXXX" is the hexadecimal value of the RAM address entered in ADR122, and "YYYY" is the hexadecimal value of the RAM address entered in ADR123.

NOTE: The value displayed is in 2's complement. Values from 0000 to 7FFF are positive magnitude (0 to + 32767). Values from FFFF to 8000 are negative magnitude (FFFF = -1, 8000 = -32768).

AI. TEST 12 - SCR TEST

WARNING

THE FOLLOWING TEST WILL CAUSE CURRENT TO CIRCULATE IN THE MOTOR, THE MOTOR MUST BE CONNECTED TO COMPLETE THE TEST.

NOTE: Test 12 requires a Programmer Module or serial link device to operate. The SCR TEST PROGRAM consists of a series of individual tests. Refer to the SCR TEST TROUBLESHOOTING section of this manual for details on specific tests.

Test 12 checks the SCRs and power components (inductors, capacitors, diodes, wiring, etc.) of the converter and inverter sections of the AC-500 controller. The motor must be connected for Test 12 to function. When Test 12 is initiated, the RUN AUXILIARY relay picks up to energize the AC output contactor (if used). The test then begins checking the converter and inverter sections, monitoring voltage and current feedbacks to detect any faults. If at any time the test determines a faulted component(s) it will stop testing and display "FL14 SCXXX" (where XXX is the code for the failed component(s). The fault is then deciphered by referring to Table 4, "Specific Fault Troubleshooting" and Table 5, "SCR Fault Codes". The component may then be repaired or replaced as necessary.

If no faults are encountered, the test will display PASSED. When no faults are encountered the entire test runs in approximately three minutes.

A8 TEST 51 - PROGRAMMER CARD SOFTWARE

Test 51 allows a check of the software **date** of the Programmer microprocessor (PUP).

A9. TEST 52 - PROGRAMMER CARD SOFTWARE REVISION

Test 52 provides a check on the software **level** of the "PUP".

A10. RAM TEST - DISPLAYS RAM LOCATION VALUE

RAM test allows a "one-shot" look at a RAM address (refer to TEST 11 for variables). The value displayed is in decimal numbers. To display a RAM address, press [RAM], the Programmer will display "***RAM***". Then press the keys corresponding to the desired RAM variable. The drive will display the following "XXX YYYY" where "XXX" is the RAM address and "YYYY" is the RAM value. Press [CLEAR/MODE] to clear the display.

All. DACO, DAC8 and DAC12 (DAC8 and DAC12 Require an Optional Application card)

WARNING

THESE OUTPUTS MAY ALREADY BE UTILIZED FOR THE APPLICATION. REFER TO THE DRIVE ELEMENTARIES AND CUSTOM SOFTWARE DESCRIPTION.

Three selectable analog outputs, DACO, DAC8 and DAC12, are available for test purposes and are programmed similar to Test 11. To program DACO, enter the Parameter mode and call up ADR090-DACO. Enter the signal to be monitored from Table 3, "RAM Variable Scaling". Putting that RAM Address in ADR090 will cause that signal to appear at the DACO test point. DAC8 and DAC12 perform in a similar manner, but the signal RAM address is loaded into ADR096-DAC8 or ADR097-DAC12 and the output appears at the DAC8 or DAC12 terminal on 4TB.

A12. STATUS BITS 1, 2 and 3 (Requires Optional Application Card)

WARNING

THESE OUTPUTS MAY ALREADY BE UTILIZED FOR THE APPLICATION. REFER TO THE DRIVE ELEMENTARIES AND CUSTOM SOFTWARE DESCRIPTION.

ADR098-BIT#1, ADR099-BIT#2 and ADR100-BIT#3 are programmed in the same manner as DAC8 and DAC12. They are different in that they are logic outputs rather than analog outputs.

Signals listed in Table 3, Logic Signals, can be used to select addresses to enter in ADR098, ADR099 and ADR100. Variables listed under Table 3, "Analog Signals", may also be used in the bit outputs to indicate the polarity of the signal (positive or negative). These outputs also appear on 4TB as "B01", "B02", and "B03" respectively.

B. OPTIONAL ADVANCED DIAGNOSTIC TESTS

B1. TEST 20 ELECTRONIC WIRE CHECK

WARNING

SIGNAL LEVEL DETECTORS AND BIT OUTPUTS MAY CONTROL EXTERNAL DEVICES. CONSULT THE SYSTEM ELEMENTARIES BEFORE USING THIS FEATURE.

This test allows five different types of wire-checking and output forcing test as follows.

1) Signal Level Detector (SLD) forcing.
2) Status Bit monitoring.
3) Fault Code testing.
4) Logic input monitoring (Application card).
5) Logic input monitoring (Control card).
6) Microapplication card input monitoring (future).

To perform all tests, it is first necessary to be in Diagnostic mode with the EEROM write enabled (write protect the EEROM after this test is completed). Then key in [TEST], [2], [0] , and [ENTER]. This causes Test 20 to be active and the display to show "1.'.00111111". The left-most digit indicates the test level in this case, the'l.'. indicates level 1 or Signal level Detector (SLD) forcing.

To change the type of test, press the [INC] or [DEC] key until the correct test number is displayed in the left-most digit. For example, if Test 20 is called for and the initial display is shown above, press [INC] once to run the next level (20-2), Status Bit monitoring. The programmer display will be similar to "2.'.11111100". Each test will be described below.

NOTE: Test 20-l temporarily changes the **value** of Parameter ADR009-SLDJP. If the AC control power is removed while keys 0 thru 5 are depressed, ADR009-SLDJP may be changed and will not contain the correct value when AC power is reapplied. The value of SLDJP should be noted prior to initiating this test.

TEST 20-1 - SIGNAL LEVEL DETECTOR (SLD) FORCING (Requlres Application card)

This display will show "'1.'.00111111". SLDO is energized if the right most digit is 1, and dropped out if it is a zero. Scanning from right to left, the status of all six SLDs (SLDO through SLD5) can be seen. That is, the second digit from the right is SLD1, the third digit from right is SLD2, etc. To change the state of SLDO, depress [0] key.

NOTE: SLDs can not be toggled while the drive is running or the drive will fault. While the [0] key is depressed SLDO will be maintained in the opposite **state** it was originally. When the [0] key is released, SLDO returns to its original state. Keys 1-5 operate in the same manner for SLD1 thru SLD5.

TEST 20-2 - STATUS BIT OUTPUTS (Requires Application card)

Status bits B01, B02 and B03 (ADR098, ADR099, and ADRIOO respectively), are programmed as previously described in Section A12. If Test 20-2 is called for, B01 will appear as the right most character in the display, B02 will be one to the left of B01 and B03 is one to the left of B02. If Bit#l is "high" the right most character will be "1"; if it is "low", it will be displayed as [O].

TEST 20-3 - FAULT CODE OUTPUTS (Requlres Applicatlon card)

To test remote fault code lines and their effects on external equipment, place the drive in Test 20-3. Next enter the desired fault code by keying in [O]. The programmer then displays "FAULT". Next key in number of fault code desired and press [ENTER]. (NOTE: Fault 63 will cause FC5 to be high).

This test does **not** cause the drive to fault, unless a number greater than 127 is entered. This test is a way of forcing the fault code output lines on the Application card to check interfaces with external equipment using the fault output lines. The Diagnostic LEDs will also display the appropriate fault code. .

TEST 20-4 - LOGIC INPUT MONITORING (Requires Application card)

If Test 20-4 is selected the display will show "'4.'.XXXXXX" where Xs will be either "1" or "0". A "I" will be displayed if a particular input is tied to REF24 and the drive is programmed for **negative** logic (JP19 and JP20 on the Control card are in 1-2 and 3-4 position respectively and JP5 and JP6 on the Application card are in the 1-2 and 3-4 position respectively). From right to left the digits are MODE 0 thru MODE 7.

TEST 20-5 LOGIC INPUT MONITORING (Control card)

Test 20-5 is similar to Test 204 except that the left four digits do not apply. Starting at the first digit from the right, the **status** of four logic inputs will be displayed as

"5.XXXXDCBA".

A = Status of RUN command inverted ("0" indicates RUN is initiated).

B = Status of JOG command inverted ("0" indicates JOG is initiated).

C = Status of POL command inverted ("0" indicates POL is initiated).

D = Status of XSTOP command ("1" indicates XSTOP is initiated).

TEST 20-6 MICRO-APPLICATION CARD INPUT MONITORING (Micro-application card requlred)

Test 20-6 is reserved for future use.

B2. TEST 21 - DIAGNOSTIC VARIABLE PAIRS

NOTE: Test 21 will cause ADR122-DIAG1 and ADR123-DIAG2 to change.

Test 21 performs in a similar manner to Test 11 discussed in Section A6 except that several pairs of RAM addresses are already pi-e-programmed in memory. Test 21 is entered from the Diagnostic mode by keying in [TEST], [2], [1], and [ENTER]. The programmer will then display "TST21". Next key in [1], [3], or [1] [0] (corresponding to Tests 21-1, 21-3 or 21-10 respectively), and [ENTER], to display the pairs of RAM addresses shown in the table below.

As in Test 11, the display will show "DGXXXX.XXXX". The left and right four characters are as selected from the listing. Test 21-3 will display the variables programmed into ADR096-DAC8 and ADR097-DAC12.

If [1], [0] is entered, calling for Test 21-10, then the display will prompt with "ADDR1". At this time, key in the first RAM address you wish to monitor and press enter (typical RAM addresses are listed in Table 3, "RAM VARIABLES'). The display will then prompt with "ADDR2". Key in the second RAM address and press [ENTER]. The display will now be "DGXXXX.XXXX", where the left four digits are the value of the signal address stored in ADDR1, and the right four digits are the value of the signal address stored in ADDR2.

B3. TEST 22 - DIAGNOSTIC VARIABLES THROUGH DAC OUTPUTS (Requires Application Card)

Test 22 operates similarly to Test 21 above, with the addition that the various signal pairs areautomatically sent to the DAC8 and DAC12 terminals on 4TB. Calling for Test 22-3 causes +5V to appear at the DAC8 and DAC12 terminals in order to quickly check the integrity of DAC8 and DAC12.

NOTE: Test 22 temporarily changes the value of ADR096-DAC8, and ADR097-DAC12. If AC control power is removed while conducting Test 22 then DAC8 and DAC12 will not contain the original value when power is reapplied. The value of DAC8 and DAC12 should be noted prior to conducting this test.

B4. TEST 25 - SQUARE WAVE TEST REFERENCE

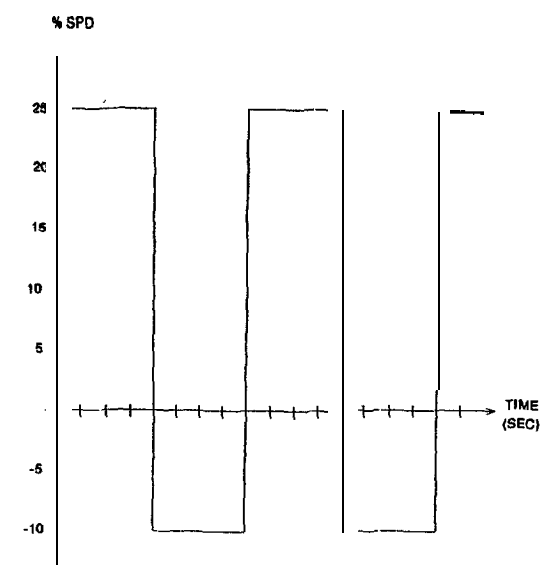
Test 25 causes the drive to run and follow an internal drive square wave reference set up by the values in ADR092-TSTR1, ADR093-TSTR2, and ADR094-TSTT1. TSTR1 and TSTR2 are run reference levels and TSTT1 is the time duration of each half cycle. Run references are 0 to 127 with 100 equal to 100% speed with positive reference. Negative reference is accomplished with values from 128 to 255 with the magnitude equal to 256 minus the setting (for example, 156 corresponds to -100%). The time period is set with TSTT1 and will have a 50% duty cycle for powers of 2 (1, 2, 4, 8, 16 etc.). The cycle time will be 262 x TSTT1 milliseconds. That is, with TSTT1 set to 8, the test will cycle between TSTR1 and TSTR2 with approximately 1 sec for each reference. Pressing [STOP] will end the test.

NOTE: Test 25 temporarily changes the value of ADR003-DGNJP. If AC control power is removed while conducting this test then DGNJP will not contain the correct value when AC power is reapplied. The value of DGNJP should be noted prior to conducting this test.

Example: Program the square wave reference to run 25% of top speed forward for approximately 4 seconds and 10% reverse for approximately 4 seconds.

ADR092-TSTR1 = 25 (25% speed)
ADR093-TSTR2 = 246 (-10% speed)
ADR094-TSTT1 = 4000 ms/262 ≈ 16

The reference will look like the sketch below:



RAM VARIABLE PAIRS FOR TEST 21 AND 22

TEST NUMBER	LEFT 4 DIGITS (RAM-Addr)		RIGHT 4 DIGITS (RAM-Addr)	
1	Speed Reference	(065)	Speed feedback	(071)
3	DAC8	(ADR096)	DAC12	(ADR097)
10	Diag Var. #1	(ADR122)	Diag. Var. #2	(ADR123)

VI. DIAGNOSTIC MODE FOR DRIVE SETUP

NOTE: Step A of the following procedure utilizes Diagnostic Test 21-10. All other steps utilize the DVM mode. These procedures allow accurate tuning of all analog pots as listed below:

Control card	OFFSET	P4
Control card	REF SCALE	P1
Process Interface card	SFB ZERO	P7
Process Interface card	PF OFF	P5
Process Interface card	PF GAIN	P4
Application card	ASR SCALE	P1

A. Place the drive in DIAGNOSTIC mode and key in [TEST], [2], [1], [ENTER], then [1], [0] and [ENTER]. The drive will display "ADDR1". Key in [4], [9], [4] and [ENTER]. The drive will display "ADDR2". Key in [0], [7], [1] and [ENTER]. The drive will display "DG.XXXX.XXXX". Use the left -four characters and P4-OFFSET on the Control card to adjust the drive offset to zero. Next, using the right four characters and P7-SFB ZERO, on the Process Interface card, adjust speed feedback for zero. Press [TEST] to clear Test 21.

B. Place the drive in DVM mode by pressing [DVM].

C. Connect the DVM test probe from TP DVM on Process Interface card (PI card) to TP SRS on the Control card. Set the input at MSR on 3TB65 to its maximum value and adjust Pl-REF SCALE on the Control card, for 4.0 VDC at TP SRS on the Control card. (Note: If 4.0 VDC cannot be obtained at SRS, then check JP1 on the Control card for proper position. If MSRs maximum voltage is 5-8 VDC, then position 2-3 is correct.) A similar procedure can be used to set up ASR on the Application card using TP ASR and Pl-ASR SCALE.

D. Calculate the tachometer voltage at top speed from the tachometer nameplate and motor top speed. Refer to Table 1 for Process Interface card SW1 setting, and determine which switch positions should be closed for desired drive top speed (SW1, position 7 should be **open** for DC tachs, and **closed** for AC tachs). Start the drive and run at half speed (2.0 VDC at TP SRS on Control card if running in manual mode or 2.0 volts at TP ASR on Application card if running in auto mode). Monitor motor speed, either with a direct reading mechanical tach or with a voltmeter between 1TB-1 and 1TB-3 on the Process Interface card. Adjust Pl-TACH SCALE on the Process Interface card for 50% of motor speed.

Run the drive at top speed and recheck the motor speed. Readjust Pl-TACH SCALE, as necessary for the correct top speed. TP SFB on the PI card should now read 4.0 VDC and be equal to TP SRS voltage (TP ASR if running in auto mode). This completes the calibration of the speed feedback circuit and setup of the drive's analog pots.

E. If the Process Follower option is used proceed as follows.

To check the setup of the Process Follower, verify that JP2 and JP3 on the PI card are set as required according to the following table:

JP2	JP3	Follower Mode
1-2	1-2	5-65 MA
1-2	2-3	5-50 MA
2-3	1-2	4-20 MA
2-3	2-3	1-5 MA or 0-10V

With the drive in DIAGNOSTIC mode, turn on the DVM (press [DVM]). Connect the test probe from TP DVM on the PI card to TP PFRF also on the PI card. Adjust P5-PF OFF, on the PI card for zero volts on the DVM. Then bring the process follower input to its maximum value and adjust P4-PF GAIN, on the PI card for 7.5 VDC on the DVM.

NOTE: It is not necessary to run the drive to calibrate the process follower, only to vary its input as described above.

MAINTENANCE AND REPAIR

Periodic Maintenance and Inspection

As with all electrical equipment, periodic inspection and maintenance of the AC-500 controller will help to assure maximum performance and reliability. The following inspection and maintenance procedures should be performed on a regular schedule as indicated or more frequently under severe service or environmental conditions.

WARNING

DANGER OF ELECTRICAL SHOCK EXISTS. ELECTRICAL SHOCK CAN CAUSEPERSONAL INJURY OR LOSS OF LIFE. PRIOR TO PERFORMING ANY MAINTENANCE, VERIFY THAT POWER TO THE DRIVE IS SWITCHED OFF AND DISCONNECTED. IT IS RECOMMENDED THAT THE POWER DISCONNECTS BE LOCKED IN THE "OFF" POSITION AND RED TAGGED DURING MAINTENANCE. ALSO, VERIFY THAT THE CAPACITORS ARE DISCHARGED AND EXTERNAL CONTROL CIRCUITS DE-ENERGIZED.

Mechanical Maintenance

Adequate cooling is essential to the proper operation of the AC-500 controller. A build up of dust, dirt, or oil residue can cause the controller to overheat. In addition, corrosive or conductive material and excessive moisture may lead to circuit, connection or component failures.

Visually inspect the power unit and control circuits regularly, performing the following maintenance every six months or as required to assure proper cooling and prevent breakdown of electrical circuits due to contamination or loss of adequate clearances.

AIR FILTERS - Replace air filters every six months or more often as required.

BLOWERS/FANS - Check that rotation and speed of the cooling blowers or fans are correct, providing proper air flow. Inspect impellers for build up of dirt or dust. Special care should be taken to clean all heatsink cooling surfaces and fins.

CIRCUIT CARDS - Clean surfaces of the cards with a soft antistatic brush, taking care not to bend or dislocate components. DO NOT use compressed air on circuit cards.

Environmental conditions may occasionally require that the power unit be disassembled to allow thorough cleaning and removal of corrosive contaminates. Care should be taken not to scratch or otherwise damage mating surfaces of SCRs, diodes or other electrical components. Re-assembly should be according to the instructions provided in the "Parts Repair and Replacement" section of this manual.

Solvents should be used sparingly and only when necessary. Care should be taken to assure that the selected solvents are suitable for the materials to be cleaned. Solvents such as Freon TE or TF should be used for SCR or diode surfaces.

It is recommended that shop compressed air NOT be used for "blowing out" dust or dirt from the controller. High pressures and airborne oils may cause damage to equipment and risk to personnel. Should it be necessary to "blow-out" the equipment, a source of clean, dry air at acontrolled low pressure should be provided specifically for this purpose. Proper safety precautions, including the use of goggles, should be taken.

Electrical Maintenance

Electrical maintenance consists of periodic visual and physical inspection of the power unit. A visual inspection should be made for signs of arcing, overheating (indicated by component discoloration), corrosive contamination and SCR condition. All signal stab-on connections should be checked by pulling on them with finger pressure to insure a good connection.

As an inspection aid, the "Reliability Inspection" checklist on the following pages may be used as a guideline when performing scheduled maintenance.

RELIABILITY INSPECTION

DATE: _____

I "POWER OFF CHECKS"

A. CONVERTER SECTION

— Power Connections at P1 and N1 module points and L1, L2, L3 bus bars.

— Snubber connections at the L1, L2, L3 bus bars

— Stab-on connections on the pulse transformer card.

— Verify physcial integrity of the line to line snubber capacitors and resistors on the Converter and Commutation cards.

— If the Converter SCR module assembly is removed or disconnected a megger reading of greater than 100K at 1000 VDC (460V drive) is expected across each SCR. If the assembly is not removed, suspect volt-ohmmeter readings of less than 50K should be meggered.

1SP	_____	ohms	1SN	_____	ohms
2SP	_____	ohms	2SN	_____	ohms
3SP	_____	ohms	3SN	_____	ohms

— Physical inspection of each SCR gate lead connector — for signs of looseness or corrosion.

B. COMMUTATION SECTION

— Inspect power connections at top and bottom of module and the MCA, MCB, MCC bus bars. (Also heatsink attatchment points 250-500 KVA).

— Check tightness of heatsink snubber connections with moderate finger pressure applied. (250-500 KVA).

— Snubber card and capacitor connections.

— Physical condition of snubber card resistors - verify solder connections and resistance value (approximately 10 ohms) of each resistor.

— Physical inspection of each SCR gate lead connection for signs of looseness or corrosion.

— If the SCR module assembly is removed or disconnected a megger reading of greater than 100K at 1000 VDC (460V drive) is expected across each SCR. If the assembly is not removed, suspect volt-ohmeter readings of less than 50K should be meggered.

CSPA	_____	ohms	CSNA	_____	ohms
CSPB	_____	ohms	CSNB	_____	ohms
CSPC	_____	ohms	CSNC	_____	ohm

C. INVERTER SCR AND DIODE SECTION

— Inspect power connections at top and bottom of modules and the T1, T2, and T3 bus bars (also heatsink attachment points on the 500 KVA package)

— Check tightness of heatsink snubber connections with moderate finger pressure applied (300-500 KVA)

— Snubber card and capacitor connections.

— Physical condition of snubber card resistors - verify solder connections and resistance (approximately 10 ohms) of each resistor.

— Physical inspection of each SCR gate lead connection for signs of looseness or corrosion.

C. INVERTER SCR AND DIODE SECTION (continued)

— Physical inspection of each SCR gate lead connection for signs of looseness or corrosion.

— If the SCR module assembly is removed or disconnected a megger reading of greater than 100K at 1000 VDC (460V drives) is expected across each SCR (reverse diode direction). If the assembly is not removed, a low volt-ohmeter reading is expected in the forward diode direction and a high reading in the reverse diode direction is required.

POSITIVE MAIN SCRS AND DIODES

ISPA	_____ ohms		DPA forward	_____ ohms
ISPB	_____ ohms		DPB forward	_____ ohms
ISPC	_____ ohms		DPC forward	_____ ohms

NEGATIVE MAIN SCRS AND DIODES

ISNA	_____ ohms	DNA	_____ ohms
ISNB	_____ ohms	DNB	_____ ohms
ISNC	_____ ohms	DNC	_____ ohms

D. COMMUTATION CAPACITORS

— Oil Leakage - note that slight oil film around terminals is normal.

— Connections for physical integrity - do not overtighten.

— Case to ground resistance of less than 5 ohms.

E. REACT0 RS

— Physical appearance - no signs of overheating or corrosion.

— Power connection integrity.

F. COMMUTATION POWER SUPPLY

— Inspect filter capacitors physcial appearance (swelling, discoloration) and vent plug (vent plugs turning white and/or pushed out of capacitor).

— Inspect all connections at capacitors, resistors and SCRs.

RESISTOR VALUES

200 KVA	500 KVA
R3 _____ (300 ohms nominal)	CPSR _____ (10 ohms nominal)
R4 _____ (4 ohms nominal)	

G. FRAME SPECIFIC CHECKS

200 KVA

DCD diode

— Power connections

— Ohmmeter readings - the diodes are forward biased from the outside terminals to the middle terminal.

500 KVA

— Auxiliary SCRs: Integrity of snubber and auxiliary gate lead connections.

— All connections on the Commutation Power Supply card.

— Verify Commutation Power Supply card component solder connections and snubber resistor ohm readings.

— If the SCR module is removed or disconnected, a megger reading ofgreater that 100K is expected. If the assembly is not removed, then a volt-ohmeter reading of less than 50K ohms is suspect, and a megger test IS required.

H. LINK CAPACITOR BANK

— Capacitor appearance - check for signs of swelling, leaking or the vent plug being discolored or pushed out.

— All capacitor connections.

— Check ohmmeter readings across the positive and negative banks, (should be balanced).

— Any suspect capacitor leakage current can be measured by applying rated DC working voltage (marked on the side of the capacitors) for 3 minutes. The resultant current should be less than 6mA/per capacitor.

I. REGULATOR CARD RACK

— Visually check all ribbon cables to insure that they are seated properly in their sockets.

— Visually check the connectors for physical damage such as bare wire or wires pulling out of the connectors.

— Check solder joints of card mounted connector ends.

— Clean as needed.

— Check common wire termination at the Power Supply/Interface card and the Control card ends.

— Check all hardware jumpers per the Custom Instruction Book List.

II. POWER ON CHECKS

— Verify commutation power supply voltage at standby (CP to CN). It should be greater than 500 VDC.

— Perform SCR (Test 12) in the diagnostic mode.

— Run the drive at mimimum speed, mid speed and full speed and measure commutation capacitor AC voltage balance at the capacitors and the Inverter Firing Generator card feedback. Record below.

		MIN SPD	MID SPD	MAX SPD
Capacitor	CCA	_____VRMS	_____VRMS	_____VRMS
Voltages	CCB	_____VRMS	_____VRMS	_____VRMS
500 to 800 VRMS	ccc	_____VRMS	_____VRMS	_____VRMS
Feedback	ACV	_____VRMS	_____VRMS	_____VRMS
Voltages	BCV	_____VRMS	_____VRMS	_____VRMS
2 to 3 VRMS	ccv	_____VRMS	_____VRMS	_____VRMS

— Verify all adjustments for speed and loads required including references and metering output.

PARTS REPAIR AND REPLACEMENT

WARNING

ELECTRICAL SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. WHEN POWER OFF MAINTENANCE IS BEING PERFORMED, VERIFY ALL POWER TO THE DRIVE IS SWITCHED OFF OR DISCONNECTED. IT IS RECOMMENDED THAT THE POWER SWITCHES BE RED-TAGGED DURING POWER OFF MAINTENANCE. VERIFY THAT THE CAPACITORS ARE DISCHARGED.

POWER SECTION REPAIR

The removal, repair and replacement instructions vary depending on the type of power section and its KVA rating. Refer to the instructions which follow under specific heading which applies to your drive.

200KVA POWER SECTION (Figure 5.1)

The 200 KVA power section is contained on one heatsink assembly located behind the Inverter and Converter Pulse Transformer card holders. The power section consists of dual cell modules containing sets of SCRs and/or diodes in various configurations. (Refer to the Power Circuits and Connection Diagrams, Figures 1.4 and 1.9 for cell names.) To access the power section, loosen the thumb screws on either of the pulse transformer card holders and swing out. The hinges are designed to allow pivoting the holder for access to all the cell connections.

Tools and materials required:

- Table
- Torque Wrench (80 in-lb)
- Socket - 13mm and 7/16" for 3/8" ratchet
  - 5/16" for 1/4" ratchet
- Ratchet - 3/8" and 1/4"
- 14" Ratchet Extension, (3/8" and 1/4")
- GE GL322 Versilube® Plus
- Lint-free cloth

MODULE REPLACEMENT

1. Determine which module has failed and open the appropriate card holder (the commutation and positive inverting modules are behind the Inverter Pulse Transformer card and the converter and negative inverting- modules are behind' Converter Pulse Transformer card).

2. Locate the module to be replaced and disconnect the gate leads, taking care not to bend the connectors and verifying that the leads are marked.

3. Remove the 13mm bolts and lock washers connecting the bus bar assembly(s) and snubber wires (where used) to the package (and 1/4" bolts and keynuts on bus bars where applicable). Brace the bus bar so that no stress is transmitted to any of the packages connected to the same pieces of bus bar.

4. Remove the four slotted hex head (5/16") screws mounting the package to the heatsink assembly and remove the defective module.

5. Clean and lubricate the heatsink before reassembling. Place a small amount of GE GL322 Versilube® Plus (Part No. 104X214AB059) in the center of the heatsink mounting surface. Polish the interface area until clean, using a combination of back and forth, sideways and circular motions with a lint-free cloth or tissue. Finally, place a small amount of Versilube Plus on the mating surfaces of the heatsinkand use a lint-free cloth or tissue to spread evenly to make a thin coating for good conduction.

6. Clean the module mating surface with GL322 Versilube Plus. Apply a thin film to the mating surface and wipe off any excess.

7. Mount the cell on the heatsink with the four 5/16" slotted hex head and snubber wires (where applicable). If a torque wrench is available, tighten to base torque specifications shown in the following table.

8. Reconnect the bus bar(s) and snubber wires (where applicable) with the 13mm bolts and washers to finger tight and re-install 1/4" bolts and keynuts to bus bar (where applicable) and tighten.

9. Support the bus bar(s) and tighten the bolts by turning each a 1/4 turn alternately, until hand tight. If a torque wrench is available, tighten to the torque specifications listed in the following table.

200KVA FRAME TORQUE RATINGS

Module Description	Catalog Number	Torque Specification
Converter SCR Module	104X125DC021	40 to 53 in-lb — base (to heatsink) 70 in-lb • top connections 60 in-lb • tongue connection
Inverter Module	104X125DC029	35 to 55 in-lb • base (to heatsink) 70 in-lb • top connections 60 in-lb • tongue connection
Commutation SCR Module	104X125DC025	40-53 in-lb • base (to heatsink) 70 in-lb • top connections 60 in-lb • tongue connection
Bypass Diode Module	104X125DC030	35-50 in-lb • base (to heatsink) 26 in-lb • top connections

10. Close card holder(s) and secure thumb screw(s).

11. Reconnect the module gate leads taking care not to bend or spread the connectors. Verify that all connections are tight before applying power.

500KVA POWER SECTION (Figures 5.2 end 5.3)

The 500KVA power section consists of four modules as shown in Figure 5.2: Converter Module, Commutation Power Supply Module and two Inverter Modules (positive and negative) The module removal-and cell replacement procedures are similar between all modules. Differences will be noted where applicable

WARNING

USE CAUTION TO AVOID DROPPING HARDWARE INTO THE POWER SECTION. PLACE A PIECE OF CARDBOARD UNDER THE INVERTER STACK TO CATCH ANY LOOSE HARDWARE. IF ANY HARDWARE IS DROPPED INTO THE POWER SECTION. IT MUST BE LOCATED AND REMOVED BEFORE REASSEMBLING AND APPLYING POWER,

Tools and materials required:
- Table
- Torque Wrench (80 in-lb)
- Socket • 516" for 3/8" ratchet
- Deep Socket • 7/16" and 9/16" for 3/8" ratchet
- Ratchet • 3/8"
- 14" Ratchet Extension
- Phillips head screwdriver
- GE. GL322 Versilube® Plus
- Lint-free cloth

INVERTER STACK REPLACEMENT

1. Open all electrical circuits to the case in which the inverter stack is located.

2. Check voltage across capacitor tray (P2 to N2) with a DC voltmeter. The capacitor discharge resistor should have reduced this voltage to 10 volts or below before work starts in the case.

3. If replacing the positive stack, disconnect NFB, NPT, the negative module gate pulse connectors (INA, INB and INC) and remote thermal switch wires (THSW3 • 4) from Inverter Pulse Transformer card mounted on the positive module. If replacing the negative module, disconnect INA, INB and INC from the Inverter Pulse Transformer card mounted on the positive module.

4. Label and disconnect three pairs of wires to the snubber cards at the left side of the module.

5. On the negative module only, remove six 9/16" bolts and washers holding the motor buses (T1, T2 andT3) in place and remove bus bars.

6. Remove three 9/16" captive nut-washer assemblies from the right side of the module connecting the module output bus bars to the outgoing motor bus bars (T1, T2 and T3).

7. Remove the three 7/16" nuts and washers from the inside top edge of the heatsink (right side of positive stack, left side of negative stack). Bend the #4 wires (DNA, DNB and DNC or DPA, DPB and DPC) away from the stack until they are clear of the retaining bolts.

8. Remove six 5/16" nuts and washers securing the center ribbon cable track and cover and move away for next step. (It may be necessary to disconnect NFB and NPT from the Commutation and Converter module to move the cable track).

NOTE: Support the stack assembly as the next step removes the supportive hardware to the stack.

9. Starting at the bottom of the stack, remove the six 9/16" captive nut-washer assemblies (three on bottom, three on top). Use caution when removing the nuts so that hardware does not fall into the stack or blower assembly.

10. Remove stack and place on table with pulse tranformer card or insulator facing up. The stack can be repaired on the table (see Press-Pak cell replacement).

NOTE: If both P2 and N2 stacks are removed, the interconnecting buss bars will be loose. In this case, remove the N2 stack first and reinstall the 9/16" nuts on the right side of the interconnecting bus bars to retain them while removing the P2 stack. Reverse the procedure during reassembly.

11. Install the repaired module or a spare module by following the above proceure in reverse. Verify that ail hardware is tight and all plugs are reconnected before re-applying power.

CONVERTER STACK REPLACEMENT

1. Open all electrical circuits to the case in which the converter stack is located.

2. C-heck voltage across capacitor tray (P2 to N2) with a DC voltmeter. The capacitor discharge resistor should have reduced this voltage to 10 volts or below before work starts in the case.

WARNING

USE CAUTION TO AVOID DROPPING HARDWARE INTO THE POWER SECTION. PLACE A PIECE OF CARDBOARD OVER THE STACKS TO CATCH ANY LOOSE HARDWARE. IF ANY HARDWARE IS DROPPED INTO THE POWER SECTION, IT MUST BE LOCATED AND REMOVED BEFORE REASSEMBLING AND APPLYING POWER.

3. Remove three pairs of 7/16" captive nut-washers connecting the N1 bus strips to the left side of the Converter module. Bend back strips slightly to give clearance when removing module.

4. Disconnect NFB, NPT, L1, L2, L3, CPSR, LCPR1, LCNR1, LCPR2, LCNR2 and remote thermal switch wires from the Converter Pulse Transformer card.

5. Remove three 9/16" captive nut-washer assemblies connecting the AC line bus bars (LI, L2 and L3) to the bus bars on the right hand side of the Converter module.

6. Remove six 5/16" nuts and washers securing the center ribbon cable track and cover and move away for the next step. (It may be necessary to disconnect NFB and NPT from the Commutation Module to move the track).

NOTE Support the stack assembly as the next step removes the supportive hardware to the stack.

7. Starting at the bottom of the module, remove the six 9/16" captive nut-washer assemblies (three on the bottom, three on top). Use caution when removing the nuts so that hardware does not fall into the modules or blower assembly.

8. Remove the module and place on a table with the Pulse Transformer card facing up. The module can be repaired on the table (see Press-Pak Cell Replacement).

9. Install the repaired module or a spare module by following the above procedure in reverse. Use caution to insure that there is clearance between the N1 bus bars and L1, L2, L3, MCA, MCB, and MCC bus bars when reconnecting. Verify that all hardware is tight and all plugs are reconnected before re-applying power.

COMMUTATION STACK REPLACEMENT

1. Open all electrical circuits to the case in which the commutation stack *is* located.

2. Check voltage across capacitor tray (P2 to N2) with a DC voltmeter . The capacitor discharge resistor should have reduced this voltage to 10 volts or below before work starts in the case.

WARNING

USE CAUTION TO AVOID DROPPING HARDWARE INTO THE POWER SECTION. PLACE A PIECE OF CARDBOARD OVER THE STACKS TO CATCH ANY LOOSE HARDWARE. IF ANY HARDWARE IS DROPPED INTO THE POWER SECTION. IT MUST BE LOCATED, AND REMOVED BEFORE REASSEMBLING AND APPLYING POWER.

3. Remove three 7/16" nuts and washers at the right side of the stack connecting the LCN wires to the stack. Bend the wires away from the stack for clearance.

4. Disconnect NFB, NPT, CCAR, CCAW, CCBR, CCBW, CCCR, CCCW, and remote thermal switch wires from the Commutation Power Supply Pulse Transformer card.

5. Label and disconnect six pairs of wires to the snubber cards at the left side of the stack.

6. Remove the three 9/16" captive nut-washer assemblies connecting to the MCA, MCB and MCC bus bars on the right side of the stack.

7. Remove six 5/16" nuts and washers securing the center ribbon cable track and cover and move away for next step. (It may be necessary to disconnect NFB and NPT from the converter and inverter stacks to move the track).

NOTE: Support the stack assembly as the next step removes the supportive hardware to the stack.

8. Starting at the bottom of the stack, remove the two 9/16" nuts and washers that retain the LCP wires and bend the LCP wires clear of the retaining studs. Remove the four captive nut-washer assemblies from the stack (one on the bottom and three on the top). Use caution when removing the nuts so that the hardware does not fall into the module or blower assembly.

9. Remove the stack and place it on a table with the Pulse Transformer card facing up. The stack can be repaired on the table (see Press-Pak Cell Replacement).

10. Install the repaired stack or a spare stack by following the above procedure in reverse. Verify that all hardware is tight and all plugs are re-connected before re-applying power.

PRESS-PAK CELL REPLACEMENT

1. Tools required:
   Deep Socket - 7/16" and 9/16" for 3/8" ratchet
   Ratchet - 3/8"
   Nut Driver - 7/16"
   Torque Wrench - 3/8"

2. Disconnect all gate plugs and stab-on connectors from the gate pulse transformer card (not required on the negative Inverter module).

3. Remove six 7/16" nylon nuts with nut driver. Remove gate pulse transformer card.

4. Remove six more 7/16" nylon nuts with nut driver. Remove Lexan insulator.

5. Determine the location of the cell to be replaced from Figure 5.3 and proceed through the following steps working on that portion of the heatsink assembly.

6. Inverter modules only - Remove the two 7/16" captive nut-washer assemblies, connecting the tie bus straps to the appropriate front heatsink (the card side of the module is the front).

NOTE: Do not remove the third nut connecting the commoning support bus bar to the heatsinks in the case.

7. All modules - Loosen the two 9/16" clamp nuts, rotating 1/4 turn each, alternating nuts, until finger loose. Hold the clamp bar while removing nuts and convex washers completely. Remove the top clamp bracket.

8. Remove the front heatsink taking care not to damage the clamp threads or pull the twisted pair(s) of voltage feedback wires attached to the heatsink (not applicable to the Converter module). Stand the front heatsink on end and support to allow replacing the cell.

9. If replacing the front cell (now on top) simply lift straight up, being careful not to damage any wiring or the alignment pin in the bus between the cells.

10. If replacing the back cell (now on the bottom) remove the front cell as described above then carefully lift up the bus bar and set aside, then remove the cell. On Inverter modules, lift up on the bus bar to give the alignment pin clearance from the cell and slide out of the reactor. Remove the cell.

NOTE: It is recommended that the following cleaning procedure be performed on both cells, even if only one is being replaced.

11. Clean and lubricate the heatsinks and bus bar before assembling. Place a small amount of GL322 Versilube Plus (Part No. 104X214BA059) in the center of the mounting surfaces. Polish the interface areas until clean, using a combination of back and forth, sideways and circular motions with a lint-free cloth or tissue. Finally, place a small amount of Versilube on the mating surfaces of the heatsink and use a lint-free cloth or tissue to spread the grease evenly to make a thin coating for good conduction.

12. Clean the mating surfaces of the cells with Versilube Plus. Apply a thin film of Versilube Plus to the mating surfaces and wipe off any excess.

NOTE: Before re-assembling the module, run the 9/16" nuts down on the heatsink clamp to verify free movement and that no burrs are present.

13. Orient the bottom cell to coincide with similarly installed cells in the module and place the cell centering hole over the roll pin in the heatsink surface.

14. Place the bus bar on the cell (the bus bar slides through the center of the choke on Inverter modules) with the roll pin in the cell centering hole.

15. Orient the top cell as originally installed, align it over the roll pin and place on the bus bar.

16. Raise the U-clamp with one hand and replace the front heatsink piece, being careful not to damage the clamp threads. Replace the clamp bracket, aligning the centering pin in the heatsink hole.

17. Reinstall the washer with the convex side down into the clamp. Replace the 9/16" nuts, turning alternately until finger tight and taking care to keep the clamp bracket level by maintaining the same number of threads above the 9/16" nuts.

18. Visually check the front heatsink to verify that it is properly aligned with the adjacent heatsinks and square with the bottom heatsink.

    Align the bus bar between the cells to an adjacent bus bar by measuring and adjusting to 6-1/2" between centers.

CAUTION

TO AVOID OVERTIGHTENING, APPROACH THE MARK WITH SMALLER INCREMENTS DURING THE LAST 1/4 TURN. VERIFY ACTUAL PACKAGE SIZE — IF THE CONTROLLER HAS BEEN DERATED ACTUAL STACK ASSEMBLY TORQUE RATINGS MAY BE FOR NEXT HIGHER PACKAGE.

19. Using a torque wrench set to 20 in-lb, alternately tighten the nuts 1/8 of a turn at a time, until 20 in-lb is-reached on both. After removing previous marks on the studs and nuts, mark the studs and nuts at 0° with a fine marker. Alternately tighten the nuts in 1/4 turn increments until the proper number of turns per the table below.

CELL	PART NUMBER	TURNS PASS 20 in-lb
Converter		
250-300 KVA	36A359102AAG04	5/8
350-500 KVA	36A359102AAG01	1 1/2
Inverter	36A359102AAG05/G03	1
Commutation	36A359102AAG02	3/4

20. If working on an Inverter module, reconnect the tie bus straps to the side studs on the module. Verify that the straps do not overlap and tighten the 7/16" nuts until snug.

21. Place the Lexan insulator on top of the stack to verify alignment of the studs (if necessary, loosen the 1/4" jam nuts to allow alignment).

22. Replace the Lexan insulating barrier and secure with six nylon nuts, hand-tightening until snug. Replace and secure the pulse transformer card (not applicable to the negative inverter stack) in the same way.

23. Reconnect all gate leads and stab-on connectors removed in step 2, being careful not to bend any of the connections. Verify that all connections are tight.

CARD REPLACEMENT PROCEDURE

CAUTION

TREAT ALL CARDS WITH STATIC SENSITIVE HANDLING TECHNIQUES. USE A "GROUNDING" STRAP WHEN CHANGING CARDS AND STORE CARDS IN THE ANTI-STATIC BAGS THEY ARE SHIPPED IN.

1. Carefully disconnect all cables.

2. Regulator cards - release all hold-down tabs and remove board.
   Pulse transformer, MOV Suppression and Commutation Power Supply cards - remove nylon mounting nuts and remove card.

3. Set all jumpers on the replacement board exactly as the jumpers on the failed board.

4. Set all potentiometers visually the same as on the original card.

5. If the Control or Microapplication card is changed, remove the EEROM from the failed card and install in the new card. If the EEROM from the failed card is suspected to be damaged, it will be necessary to reprogram the new EEROM to the custom parameter list. Use appropriate chip removal tools and techniques for static sensative devices.

6. Regulator card section - install the new board, verifying that all tabs snap tightly into position.
   All other cards - tighten nylon mounting nuts till hand tight. **DO NOT OVERTIGHTEN.**

7. Reconnect all cables, verifying that they are properly seated at both ends.

8. Fine tune all pot adjustments as required per instructions in Table 1 of the Adjustments section and the Start Up Section. Note any special engineering adjustment procedures indicated in the control elementaries.

SPARE AND RENEWAL PARTS

Renewal parts information for this equipment is included in the **CUSTOM INSTRUCTION BOOK** provided for the drive. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated in the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting the factory with the Requisition and Model numbers for the controller (found on the nameplate located on front of the controller).

It is recommended that the parts list be reviewed and a set of spare parts be maintained to minimize down time should repairs be required. Renewal parts may be ordered by contacting your local **GE Sales Office** or an **Authorized GE Parts Distributor in your area.**

WARRANTY parts replacements may be ordered direct from the factory by calling:

GE DRIVES DEPARTMENT
PRODUCT SERVICE
(814) 875-3851

PART NUMBER IDENTIFICATION

The parts listed are divided into four catagories, each with its own part number structure:

Order-Specific Assemblies
Common Assemblies
Component Parts
Circuit Cards

Order-Specific Assemblies are generally major assemblies consisting of a group of common assemblies that make up the specific controller provided. Other items which are obtained specifically for the order may also be provided with a similar part number structure.

Example:

7VXYZ999CI01

W-assembly provided by GE DRIVES DEPARTMENT
XYZ999 — identification number (original order)
CI-assembly type (CI = Core Inverter)
01- unit identification number

Common Assemblies are subassemblies, consisting of a group of component parts, which are in common use in **GE DRIVES** products and are not specific to a particular order. The part number consists of the number "36" followed by an alpha character and may be up to 14 characters in total length.

Example:

36C774505AAG02 is the part number for a 500KVA Commutation SCR Stack Assembly for an AC-500 controller.

Component Parts are the basic components used in the assemblies and generally represent the lowest discrete level. Component part numbers consist of a combination of alphanumeric characters, defining the class of component and the specific item.

Example:

104X109AD038

104X- component part
109 — component type (fuse)
AD038 — specific item (500 V, 3.2A FNQ)

Circuit Cards are identified by a catalog number giving the product class, card function and revision levels. It should be noted that some cards contain software which is designated by the suffix "Mn" where "n" is the software number. These cards are also available less software with suffix "Gn" designation. The factory may substitute later versions of cards based on availability and design enhancements.

531X140CCHALM2

531X- card for GE DRIVES DEPARTMENT microprocessor based controller
140CCH — Control card for AC-500 controller
ALM2 -revision level, including software

As an aid in expediting card identification, a partial listing of the cards, functions and associated root part numbers follows:

PRINTED CIRCUIT CARDS — ALL KVA RATINGS

DESCRIPTION	
Control card	531X140CCH—
Power Supply/Interface card	531X113PSF—
Inverter Firing Generator card	531X182IFG—
Terminal Board card (3TB)	531X170TBS—

PRINTED CIRCUIT CARDS - up to and Including 200KVA ratings

Inverter Pulse Transformer card	531X183XPT—
Converter Pulse Transformer card	531X195CPT—
Snubber card (w/short capacitor bracket) - Qty 13/controller	531X190SRC—

PRINTED CIRCUIT CARDS — up to and Including 500 KVA ratings

Converter Pulse Transformer card	531X185CPT—
Inverter Pulse Transformer card	531X184IPT—
Commutation Pulse Transformer card	531X199SPT—
Commutation Power Supply card	531X186CPS—
Snubber card (w/no capacitor bracket) - Qty 10/controller	531X190SRC—
Snubber card (w/long capacitor bracket) - Qty. S/controller	531X190SRC—

PRINTED CIRCUIT CARD (OPTIONAL) — all ratings

Programmer card	531X135PRG—
Application card	531X139APM—
Terminal Board Relay card (4TB)	531X171TMA—
Process Interface card	531X133PRU—
MOV card	531X128HMS—

RIBBON CABLES

Circuit interconnection is facilitated by the use of multi-conductor ribbon cables. Replacement cables may be ordered by identifying the cable plug designation silk-screened on the circuit card and ordering the appropriate cable as follows:

CABLE PLUG	CABLE PART NUMBER	
	500KVA	200KVA
1PL	36A359104AAG01	36A359104AAG01
2PL	104X154CB119	104X154CB119
3PL	36A359104AAG03	36A359104AAG03
4PL	36A359104AAG04	36A359104AAG17
5PL	36A359104AAG05	36A359104AAG05
6PL	36A359104AAG06	36A359104AAG18
7PL	36A358208AAG05	36A358208AAG05
9PL	36A359104AAG09	36A359104AAG09
10PL	36A359104AAG10	36A359104AAG19
11PL	36A359104AAG11	36A359104AAG20
12PL	36A359104AAG12	36A359104AAG21
13PL	36A359104AAG13	36A359104AAG22
14PL	36A359104AAG14	36A359104AAG23
18PL	36A358208ABG01	36A358208ABG01
APTPL	36A359104AAG08	none used
PFPL	36A359104AAG15	36A359104AAG15
CPTPL	36A359100BAG02	36A359100BAG05
OPTPL	36A359100ABG07	36A359100ABG16
3TBXPCOM	36A359100ARG03	36A359100ARG03

GLOSSARY OF TERMS AND NOMENCLATURE

A(1-6)F	— Converter SCR 1 thru 6 Firing Signals
A/D	— Analog to Digital Converter
ACOM	— Analog Common
(A,B,C) COV	— Commutation Overvoltage (A,B or C phase)
(A,B,C) CV	— Commutation Capacitor Voltage Test Point Phase A, B or C
AS(A,B,C)	— Auxiliary CPS SCR for Phase A,B,C (500KVA frames only)
BM (1,2)	— Blower Motor #1 (left) or #2 (right)
CC (A,B,C)	— Commutation Capacitor for Phase A,B or C
CCV (A,B,C)	— Commutation Voltage Feedback for Phase A,B or C
CFB	— Current Feedback
CFU1	— Control Fuse for -24VDC Power
CFU2	— Control Fuse for 115VAC Power
CFU3	— Control Fuse for +24VDC Power
C (N,P)	— Commutation Negative or Positive Bus
CPS	— Commutation Power Supply
CPSR	— Commutation Power Supply Output Resistor (500KVA frame only)
CPT	— Control Power Transformer
CPTFU(1,2)	— Control Power Transformer Fuse 1 or 2
CREF	— Current Reference Test Point
CS (N,P) (A,B,C)	— Commutation SCR (Negative or Positive) for Phase A,B or C
CT	— Current Transformer
CT (A,B,C)	— Current Transformer Phase A,B or C
CVF	— Converter Voltage Feedback
CVCO	— Converter Voltage Controlled Oscillator
CVRDB	— Converter Voltage Reference for Dynamic Brake Control
C1 (N,P)	— Link Capacitor Bank (Negative or Positive)
C2 (N,P)	— CPS Capacitor Bank (Negative or Positive)
D/A	— Digital to Analog Converter
DBT	— Dynamic Brake Trip
DBPL	— Dynamic Brake Control Plug
DCOM	— Digital Common
D(N,P) (A,B,C)	— Freewheeling Diode (Negative or Positive) for Phase A,B or C
EEROM	— Electronically Erasable Read-Only Memory, also called EE or E^2 (E squared). Contains controller parameters.
EE ENABLE DEVICE	— Device used to “Enable” or “Protect” EEROM programming. JP13 on the Control card.
EN	— Enable Signal
FC1	— Frequency Voltage Signal
FCC	— Filter Capacitor CT
FCMET	— Frequency Meter Voltage Output
FF	— Force Fire Command Signal
FU	— Fuse
FX1	— Fused 115VAC for Control
GFT	— Ground Fault Trip
“Hard Reset”	— Hardware Reset, resets the drive cards. Used to clear some faults and allow certain parameter changes. Activated by pressing SW2 on the Control card or connecting 3TB66 to 3TB69.
IFG	— Inverter Firing Generator (card)
IFT	— Inverter Fault Trip
IMET	— Current Meter Voltage Output
IS	— Inverter Start Relay (K1)
IS (N,P) (A,B,C)	— Inverter SCR (Negative or Positive) for Phase A,B or C
IOC	— Instantaneous Overcurrent
JOGR	— Jog Reference
JP	— Jumper

GLOSSARY OF TERMS AND NOMENCLATURE (continued)

K1	— Inverter Start (IS) relay located on Power Supply/Interface card
K2	— Run Relay located on Power Supply/Interface card
K3	— Fault Relay located on Power Supply/Interface card
LA	— Commutation Power Supply Output Reactor
LC (A,B,C)	— Commutation Reactor or Phase A,B or C
LC (N,P)	— 'Commutation Saturable Reactor (Negative or Positive)
LED	— Light Emitting Diode
LI (N,P) (A,B,C)	— Inverter Saturable Reactor (Negative or Positive) for Phase A,B or C
LK	— Link Capacitor Reactor
LVF	— Link Voltage Feedback
LVCO	— Link Voltage Controlled Oscillator
L(1,2)	— Link Reactor (1 is Positive, 2 is Negative)
L(1,2,3) C	— Input 3 phase - Line 1,2 or 3
MOV	— Metal Oxide Varistor
MSR	— Manual System Reference
MUP1,2,3	— Multiple Use Point on Control card
N1	— Negative Converter Output
N2	— Negative DC Link Output
PARAMETER	— Adjustable software setting used to program and tune the controller regulators.
PCN,P	— Process Reference Input (- and +)
PCOMX	— Power Supply Common Point
PI	— Process Interface (card)
P1	— Positive Converter Output
P2	— Positive DC Link Output
Q	— Link Capacitor Bank Midpoint
RAM	— Random Access Memory
RS232C	— Serial Link Communication Standard
RS422	— Serial Link Communication Standard
RT1,2,3	— AC Input Line Voltage Feedback - Line 1,2 or 3
R1(N,P)	— Link Capacitor Bank Resistor (Negative or Positive)
R2(N,P)	— CPS Capacitor Bank Resistor (Negative or Positive)
R (3,4)	— CPS Resistors (200 KVA frame only)
SC(N,P) (A,B,C)	— Commutation (Negative or Positive)SCR Snubber for Phase A,B or C
SCR	— Silicon Controlled Rectifier
SDC	— DC Link Snubber
SI (N,P) (A,B,C)	— Inverter (Negative or Positive) SCR snubber for phase A,B or C.
SLD	— Signal Level Detector, SLDO thru SLD4 (RX1 thru RX5) on Application card and SLD5 on 3TB.
"Soft Reset"	— Software Reset, initiated by pressing [RESET] key on the Progammer.
SRS	— System Reference test point on Control card.
THSW	— Thermal Switch
TP	— Test point
T1,2,3	— Output 3 Phase - Line 1,2 or 3
u v	— Under Voltage
v c o	— Voltage Controlled Oscillator
VMET	— Voltage Meter Output
12XIF	— 12 Times Inverter Frequency Pulse
(1,2,3)S	— Converter SCR Module Phase (1,2 or 3)
(1,2,3)SN	— Line (1,2 or 3) Negative Converter SCR
(1,2,3)SP	— Line (1,2 or 3) Positive Converter SCR
(1,2...)PL	— Ribbon Cable Harness



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